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Sampling Geometries and Band-Limits for Radiance Fields

*The size limit on a splatting primitive should be the inverse of the
information the capture accumulated about it, floored at the per-view Nyquist
limit — a matrix, not a scalar.*

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Baselines 3D Gaussian Splatting (2023) · Mip-Splatting (2024)

Abstract

3D Gaussian Splatting builds a radiance field out of several hundred thousand small translucent ellipsoids, fitted by gradient descent to a set of photographs. Nothing in that procedure sets a lower bound on how small a primitive may become. Mip-Splatting supplies one: a scalar variance floor, $f_k^2 = 0.2 (d_{\min}/f_{\max})^2$, added in quadrature to all three axes of every primitive. This thesis argues that the correct bound is a matrix, with a separate limit along each direction.

The reasoning treats a primitive’s position as something the photographs measure. Each camera that sees it contributes one noisy observation, and the precision of that observation is direction-dependent: a camera pins a point down well across its line of sight and poorly along it. Accumulating over all the cameras that saw the primitive gives a 3×3 matrix,

$$\Lambda_k = \sum_n w_{nk} (J_n W_n)^\top \Sigma_{\text{pix}}^{-1} (J_n W_n),$$

the Fisher information of that measurement model at unit weights. Its eigenvectors are the directions the capture constrained well and badly, and its eigenvalues say by how much. The claim tested is that structure finer than $1/\sqrt{\lambda_i}$ along eigendirection i is not identifiable from these images.

Three results follow. First, Mip-Splatting’s filter is a special case of this expression, obtained by keeping one view, discarding the depth direction and averaging what remains; its constant $\sqrt{0.2}$, introduced without derivation, is exactly $s \sigma_{\text{pix}}$, a confidence radius of 0.447 pixels. Second, the anisotropy has a closed form in three regimes: within a single view it falls off as $\cos^2 \varphi$ away from the optical axis; across two views separated by θ the spectrum is triaxial, $\{2, 2 \cos^2(\theta/2), 2 \sin^2(\theta/2)\}$; and across a distribution of views over an angular window, which is what a capture protocol is, the correct expression is a spherical-cap form that the two-view result understates by $\sqrt{2}$. Measured at each protocol’s own width, the observed conditioning of the 5 parallax protocols agrees with the cap form to 14–27 %, where the two-view expression falls short by up to 42 %. Third, flooring the anisotropic limit at Mip-Splatting’s own value makes the two methods coincide exactly wherever that value dominates, so parity on a well-conditioned benchmark is structural and an improvement there would indicate a bug.

A second proposal applies the same reasoning on the sphere of directions, masking the spherical-harmonic coefficients a capture cannot determine. Those are 76 % of the model’s parameters. Chapter 7 reports the controlled test: at matched structure and matched budget the criterion loses to magnitude pruning, and it is reported as refuted.

Concurrent work narrows what can be claimed for the main idea, and the narrowing is stated here. AAA-Gaussians (2025) already makes the 3D filter a full 3×3 , though derived from the view being rendered. PUP 3D-GS (2025) already builds a per-primitive second-order spatial sensitivity, though it collapses that to a scalar and uses it to prune. What remains new is where the anisotropy comes from: the conditioning of the training capture, fixed once per primitive, with its whole spectrum kept.

Current Status

Reproduction. Both baselines are reproduced end to end on free cloud compute. Mip-Splatting lands within 0.09 dB of its published figures at all four test scales, over the 8 scenes the published mean covers, which meets the numerical-reproduction standard set in advance. So does the gap between the two arms, within 0.20 dB, and that gap is the quantity the two-arm construction exists to produce. The 3DGS column claims nothing: one scene will not train on that arm, leaving a seven-scene mean standing against an eight-scene published figure.

Falsification Gates. Two were run before any method code existed, since either could have ended the project: does the resolution floor already bind everywhere, and is real capture geometry anisotropic enough to matter? Both passed on measurement.

The Anisotropic Filter. Implemented, and it reduces to its baseline where the theory requires: to 1.4×10^{-21} in the unit test, and to 0.028 dB over 7 scenes in paired training runs, below the run-to-run spread. It costs $2.8\times$ the peak memory and $1.32\times$ the training time of the baseline it equals on a well-conditioned capture, so on such a capture it should not be used.

The Stress Suite. On the 5 protocols where the resolution floor dominates, the estimation term exceeds it on zero primitives, the filter is isotropic to two decimals, and the difference from Mip-Splatting runs between -0.041 and $+0.017$ dB. On the sixth, a three-camera pencil spanning 29° , the estimation term exceeds the floor on 93 % of primitives, the filter becomes anisotropic at $1.47\times$, and the anisotropic filter leads by $+0.197$ dB on the mean and on every one of eight paired cells. The step falls where the estimation term overtakes the floor, as predicted and recorded beforehand. Two qualifications: the pencil reconstructs at 14 dB against the full orbit's 34, so what is established is the mechanism and not the method's usefulness at that operating point; and eight consistent pairings are eight cells of one run on two scenes.

The Angular Mask. Refuted. Against a control carrying its own nested-degree structure and its own budget, so that only the criterion differs, magnitude pruning wins by up to 0.415 dB. Chapter 5 set the standard that a criterion unable to beat magnitude pruning at matched size is not worth the machinery, and by that standard it fails.

Corrections Made During The Work. Four, each reported where it happened. The arm labelled 3DGS had kept Mip-Splatting's screen-space opacity compensation, so it was anti-aliasing better than 3DGS at the scales the gate was scored on; rebuilt and re-measured, the gate passes on all three criteria. Rows that a re-run had replaced were never retired, so every cell of one column became an average of two builds. Both measured columns of Table 7.1 were means over seven scenes compared against published figures that are means over eight, which inflated the apparent reproduction error fivefold. And the residual offset had been attributed to a change in the densification rule without being tested; measured against the commit that predates that change, the effect is 0.11 dB and is not consistently signed.

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Part I

Problem

Capture-Limited Resolution

Suppose you are handed eighty photographs of an object and asked to produce a picture of it from a viewpoint nobody stood at. This is the problem a radiance field solves. Before any of the machinery, consider what the photographs can tell you, and where they run out.

They sample the scene, and they sample it unevenly. A surface close to the cameras is covered by many more pixels than one far away. Some parts of the object are seen from many directions, others only from a narrow range, and a few may be seen from a single grazing angle or not at all. Different cameras may not even have the same focal length, so their pixels do not correspond to the same patch of surface. Whatever you reconstruct from such a set, there is a limit on how fine the detail can be, and that limit belongs to the capture. It follows the photographs into whatever representation stores the answer, and no choice of representation lifts it.

NeRF acts on this directly. Its positional encoding uses L frequency bands, and L is chosen from, in the authors’ words, “the maximum frequency present in the sampled input images”. The quantity is the right one; the instrument is one number, picked by hand, applied to an entire scene.

3D Gaussian Splatting drops even that. Its authors state that no regularisation is applied, and a primitive may therefore shrink without bound: nothing in the objective opposes it. As long as the model is evaluated at the resolution it was fitted at, this does no visible harm, which is why it went unnoticed for so long. Change the resolution and the damage appears at once. Section 3.2 puts a number on it: 15.64 dB in the published figures, and 15.61 dB in this project’s own reproduction.

Mip-Splatting restored a limit, and the restoration works; the numbers in Chapter 7 confirm it comfortably. But the limit it restores is a single scalar per primitive, added in quadrature to all three axes of the covariance. This thesis asks what that scalar ought to have been, and answers that the right object is a matrix.

1.1 The Claim

THESIS

“The maximum frequency present in the sampled input images” varies from place to place and from direction to direction. It falls with depth; it differs across the ray from along it; and it varies over the sphere of viewing directions. The accumulated sampling Fisher information is its local, directional generalisation. Floored at the per-view Nyquist limit, it reduces to Mip-Splatting exactly where the capture is well conditioned, and departs from it precisely where the capture is ill conditioned.

1.2 The Signal-Processing Problem

Stripped of its graphics vocabulary, the problem is reconstruction from non-uniform samples without a matched prefilter — a classical signal-processing failure mode. The objects involved are familiar ones.

Table 1.1: The same objects, in two vocabularies.

graphics term	signal-processing object
Anti-aliasing filter (2D Mip filter)	Reconstruction prefilter matched to the display sampling rate
3D smoothing filter	Prefilter matched to the <i>acquisition</i> sampling rate
Primitive covariance Σ_k	Support of a basis function in a non-orthogonal expansion
“Too many small Gaussians”	Over-parameterisation relative to the information in the measurements
Floater and depth artefacts	Fitting components along directions the data does not constrain
Spherical-harmonic view dependence	Basis expansion whose Gram matrix may be rank-deficient

The contribution is to compute the acquisition-side prefilter from the conditioning of the capture instead of choosing it by hand, and to keep it anisotropic because the information is. AAA-Gaussians (2025) already has an anisotropic 3D filter, derived from the view being rendered, and PUP 3D-GS (2025) already builds a per-primitive second-order information matrix in order to prune. Section 3.4 states the line of separation precisely and Chapter 8 returns to it; the short form is that what is new here is the *source* of the anisotropy and the retention of its *spectrum*.

1.3 Contributions

1. **A band-limit derived from the capture.** The finest structure a primitive may carry, per direction, computed from an observability matrix accumulated over the cameras that saw it, using the same Jacobian the renderer already forms (§4.2–§4.3). The limit is a property of the capture, fixed once per primitive, and its whole spectrum is kept.
2. **Mip-Splatting’s filter identified as a special case.** It is the single-view, lateral-only, isotropised reduction of that expression, and its undocumented constant $\sqrt{0.2}$ is $s\sigma_{\text{pix}}$: a confidence radius of 0.447 pixels (§4.5).
3. **Closed forms for the anisotropy in three regimes.** Within one view, $\lambda_2/\lambda_1 = \cos^2 \varphi$ off-axis (§4.6). Across two views, the triaxial spectrum $\{2, 2\cos^2(\theta/2), 2\sin^2(\theta/2)\}$ (§4.7). Across a distribution of views, the spherical-cap form (§4.7.1), which a real protocol obeys and which the two-view expression understates by $\sqrt{2}$. Together they predict where the method acts and where it cannot.
4. **An exact-reduction theorem.** Flooring the anisotropic limit at Mip-Splatting’s own value makes the two coincide wherever that value dominates, so parity on a well-conditioned benchmark is structural (§4.8).
5. **A criterion for angular identifiability of spherical harmonics**, which are 76 % of

the model, and a controlled test of it against magnitude pruning at matched structure and matched budget (Chapter 5, §7.13).

6. **An executed reproduction of both baselines** on free compute, with twelve pre-flight assertions and the per-scale recovery the published evaluation does not provide (Chapters 6 and 7).
7. **A measured cost for the method**, paired in one session on one card: $2.8\times$ peak memory and $1.32\times$ training time for a model whose primitive count matches the baseline's to 0.2% (§7.11).
8. **Checks that catch measurement errors which read as plausible.** A build that refuses any table whose cells span two code versions, a test that holds the thesis, the figures and the tabulated results to the same numbers, and a record that keeps every superseded run. Four errors in this project's own measurement were found this way, each reported with the correction it forced (Appendix G).

Splatting And Existing Band-Limits

THE IDEA

A radiance field is a model of a scene that you can photograph from a new angle. You give it a few dozen real photographs; it fits something to them; you then ask it for a picture from a viewpoint nobody stood at.

3D Gaussian Splatting fits a few hundred thousand small translucent blobs — ellipsoids with a colour and an opacity. Rendering is cheap because a blob projects to an ellipse, and drawing a few hundred thousand ellipses in depth order is something a GPU does very well. Fitting is gradient descent: render, compare to the photograph, nudge every blob.

Nothing in that procedure stops a blob from becoming smaller than anything the photographs can resolve. If a blob ends up a tenth of a pixel across, it still gets drawn — at full opacity, into whichever pixel contains its centre. At the resolution you trained on this stays invisible, because the fit compensates. Change the resolution and it shows: the sub-pixel blobs land somewhere other than where the fit assumed, and the image acquires energy it should not have. That is what you are seeing in Figure 7.1, where the same model is fine at full resolution and a bright smear at one-eighth.

The fix, in general terms, is a *band-limit*: a floor on how small a blob is allowed to be. This chapter sets out where each published method puts that floor and what it is made of. The rest of the thesis argues that the floor should be a matrix, with a separate limit along every direction, and measures what happens when you make it one.

2.1 The Emission–Absorption Model

Both baselines assume the same emission–absorption model. Along a ray $r(t) = o + td$ with density σ and colour c , the rendered radiance is

$$C = \int T(t) \sigma(r(t)) c(r(t), d) dt, \quad T(t) = \exp\left(-\int_0^t \sigma du\right). \quad (2.1)$$

Discretised, this is the α -blending sum both methods use. 3DGS states the equivalence between its α -blending and NeRF’s ray marching directly; the two differ in how the field is parameterised and evaluated; what is being rendered is the same in both. This is what allows a single argument to span both.

2.2 NeRF’s Frequency Limit

NeRF stores the field in an MLP applied to a positional encoding $\gamma(x)$ with L frequency bands. The MLP’s spectral bias means the encoding band sets the finest structure representable. The authors set L by hand from the input images. The band-limit is therefore present, explicit, global, and manually chosen — one scalar for the entire scene.

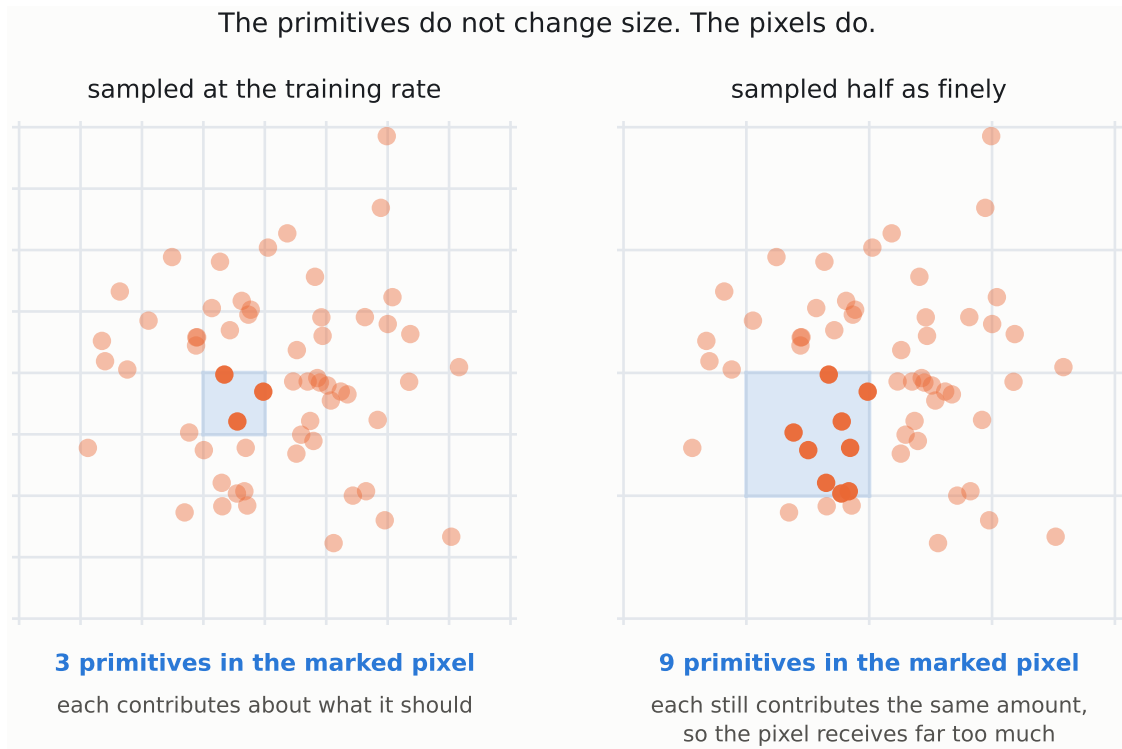


Figure 2.1: The failure in one picture. The orange discs are primitives; they are a property of the fitted model and do not change when you change the camera. The grid is the pixel grid, and it does. At the rate the model was fitted at, the marked pixel receives three primitives and the fit has set each one's opacity so that the total is right. Sample half as finely and the same pixel receives nine — each still contributing what was correct when it had a pixel to itself. The pixel gets roughly three times the light it should. Repeat over the whole image and the reconstruction brightens and thickens as well as blurring, which is what Figure 7.1 shows happening to a real scene.

2.3 3D Gaussian Splatting And EWA Projection

Splatting in three stages. This thesis is about a limit at stage 1; both baselines already have one at stage 2.

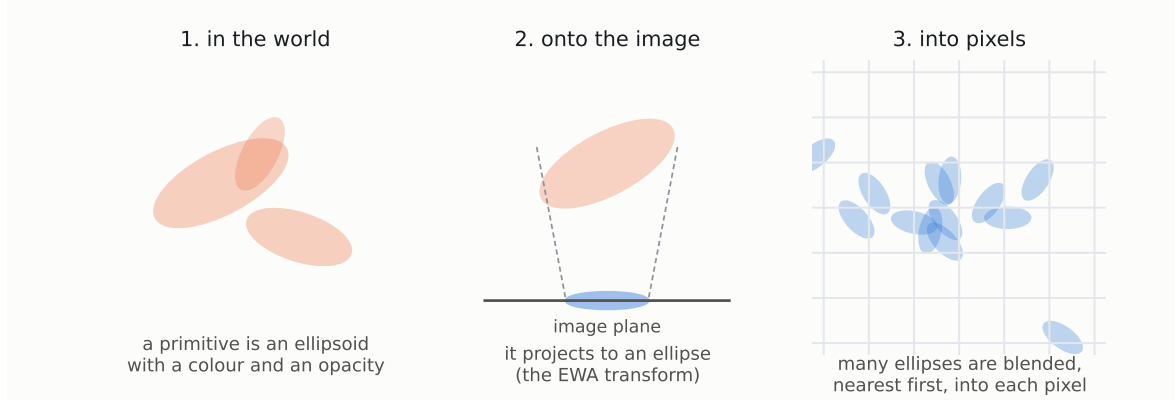


Figure 2.2: Splatting, in the three stages a reader needs before anything else in this thesis makes sense. A primitive is an ellipsoid sitting in the world, carrying a colour and an opacity. To render it, the ellipsoid is projected to an ellipse on the image plane, and the EWA transform of Equation (2.2) is what performs that projection. Many such ellipses are then blended into each pixel, nearest first. Both baselines already place a limit at stage 2, on how small a primitive’s *image* may be. This thesis is about a limit at stage 1, on how small the primitive *itself* may be, and the two are not interchangeable: a stage-2 limit is recomputed for every view, whereas a stage-1 limit is a property of the model.

3DGS represents the field as anisotropic Gaussians with mean p_k and covariance $\Sigma_k = RSS^\top R^\top$. Projection follows EWA volume splatting: with viewing transform W_n and the Jacobian J_n of the affine approximation to the projective map,

$$\Sigma' = J_n W_n \Sigma_k W_n^\top J_n^\top. \quad (2.2)$$

This pair (J_n, W_n) is the object Chapter 4 reuses. No regularisation is applied to Σ_k ; nothing bounds how small a primitive may become.

2.4 Mip-Splatting’s Scalar Floor

Mip-Splatting adds two filters, and it matters for Chapter 7 that they are separate objects.

The 2D Mip Filter is a render-side reconstruction prefilter matched to the display sampling rate. It dilates the projected covariance by kI and compensates the opacity by $\rho = \sqrt{|\Sigma'|/|\Sigma' + kI|}$, so that widening the splat does not brighten it. Vanilla 3DGS performs the same fixed dilation, with $k = 0.3$, and applies *no* compensation at all. The proposed method leaves this filter alone. The distinction matters, because an arm that keeps ρ is anti-aliasing better than 3DGS in screen space whatever its 3D filter does, and §7.3 reports the price this project paid before the difference was noticed.

The 3D Smoothing Filter is the acquisition-side limit at issue here: a scalar variance

$$f_k^2 = 0.2 \left(\frac{d_{\min, k}}{f_{\max}} \right)^2, \quad \Sigma_k \leftarrow \Sigma_k + f_k^2 I, \quad (2.3)$$

with $d_{\min,k}$ the smallest depth at which any camera sees primitive k and $f_{\max}$ the largest focal length over cameras, taken independently. The opacity is compensated so that total mass is preserved under the convolution,

$$\alpha_k \leftarrow \alpha_k \sqrt{|\Sigma_k| / |\Sigma_k + f_k^2 I|}. \quad (2.4)$$

Equations (2.3) and (2.4) are the baseline this thesis generalises, and Proposition 2 recovers them exactly.

CHAPTER 3

Limitations Of A Scalar Band-Limit

THE IDEA

Both baselines limit how small a primitive may be, and both use a single number to do it. A number says “no finer than this, in every direction”. A capture constrains some directions far better than others. Photograph an object from a narrow wedge of angles and you will know a great deal about where its surfaces are side-to-side and very little about how far away they are — the classic weak-parallax situation, familiar from stereo. The correct limit in that case is generous sideways and strict in depth. A scalar cannot say that. This chapter shows precisely where each method’s scalar lives, what it is computed from, and what information it throws away in becoming a scalar.

3.1 World-Space And Screen-Space Limits

	world-space limit (how small may a primitive be?)	screen-space limit (how small may its image be?)
3DGS	no world-space floor at all	a fixed dilation of the projected ellipse, applied at render time
Mip-Splatting	one number per primitive, from the nearest camera	the same fixed dilation, plus an opacity correction
this work	a 3×3 matrix per primitive, from every camera that saw it	unchanged — the screen-space term is not what this touches

The column on the left is the subject of this thesis. Nothing here changes the column on the right.

Figure 3.1: There are two places a limit can sit, and they answer different questions. A screen-space limit asks how small a primitive’s *image* may be, and is applied afresh for every view; both baselines have one and this thesis does not touch it. A world-space limit asks how small the primitive *itself* may be, is fixed once, and is the subject of everything that follows. 3DGS has no world-space limit at all. Mip-Splatting has one, and it is a single number derived from one camera. The proposal is to make it a matrix derived from all of them.

One quantity orders the progression: how much of the capture geometry each method reads. NeRF takes a global glance at the images; 3DGS reads nothing; Mip-Splatting reads one view’s worth; this work reads all of it.

Table 3.1: One quantity, the finest structure the capture can determine, estimated with successively more of the available geometry.

method	where the limit lives	how it is set	granularity
NeRF (2020)	positional-encoding band L	by hand, from “the maximum frequency present in the sampled input images”	one scalar, whole model
3DGS (2023)	nowhere	not set — no regularisation is applied	absent
Mip-Splatting (2024)	variance added to the covariance	f_k^2 from nearest depth and largest focal, $s = 0.2$	one scalar per primitive
this work	covariance added to the covariance, floored at the above	accumulated sampling Fisher information	per primitive, per direction

3.2 The Cost Of A Missing Limit

The left panel is the experiment this project inherits, and its magnitude is the reason the direction is worth pursuing: an 11 dB gap between a method with a band-limit and a method without one says the band-limit is doing most of the work. The question is then whether the *right* band-limit does more. Chapter 7 reports this project’s own measurement of the same quantity, which is the control for everything that follows.

3.3 Three Defects Of The Scalar Filter

1. **It ignores parallax.** f_k depends on one view. A primitive seen from fifty directions is assigned the same limit as one seen from a single direction at the same distance, although the former is far better localised.
2. **It is isotropic.** The lateral limit is applied along depth, where a single view supplies no constraint at all. Section 4.7 shows this under-smooths depth by a factor $1/\sin(\theta/2)$.
3. **It mixes cameras.** $d_{\min}$ and $f_{\max}$ come from different views, so the quotient corresponds to no physical pixel footprint. The repository itself carries the comment `#TODO consider focal length and image width` on the current branch.

3.4 Concurrent Work

A due-diligence review was run before the stress suite, with one question: has anyone already made the 3D filter anisotropic or per-direction? The answer is partly yes, and the contribution stated in §1 has been narrowed accordingly. Two of the five papers below change what this thesis may claim; and the change is recorded here, in the chapter that sets up the gap.

THE ACTUAL CLAIM

AAA-Gaussians already has an anisotropic 3D filter, and PUP 3D-GS already builds a per-primitive second-order information matrix. This work claims neither. What it claims, and all the experiments are asked to support, is a *per-primitive band-limit read off the conditioning of the training capture*: a quantity the cameras that saw the primitive fix once, ahead of

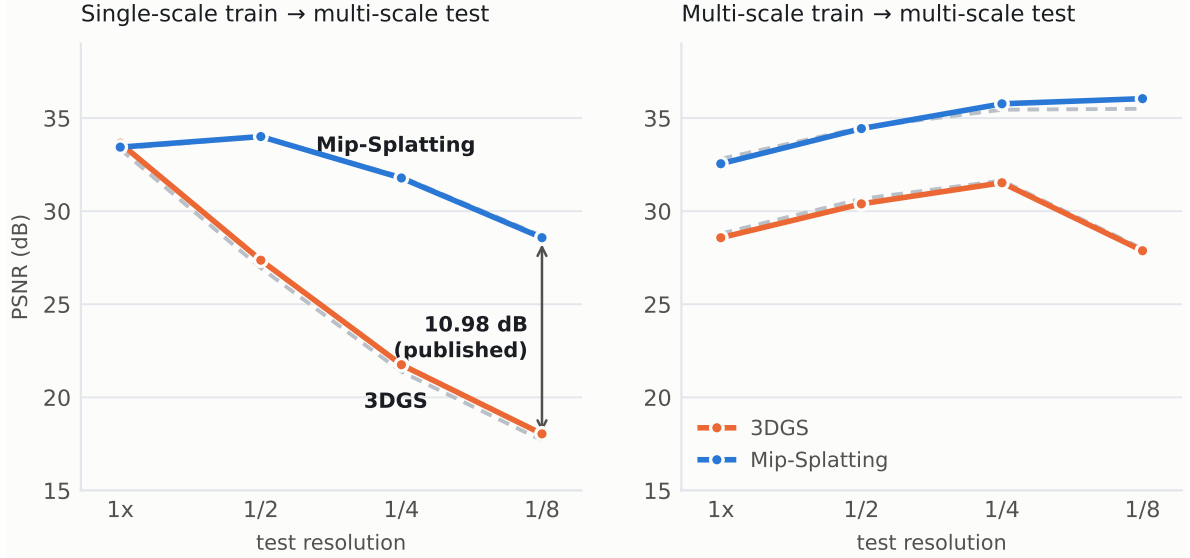


Figure 3.2: The failure the band-limit prevents. Blender synthetic, mean PSNR, from Mip-Splatting Tables 1–2 (faint dashed) with this project’s own reproduction overlaid where measured. Each measured curve is a mean over the scenes that arm has, matching the eight-scene denominator of the published curve it is drawn against; the $\frac{1}{8}$ arrow is annotated from the *published* pair, because the two measured curves are means over different scene sets (*ship* trains only on the Mip-Splatting arm, §G.2) and the distance between them therefore spans two different scene sets. *Left:* trained at one scale, tested at four. 3DGS loses 15.64 dB in the published numbers as the test sampling rate departs from the training rate, because primitives have shrunk below what any view resolves; Mip-Splatting loses 4.69 dB. *Right:* when the training set already spans all four scales, 3DGS partly recovers but still trails everywhere — the representation absorbs the scale mismatch into the fit instead of into a prefilter.

Table 3.2: The five papers that could have pre-empted this work, and the line of separation in each case. Each was read in full and recorded the same way — the claim, the mechanism, and the evidence that settled whether it pre-empts this one.

work	what it does	relation to this thesis
AAA-Gaussians (arXiv:2504.12811, 2025)	Replaces the 2D Mip filter with a full 3×3 3D filter, whose anisotropy is decomposed about the <i>current view ray</i> and recomputed every rendered frame.	Removes the novelty of “an anisotropic 3D filter”. What remains is the <i>source</i> of the anisotropy: theirs is the view being rendered, this thesis’s is the conditioning of the capture that was available. The two compose: a system could carry both.
PUP 3D-GS (CVPR 2025)	A second-order sensitivity of the photometric error in each primitive’s spatial parameters, accumulated over training views, collapsed to a scalar and thresholded to prune 90% of primitives.	The matrix has a precedent. It is built from the same ingredients as Λ_k . The divergence is what survives: PUP keeps a scalar and discards the eigenstructure, which is precisely what the anisotropic filter uses.
FisherRF (ECCV 2024)	Fisher information over model parameters as an acquisition function for next-best-view selection.	Same statistical object, opposite direction of use: it chooses which camera to <i>acquire</i> ; the anisotropic filter takes the cameras as given. Changes the dataset and leaves the primitive alone.
3DGS-MCMC (NeurIPS 2024)	Recasts densify/prune/reset as MCMC sampling, removing the heuristics and the dependence on initialisation.	Orthogonal and composable. No filter, no band-limit, no covariance floor. Cited as the nearest principled density-control work.
Analytic-Splatting (ECCV 2024)	Integrates the Gaussian response over the square pixel window in closed form instead of approximating the window by a 2D Gaussian.	Improves the <i>2D</i> half of Mip-Splatting; this thesis changes the <i>3D</i> half. Directly composable.

any render, and that keeps its whole spectrum. A primitive seen from a narrow baseline is under-determined along depth no matter which view renders it, and a filter derived from the render-time ray carries no memory of that. The `cone` and `arc` protocols of §6.4 exist to isolate that regime, and nobody has tested it, because nobody else's filter is a function of the training-camera spread.

Part II

Method

An Anisotropic Band-Limit

THE IDEA

Think of each blob’s position as something the photographs *measure*. Every training photograph that sees the blob is a measurement of where it is, and every measurement has noise — roughly a pixel’s worth, in the image plane.

Now ask the standard estimation question: given all those measurements, how precisely do you know the position? The answer varies with direction. A camera localises a point very well within its image plane and poorly along its own viewing direction; a second camera at a different angle fixes the first camera’s weak direction. Accumulate over all the cameras that saw the blob and you get a 3×3 matrix — the familiar Fisher information of the measurement model, written Λ_k throughout. Its eigenvectors are the directions the capture constrains well and badly; its eigenvalues say how well.

The proposal is then almost forced: *a primitive should not be finer, in any direction, than the capture can determine it in that direction*. Invert the matrix, and you have a covariance floor with the right shape. Then floor that at the per-view Nyquist limit, since no amount of accumulated information lets you resolve structure finer than a pixel, and you have the method.

Three things follow, and this chapter proves them. Mip-Splatting’s scalar filter is what you get by collapsing this matrix to a single view and throwing away direction. Its mysterious constant $\sqrt{0.2}$ turns out to be a confidence radius of 0.447 pixels. And when the Nyquist floor dominates, which on a normal benchmark capture it always does, the new method becomes the old one exactly. That last point is why the standard benchmarks in Chapter 7 show no improvement, and why a gain there would have indicated a bug.

4.1 The Estimation Problem

Regard the position p_k of a primitive as a parameter to be estimated from image measurements. Camera n observes the primitive’s projection at

$$u_n = \pi_n(p_k) + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \Sigma_{\text{pix}}), \quad \Sigma_{\text{pix}} = \sigma_{\text{pix}}^2 I_2, \quad (4.1)$$

where σ_{pix} is the localisation noise of an image feature, in pixels. The log-likelihood of the measurement from view n is, up to constants, $-\frac{1}{2}(u_n - \pi_n(p))^\top \Sigma_{\text{pix}}^{-1}(u_n - \pi_n(p))$.

4.2 The Measurement Jacobian

The Fisher information carried by view n about p is

$$I_n = \left(\frac{\partial \pi_n}{\partial p} \right)^\top \Sigma_{\text{pix}}^{-1} \left(\frac{\partial \pi_n}{\partial p} \right), \quad \frac{\partial \pi_n}{\partial p} = J_n W_n. \quad (4.2)$$

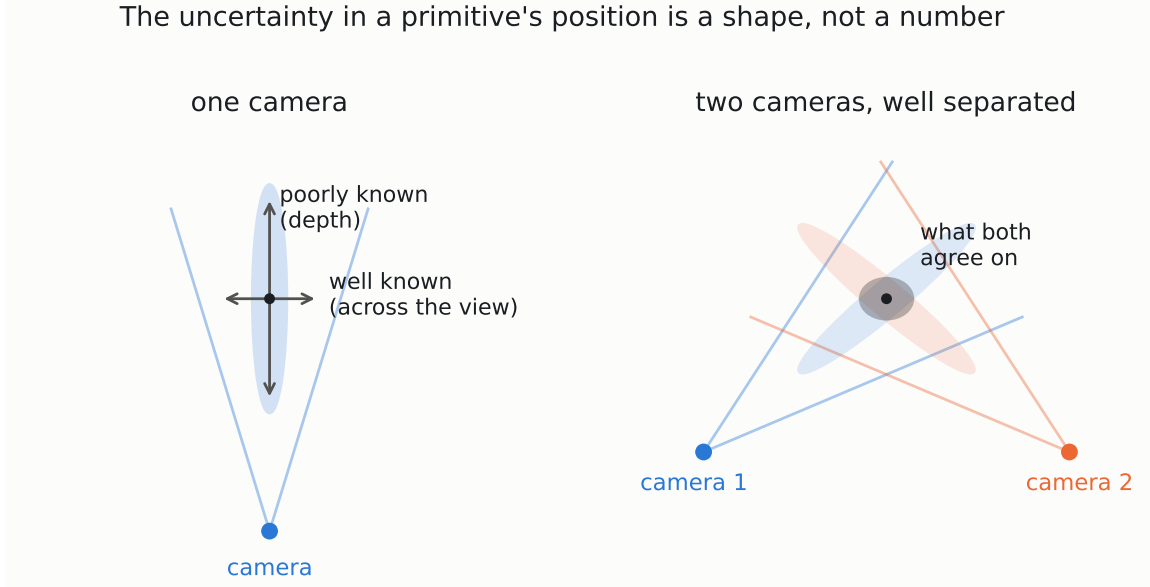


Figure 4.1: Why the limit has to be a matrix. A single camera fixes a point well across its line of sight and badly along it, so the region consistent with its measurement is a long thin cigar pointing back at the lens — the familiar fact that one photograph gives you direction but not distance. A second camera from elsewhere is precise where the first is vague, and the region consistent with both is compact. Accumulating this over every camera that saw the primitive is what Λ_k does, and its eigenvectors are these axes. A scalar limit cannot distinguish the long axis from the short one; that is the whole of the argument.

The derivative of image position with respect to world position is the EWA pair of Equation (2.2). *No new geometry is introduced:* the quantity the renderer forms to project a covariance is the same quantity estimation theory requires to bound a position. This identification is the technical core of the thesis, and it is what makes the method cheap.

Rank. $J_n W_n \in \mathbb{R}^{2 \times 3}$, so I_n has rank at most 2. A single view constrains the two directions transverse to the ray and says nothing about depth. For a pinhole camera the eigenvalues of I_n are exactly $(0, (f/z)^2/\sigma_{\text{pix}}^2, (f/z)^2/\sigma_{\text{pix}}^2)$ (Appendix B.1). Depth information exists only through parallax.

4.3 The Observability Matrix

Summing over views, weighted by each view’s visibility and blending contribution w_{nk} , which the rasteriser already accumulates,

$$\Lambda_k = \sum_n w_{nk} (J_n W_n)^\top \Sigma_{\text{pix}}^{-1} (J_n W_n) = V \text{diag}(\lambda_1, \lambda_2, \lambda_3) V^\top. \quad (4.3)$$

The Weight, Defined. $w_{nk} = T_{nk} \alpha_k \in [0, 1]$ is the primitive’s blending contribution in view n : the transmittance accumulated in front of it times its own opacity, which is the coefficient the rasteriser already multiplies its colour by. It is zero for a primitive the view does not see and near one for an unoccluded opaque one. This weighting is a *heuristic* down-weighting of occluded views, standing outside any likelihood: a view that cannot see a primitive carries no

information about it, and w_{nk} is the cheapest available statement of that fact. It is stated as a heuristic because calling it anything more would be a claim the derivation does not support.

What The Matrix Is, Precisely. With unit weights, Equation (4.3) *is* the Fisher information of the measurement model in Equation (4.1). With the weights it departs from that, so the honest name for Λ_k is an **observability matrix** — a view-weighted accumulation of the per-view measurement information, coinciding with the Fisher information exactly when $w_{nk} \equiv 1$. Under the point-feature approximation of Equation (4.1), Equation (4.3) is also the Gauss–Newton approximation to the Hessian of the photometric cost in p_k : the residual is linearised through $J_n W_n$ and second-derivative terms are dropped, as every bundle adjustment does and for the same reason.

Why This Matters For The Claim. The Cramér–Rao bound $\text{Cov}(\hat{p}_k) \succeq \Lambda_k^{-1}$ holds for an unbiased estimator of the measurement model as written. A splatting optimiser is not that estimator: it fits a photometric loss over a densifying set of correlated primitives, not independent point features with Gaussian localisation noise. Cramér–Rao is therefore kept as the *motivation* for the form of Equation (4.4): it says why the inverse of an information matrix is the right shape for a resolution limit. The claim the experiments actually test is weaker, and sufficient: along eigendirection i the capture constrains position to order $1/\sqrt{\lambda_i}$, so structure finer than that is not identifiable from these images.

PRINCIPLE

A primitive should not be narrower, along a given direction, than the uncertainty with which the capture determines its position along that direction. Structure finer than $s/\sqrt{\lambda_i}$ is not identifiable from the measurements; fitting it is fitting noise.

4.4 The Covariance Floor

Hence a per-direction variance, floored at Mip-Splatting’s own value and capped at a scene-scale fraction:

$$\sigma_i^2 = \min\left(\max(f_k^2, s^2/\lambda_i), (\beta z_{\min,k})^2\right), \quad (4.4)$$

$$\Sigma_{\text{filt}} = V \text{diag}(\sigma_1^2, \sigma_2^2, \sigma_3^2) V^\top; \quad \Sigma_k \leftarrow \Sigma_k + \Sigma_{\text{filt}}; \quad \alpha_k \leftarrow \alpha_k \sqrt{|\Sigma_k|/|\Sigma_k + \Sigma_{\text{filt}}|}. \quad (4.5)$$

Two hyperparameters are introduced and none removed. β caps the filter in degenerate directions where $\lambda_i \rightarrow 0$; without it an unconstrained direction would produce an unbounded primitive. The opacity compensation is Mip-Splatting’s own, unchanged, and Appendix B.4 shows it carries over to an anisotropic Σ_{filt} without modification.

s And σ_{pix} , Disentangled. These two appear only as the product $s \sigma_{\text{pix}}$, so quoting a default for one and a value for the product over-determines them. The convention used throughout, and the one the implementation follows:

- $\sigma_{\text{pix}} = \sqrt{0.2} = 0.447$ px is *inherited* from Mip-Splatting’s own constant. Proposition 1 shows it is the localisation noise implicit in Mip-Splatting’s constant, and taking it verbatim is what makes Proposition 2 exact instead of merely close.

- s is the dimensionless confidence multiplier, default **1** (one standard deviation). It is the only free knob of the pair, and every result in this thesis is at $s = 1$.

So $s\sigma_{\text{pix}} = 0.447$ px at the default, and a reader changing s changes the confidence level and nothing else.

Consistency Of The Floor And The Cap. The floor must never exceed the cap, or Equation (4.4) would be ill-posed. Requiring $f_k \leq \beta z_{\min,k}$ and substituting Equation (2.3) gives $\beta \geq \sqrt{0.2}/f_{\max}$. For $f_{\max} \approx 1000$ px this is $\beta \geq 4.5 \times 10^{-4}$; the default $\beta = 0.01$ carries $22\times$ headroom.

4.5 Mip-Splatting As A Special Case

Take a single camera with focal length f viewing a primitive at depth z on the optical axis. From §4.2 the lateral eigenvalue is $\lambda_{\text{lat}} = (f/z)^2/\sigma_{\text{pix}}^2$, so the estimation-limited lateral scale is $s/\sqrt{\lambda_{\text{lat}}} = s\sigma_{\text{pix}} z/f$.

Proposition 1 (Identification of the constant). *Mip-Splatting’s 3D smoothing filter is exactly Equation (4.4) evaluated for a single view, restricted to the lateral directions, and applied isotropically — provided*

$$s \cdot \sigma_{\text{pix}} = \sqrt{0.2} \approx 0.447 \text{ pixels.}$$

Their constant therefore carries a meaning: a sub-pixel localisation confidence of 0.447 px, slightly under half a pixel. The present method keeps that calibration and generalises the geometry around it. This is the strongest single argument in the thesis: it converts the relationship between the proposed method and its baseline from “similar in spirit” into “the baseline is a corollary”, and it explains a magic number in a published paper. Appendix B.5 gives the derivation in full.

4.6 Off-Axis Anisotropy

Before parallax enters at all, there is anisotropy the scalar filter discards. Section 4.2 quotes the single-view eigenvalues $(0, (f/z)^2, (f/z)^2)/\sigma_{\text{pix}}^2$, and that is exact — *on the optical axis*. The convenient rank-1 form $(f/z)^2(I - dd^\top)/\sigma_{\text{pix}}^2$ suggests otherwise, because $I - dd^\top$ is a projector and a projector’s non-zero eigenvalues are equal by construction, whatever d is. Off-axis, the true $J_n^\top J_n$ departs from a projector, and the two lateral eigenvalues separate.

Writing $a = (x/z, y/z)$ for the normalised off-axis displacement, $J = (f/z)B$ with $B = \begin{bmatrix} 1 & 0 & -a_1 \\ 0 & 1 & -a_2 \end{bmatrix}$, so $BB^\top$ has trace $2 + |a|^2$ and determinant $1 + |a|^2$. Its eigenvalues are therefore $1 + |a|^2$ and 1, giving

$$\frac{\lambda_2}{\lambda_1} = \frac{1}{1 + |a|^2} = \cos^2 \varphi, \quad \varphi = \arctan |a| \text{ the off-axis angle.} \quad (4.6)$$

At the corner of a 50° -field camera φ approaches 25° and the two lateral eigenvalues already differ by 20 %; the sampled poses of Table 4.1 run from 0.93 down to 0.47. So **perspective contributes anisotropy independently of parallax**, and a scalar filter discards it before the parallax argument of §4.7 has even begun. The implementation uses the full $J_n W_n$ and never the rank-1 shortcut, so this term survives into the filter, where the shortcut would

Table 4.1: M-4 — a single view is already anisotropic off-axis. The rank-1 shortcut $(f/z)^2(I - dd^\top)$ is a projector, so its two non-zero eigenvalues are equal by construction; the true $J^\top J$ separates them by exactly $\cos^2 \varphi$. Four randomly drawn poses, and the law verified over 2000 more.

φ (off-axis)	λ_2/λ_1 measured	$\cos^2 \varphi$	diff
14.9°	0.9341	0.9341	$< 10^{-15}$
24.6°	0.8272	0.8272	$< 10^{-15}$
26.6°	0.7990	0.7990	$< 10^{-15}$
46.9°	0.4660	0.4660	$< 10^{-15}$

Maximum error of the law over the sweep: 8.9×10^{-16} . At a representative off-axis point the shortcut differs from the truth by 17% of the largest entry — and differs in *shape*, not by a scalar, so no choice of scalar filter absorbs it.

Table 4.2: M-5 — the exact two-view spectrum. $\Lambda \propto 2I - d_1 d_1^\top - d_2 d_2^\top$ is diagonal in the frame of the two rays with eigenvalues $\{2, 2\cos^2(\theta/2), 2\sin^2(\theta/2)\}$. Three distinct values: the filter is *triaxial*, and the in-plane lateral direction is degraded as well as depth.

θ	λ_1	λ_2	λ_3	$\lambda_{\min}/\lambda_{\max}$	$\sin^2(\theta/2)$
2°	2.0000	1.9994	0.000609	0.000305	0.000305
6°	2.0000	1.9945	0.005478	0.002739	0.002739
15°	2.0000	1.9659	0.034074	0.017037	0.017037
45°	2.0000	1.7071	0.292893	0.146447	0.146447
90°	2.0000	1.0000	1.000000	0.500000	0.500000
120°	2.0000	1.5000	0.500000	0.250000 [†]	0.750000
179°	2.0000	1.9998	0.000152	0.000076 [†]	0.999924

Measured against numerical eigendecomposition; maximum absolute error 4×10^{-16} . [†] past $\theta = 90^\circ$ the depth direction is better constrained than the in-plane lateral one and the two exchange roles, so the ratio of smallest to largest is $\min(\sin^2(\theta/2), \cos^2(\theta/2))$ and not $\sin^2(\theta/2)$. Equation (4.7) is stated for $\theta \leq 90^\circ$, which is the whole of the regime the stress protocols occupy.

Table 4.3: M-5b — a protocol is a distribution, not two views. For cameras spread over a cap of half-angle α , $\lambda_3/\lambda_1 = 2(1 - E)/(1 + E)$ with $E = (1 + c + c^2)/3$, $c = \cos \alpha$ — exact, and asymptotically *half* the two-view value, because a cap’s cameras are spread across the interval, where the two-view form places them at its two ends. The two-view form therefore *understates* a real capture’s depth anisotropy by $\sqrt{2}$ in σ .

span	cap σ_D/σ_L	two-view σ_D/σ_L	measured
6°	27.01×	19.11×	
20°	8.10×	5.76×	9.42× (narrow pencil)
41°	3.95×	2.86×	
60°	2.70×	2.00×	
80°	2.03×	1.56×	
93°	1.76×	1.38×	2.10× (one-sided arc)
120°	1.38×	1.15×	
168°	1.04×	1.01×	1.42× (grazing)
179°	1.00×	1.00×	1.14× (full orbit)

Closed form checked against 200 000 sampled directions per span, maximum relative error 3×10^{-03} . The measured column is I-2 on the protocol whose median span matches the row, from Table 7.9.

discard it for good; the shortcut appears in this document only where it makes an argument shorter, and Equation (4.6) records what it costs. This is also the most likely explanation for the residual in §7.10: the grazing protocol, which is a full orbit tilted towards obliquity, measures more anisotropy than any parallax-only model predicts.

4.7 The Two-View Spectrum

Take the simplest case: two cameras at the same distance z , placed symmetrically about the primitive, so that the angle between their lines of sight is θ . Evaluating Equation (4.3) for this arrangement (the algebra is in Appendix B.2) gives $\Lambda \propto 2I - d_1 d_1^\top - d_2 d_2^\top$, and this has the exact spectrum $\{2, 2 \cos^2(\theta/2), 2 \sin^2(\theta/2)\}$.

These are three distinct numbers. The right filter is therefore triaxial, with a different limit along each of three axes: the lateral direction lying in the plane of the two rays is degraded as well, and only the direction perpendicular to that plane is fully constrained. Taking the extreme pair of eigenvalues,

$$\frac{\lambda_3}{\lambda_1} = \sin^2(\theta/2), \quad \frac{\sigma_{\text{depth}}}{\sigma_{\text{lat}}} = \frac{1}{\sin(\theta/2)} \quad (\theta \leq 90^\circ). \quad (4.7)$$

This is exact at every angle — verified against pinhole Jacobians to 4×10^{-16} over $\theta = 2^\circ$ to 179° (Table 4.2). The restriction $\theta \leq 90^\circ$ earns its place: past 90° the depth direction becomes better constrained than the in-plane lateral one and the two exchange roles, so the ratio of smallest to largest is $\min(\sin^2(\theta/2), \cos^2(\theta/2))$. At $\theta = 120^\circ$ the naive form reads 0.75 where the truth is 0.25 — a factor of three, in the optimistic direction. Every protocol this thesis stresses lies well below 90° , and the full orbit lies so far above it that the ratio is ≈ 1 either way, so the restriction costs nothing here and guards against a real error elsewhere.

4.7.1 The Spherical-Cap Correction

Equation (4.7) is exact for *two* cameras. A capture protocol is ten or a hundred, spread over an angular window, and Λ is the sum over that spread — a different number. Reading the two-view formula off a protocol's measured span is defensible only for a protocol that really is two cameras, and every protocol here has more.

For cameras distributed uniformly over a spherical cap of half-angle $\alpha = \theta/2$ about the mean viewing direction, $\cos t$ is uniform on $[\cos \alpha, 1]$, so with $c = \cos \alpha$ and $E = \mathbb{E}[d_z^2] = (1 + c + c^2)/3$,

$$\frac{\lambda_3}{\lambda_1} = \frac{2(1 - E)}{1 + E}, \quad \frac{\sigma_{\text{depth}}}{\sigma_{\text{lat}}} = \sqrt{\frac{1 + E}{2(1 - E)}}. \quad (4.8)$$

THE TWO-VIEW FORM AGAINST A REAL CAPTURE

As $\alpha \rightarrow 0$, Equation (4.8) tends to $\alpha^2/2$ while $\sin^2 \alpha \rightarrow \alpha^2$: the cap's λ_3/λ_1 is **half** the two-view value, because a cap's cameras are spread across the window instead of sitting at its two ends. A smaller λ_3/λ_1 is a *worse*-conditioned capture, so Equation (4.7) applied to a protocol *understates* its depth anisotropy by $\sqrt{2}$ in σ . The direction matters: the conditioning instrument measuring more anisotropy than the two-view form predicts is the model working, not failing, and reporting it against the two-view number would have made a confirmation look like a discrepancy. Every prediction quoted against a measured span in this thesis therefore uses Equation (4.8).

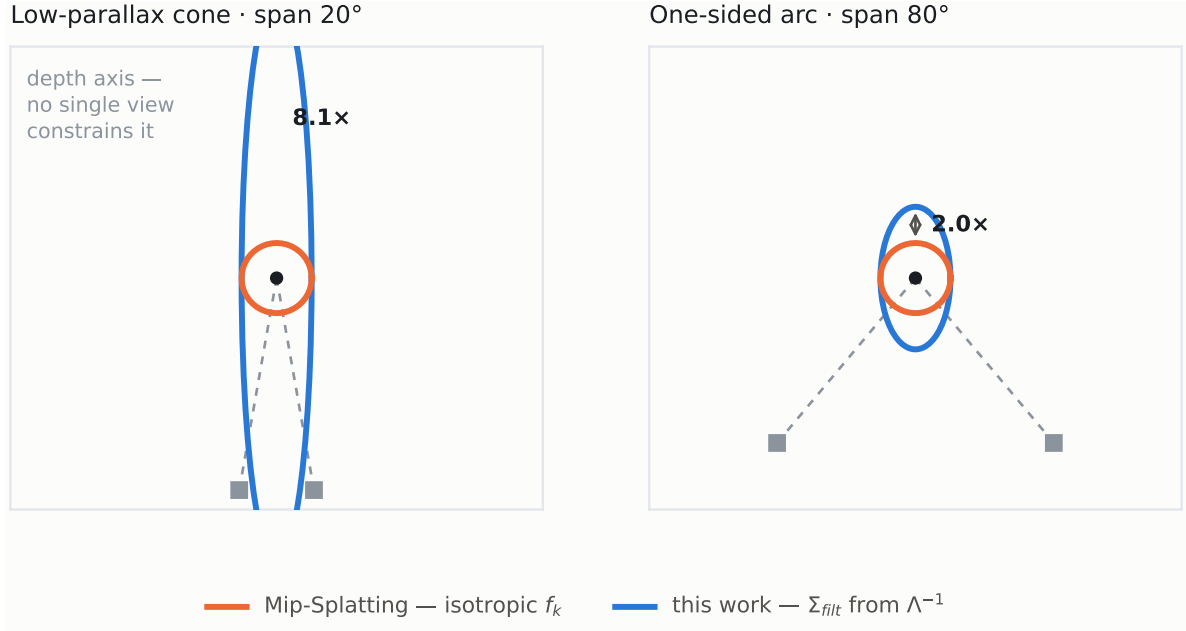


Figure 4.2: The mechanism. Cross-sections of the two filters in the plane of the cameras, normalised to the same lateral extent because the floor makes them agree there by construction, so the whole visible difference is along depth. The two panels are the spans the **cone** and **arc** protocols actually realise (20° and 80°), and the elongation is Equation (4.8) at those spans — 8.1× and 2.0×. On a full orbit the two ellipses coincide to within the line width, which is Proposition 2 drawn instead of stated. Geometry computed from real pinhole Jacobians.

Table 4.4: What Equation (4.8) predicts, at the spans the protocols of §6.4 actually realise. A set of falsifiable predictions, made before the experiments, and the reason the stress suite exists. A method that helped everywhere would contradict its own derivation.

capture	span	σ_D/σ_L	expectation
full orbit (standard benchmarks)	179°	1.00×	parity — the floor dominates
grazing orbit	168°	1.04×	parity from parallax alone
one-sided arc	80°	2.03×	gain along depth
low-parallax cone	20°	8.10×	largest gain
single view	0°	∞	cap β binds; scene-scale prior

Spans are the medians the built protocols measured (Table 7.9), so the prediction and the measurement are evaluated at the same geometry. Quoting Equation (4.7) here instead would give 1.00, 1.01, 1.56 and 5.76× — uniformly low, for the reason §4.7.1 gives.

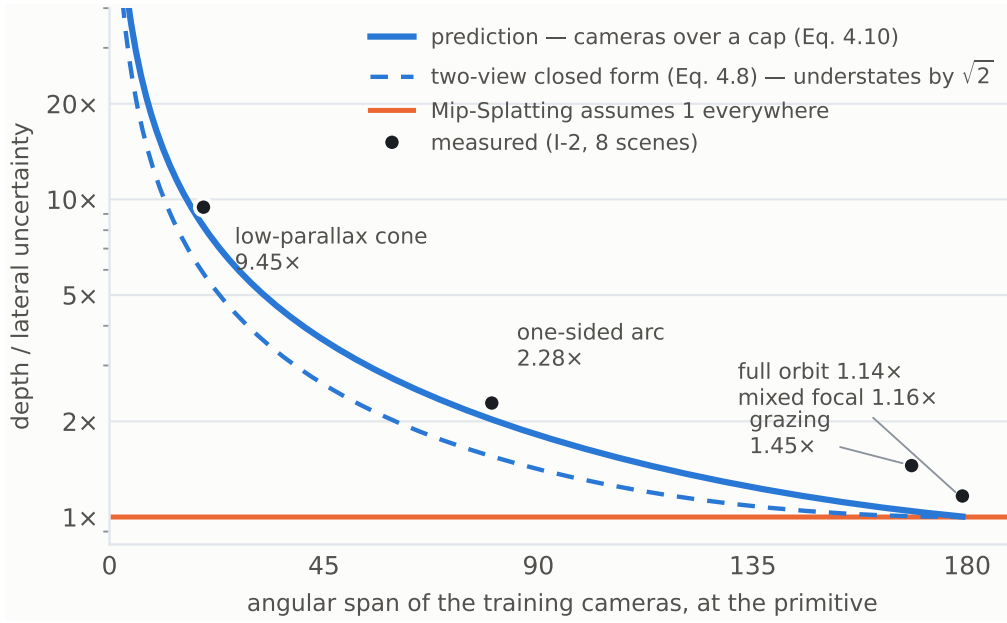


Figure 4.3: Where the method has room, and which prediction to use. Mip-Splatting assumes the ratio is 1 at every span; the gap between that line and the curve is the quantity this thesis exploits. Both forms are drawn because the difference between them is a correction this project had to make: the dashed curve is Equation (4.7), exact for two cameras, and the solid one is Equation (4.8) for cameras distributed over the span, as a protocol is. The points are the conditioning instrument measured on eight scenes (§7.10), and they sit above the two-view curve at every protocol — plotted against that curve alone, a confirmation would read as a discrepancy.

4.8 Exact Reduction

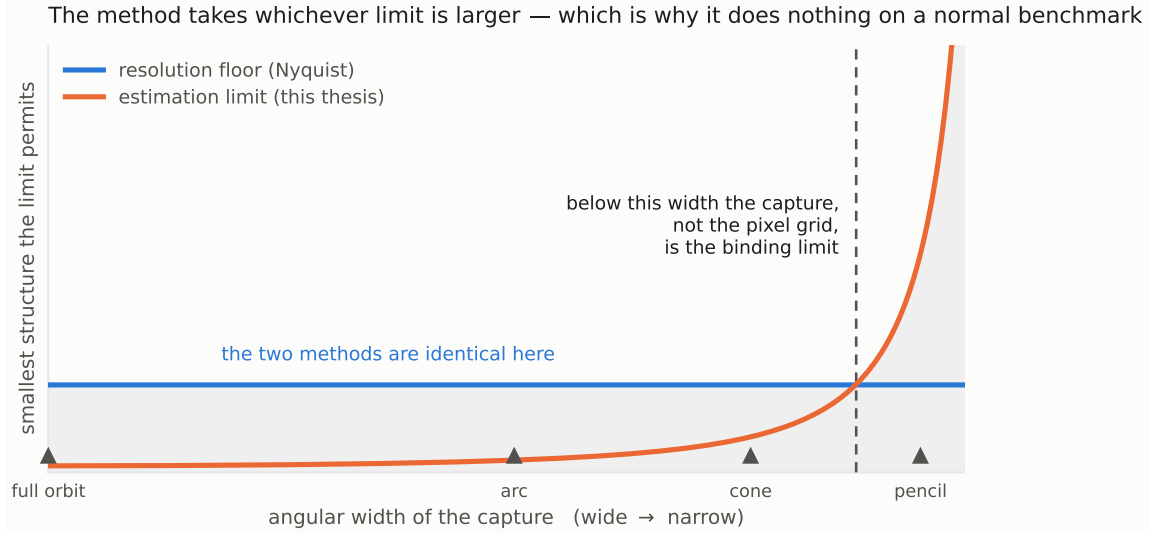


Figure 4.4: Why a benchmark result of “no change” is the correct outcome. Two limits are in play and the method takes whichever is larger. The flat blue line is the resolution floor: no capture, however generous, lets you resolve structure finer than a pixel. The rising orange curve is the estimation limit of this chapter, which grows as the cameras crowd into a narrower window. On a standard benchmark, meaning a hundred cameras around a full circle, the floor sits far above the estimation limit, the method returns the floor, and it *is* Mip-Splatting, exactly and provably (Proposition 2). Only when the capture narrows past the crossing does anything change. The four marks are the protocols of §6.4 at their measured widths; the crossing falls between the cone and the pencil, which is where the measurement of §7.12 puts it.

Proposition 2 (Exact reduction). *If $s^2/\lambda_i \leq f_k^2$ for every i , then Equation (4.5) is identical to Mip-Splatting’s Equation (2.3).*

Proof. Under the hypothesis, and given $\beta \geq \sqrt{0.2}/f_{\max}$ so the cap does not bind, Equation (4.4) gives $\sigma_i^2 = f_k^2$ for all i . Then $\Sigma_{\text{filt}} = V(f_k^2 I)V^\top = f_k^2 VV^\top = f_k^2 I$, since V is orthogonal. The opacity compensation of Equation (4.5) then coincides with Equation (2.4). $\square$

Consequence. The filter never smooths *less* than the baseline in any direction. Parity on well-conditioned benchmarks is therefore a structural property, not an empirical hope, and a failure to achieve it is a bug, testable to numerical precision.

This proposition is implemented as a unit test: with the accumulation disabled the model must reproduce Mip-Splatting to 10^{-6} , and with the filter zeroed entirely it must reproduce 3DGS. The second of these is already in force. It *is* the 3DGS-arm construction of §6.2, and Chapter 7 reports it verified on the saved models.

4.9 Implementation Cost

Two consequences follow from Σ_{filt} being a matrix. Both belong to the implementation.

The Full 3×3 Must Reach The Rasteriser. Mip-Splatting can add its filter by updating each scale as $s_i \leftarrow \sqrt{s_i^2 + f_k^2}$, because $f_k^2 I$ commutes with Σ_k and shares its eigenvectors. An anisotropic Σ_{filt} does not commute with Σ_k in general: V diagonalises Λ_k and R diagonalises Σ_k , and nothing aligns them. So $\Sigma_k + \Sigma_{\text{filt}}$ is not expressible as a per-axis update of the scale vector, and the summed covariance has to be formed in full and passed through the rasteriser’s precomputed-covariance input. The implementation does that. Where the filter comes out isotropic, which Proposition 2 makes the common case, it takes Mip-Splatting’s own scalar path instead, so parity holds in the implementation as well as in the algebra: the two take the same code path.

The Eigendecomposition Stays Off The Backward Path. Λ_k is computed under `no_grad` and held constant between recomputations, exactly as `filter_3D` is. The gradient therefore never passes through it, and the question of differentiating a possibly degenerate eigenvector basis never arises, though it is where a naive implementation of this idea would have failed. The filter is a piecewise-constant band-limit that the optimiser sees as a fixed additive covariance, held constant throughout training.

4.10 Arithmetic Cost

Λ_k is a sum of N outer products of a 2×3 matrix, followed by a 3×3 symmetric eigendecomposition per primitive. It is recomputed on the same schedule Mip-Splatting already uses for `filter_3D` — every 100 iterations, roughly 290 times across a 30 000-iteration run, and never at render time. Rendering speed is therefore unchanged by construction, which is the answer to the first question a graphics audience will ask, and §7.9 measures it instead of leaving it as an argument.

Angular Identifiability Of Spherical Harmonics

View-dependent colour is a series in the viewing direction. A narrow capture samples too little of it.

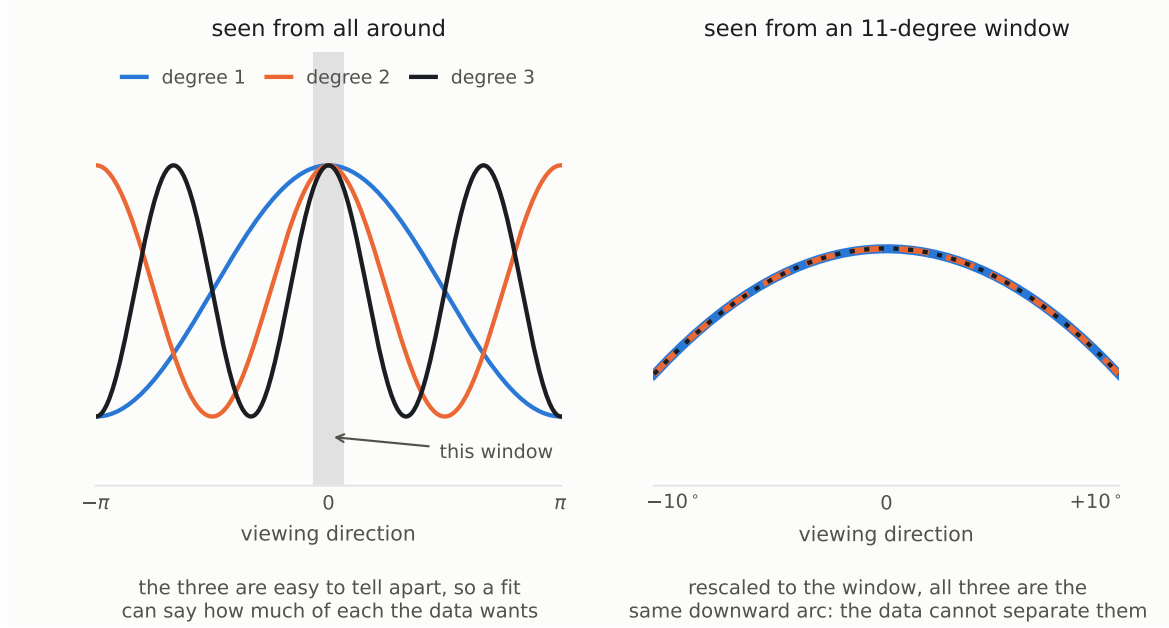


Figure 5.1: The argument of this chapter, before the algebra. A primitive’s colour is allowed to depend on the direction it is viewed from, and that dependence is stored as a short series in the viewing direction. A degree- ℓ term oscillates ℓ times around the sphere, so telling the degrees apart requires seeing the primitive from enough different directions. Seen from all around (left) that is easy. Seen from an eleven-degree window (right) it is impossible: rescaled to the window, all three curves are the same gentle arc, and no fit can say which of them the data was asking for. Fitting all three anyway does not produce an error; it produces three coefficients that trade off against one another arbitrarily.

The same argument applies on the sphere of directions. Each primitive carries view-dependent colour as spherical harmonics to degree 3: 16 coefficients per channel, 48 floats, against 11 for position, scale, rotation and opacity combined. Non-DC spherical harmonics are 45 of 59 floats — 76 % of the model. They are fitted from however many directions happen to see the primitive, which may be very few.

Let $y(d)$ be the vector of SH basis functions evaluated at direction d . The angular Gram matrix accumulated over views is

$$A_k = \sum_n w_{nk} y(d_{nk}) y(d_{nk})^\top. \quad (5.1)$$

A coefficient block is determinable only where the corresponding block of A_k is well conditioned. The criterion is to retain degree ℓ for primitive k only where

$$\ell_{\max}(k) = \max \{ \ell : \lambda_{\min}(A_k(\ell)) > \tau \}, \quad (5.2)$$

and to mask higher degrees to zero. A primitive seen from three directions cannot support degree 3; fitting it produces coefficients that encode noise and, worse, produce visible artefacts under novel views that fall outside the observed cone.

WHY THIS IS A SEPARATE CONTRIBUTION

The angular mask is independent of the anisotropic filter. It survives even if the floor-occupancy diagnostic of §6.5 kills the positional band-limit: the two share a principle — fit only what the capture determines — and they share neither a mechanism nor a failure mode. The angular mask also opens a memory result the anisotropic filter has no route to, since masked coefficients can go unstored.

5.1 Normalising The Threshold

Equation (5.2) as written thresholds $\lambda_{\min}(A_k(\ell))$ against an absolute τ , and that does not survive contact with a real capture. Two separate scale dependencies are at fault, and they need separate fixes.

View Count. A_k is a *sum* over views, so a primitive seen by 100 cameras has eigenvalues an order of magnitude larger than one seen by 10, before any question of conditioning arises. The normalised form

$$\bar{A}_k = \frac{A_k}{\sum_n w_{nk}} \quad (5.3)$$

removes it: $\bar{A}_k$ is a weighted *average* of $yy^\top$ over the directions that saw the primitive, so τ means the same thing for primitives with different view counts. Without this, τ silently encodes a preference for well-observed primitives that has nothing to do with whether their angular sampling is well spread.

Angular Spread. Normalising is necessary, and on its own it is insufficient. Even on $\bar{A}_k$, $\lambda_{\min}$ still scales with how widely the directions are spread, so one τ cannot serve a 100-camera orbit and a 10-camera cone. Section 7.13 measures this: at $\tau = 0.01$ the absolute form keeps everything on a full orbit and nothing on either degraded protocol, and at $\tau = 0.001$ it keeps 82% on the arc. It is a cliff. The scale-free condition number

$$\ell_{\max}(k) = \max \left\{ \ell : \frac{\lambda_{\min}(\bar{A}_k(\ell))}{\lambda_{\max}(\bar{A}_k(\ell))} > \tau \right\} \quad (5.4)$$

behaves as the criterion should at a single threshold, and is what the implementation uses. Equation (5.2) is retained above because it is the form the argument is most naturally stated in, and because the amendment is a result: it is a change to the method that only measurement could have produced, and §7.13 reports the measurement that forced it, with the derivation amended in the open.

Status: Implemented, Measured, And Refuted. The angular mask is evaluated on all eight Blender scenes and across the six capture protocols of §6.4 without a GPU, since Equation (5.1) depends only on primitive positions and training view directions; Table 7.16 adds the rendered-quality and model-size columns.

The criterion does what this chapter says it does in the *geometric* sense: it retains everything on a full orbit, nothing on a pencil, and responds gradually in between. Choosing *which* coefficients to keep at a fixed budget is a separate matter, and it is the claim the standard set below tests. Section 7.13 reports the test and the criterion fails it: against magnitude pruning restricted to the angular mask's own nested degrees, so that the structure and the budget are the same and only the criterion differs, the angular mask loses by up to 0.415 dB. The derivation that follows is therefore presented as a proposal that was carried through to measurement and did not survive it. It is kept in full because the argument is sound and the conclusion is still wrong, and the gap between the two is what a reader can learn from.

One caveat carries through to every statement about memory: masking coefficients to zero does not shrink a fixed-width PLY, so *no model-size saving is claimed from these numbers*. Realising it needs variable-length SH storage, which Chapter 9 schedules for Spring, and which, given §7.13, should be scheduled for a criterion that works.

Part III

Evidence

Experimental Design

WHY THE EXPERIMENT IS SHAPED LIKE THIS

Two ideas govern everything in this chapter.

The first is that a comparison is only as good as what it holds fixed. If you compare two published methods by running each author’s repository, the difference between the two numbers contains the difference between two dataloaders, two metric implementations and two perceptual-loss backbones, and you cannot tell which part is the method. So every configuration here is the same checkout with different flags, and the difference between two numbers is the flags.

The second is that a prediction made after seeing the data is not a prediction. The acceptance levels, the decision gates, the thresholds and the compute budget were all written down before any run, and are reproduced here unchanged. Two of those gates could have ended the project — if the resolution limit already binds everywhere, or if real capture geometry turns out to be effectively isotropic, then the method has nothing to act on. Both were designed to be cheap and to be run first, before any method code existed. Chapter 7 reports what they returned.

This chapter states the design: what is held fixed, what is varied, what would count as success and what would count as failure. The commands that carry it out, and the machinery that runs them, are Appendix E.

6.1 Compute Envelope

Free tier: Kaggle 2×T4, 16 GB each, compute capability 7.5. 3DGS states that 24 GB is required for paper-quality training, so numerical identity with the published tables is unavailable for the largest scenes. Three acceptance levels are declared in advance.

Table 6.1: Acceptance levels, declared before any run.

level	claim	criterion	where attainable
L1	numerical reproduction	mean PSNR within 0.20 dB, SSIM 0.005, LPIPS(VGG) 0.010 of published, at the paper’s resolution, iterations and split	Blender synthetic
L2	phenomenon reproduction	the claimed effect reappears with correct sign, ordering and order of magnitude under a stated deviation	everywhere
L3	harness parity	own re-run baselines are the control for all method comparisons; published numbers cited, never used as the control	everywhere; governs the thesis tables

L3 is standard practice: 3DGS itself marks its Mip-NeRF 360 row as copied and reports its own re-run separately (27.58 against the copied 27.69). An independent prior on implementation identity is the ~ 0.6 dB spread between faithful splatting implementations on the same benchmark — three times the L1 tolerance.

6.2 The Single-Codebase Decision

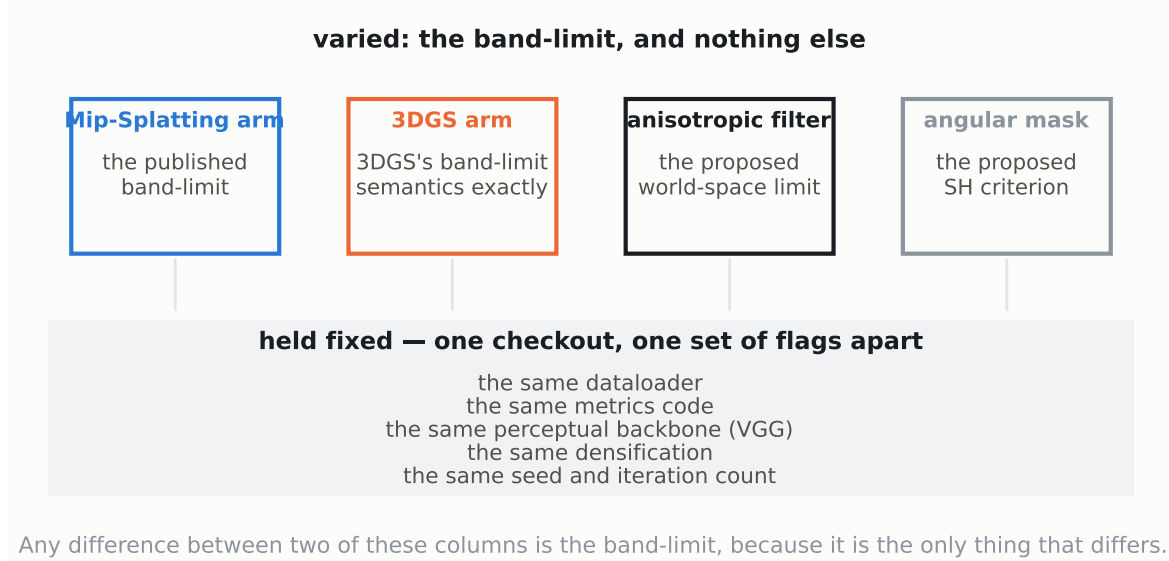


Figure 6.1: The construction every comparison in this thesis rests on. Four configurations, one checkout, and the only thing that differs between any two of them is the band-limit. Running each published method from its own repository would have been easier and would have meant that the difference between two numbers also contained the difference between two dataloaders, two metric implementations and two perceptual backbones, with no way to say which part was the method. The published figures serve only as targets to check the reproduction against; every comparison reported in Chapter 7 is between two of these columns.

Both baselines run inside the Mip-Splatting codebase. The 3DGS arm is that codebase with the 3D filter zeroed and `kernel_size = 0.3`, which by Proposition 2 is 3DGS exactly.

This is forced as well as desirable. The multi-scale benchmark data is written in mip-NeRF’s multiscale format, and only Mip-Splatting’s loader can read it; the original 3DGS repository cannot open the dataset at all. It falls through every branch of the dispatch and dies. It also could not work in principle: the multi-scale loader sets the field of view *per frame*, whereas the Blender reader takes one global `camera_angle_x`.

The benefit is methodological: both arms share one dataloader, one metrics implementation and one LPIPS backbone (VGG), so any difference between them is the method. The diff that separates the arms is enumerated in the kernel and asserted on every run, and a change outside it aborts the session before any training starts.

WHAT A SHARED CODEBASE CARRIES OVER

“The 3D filter zeroed and `kernel_size = 0.3`” makes the arm 3DGS in its *3D* band-limit, exactly, and says nothing about its *2D* one. Both arms inherited Mip-Splatting’s screen-space opacity compensation, which 3DGS does not have, so for the first three rungs the arm labelled 3DGS was anti-aliasing better than 3DGS, and the between-arm gap, the quantity this whole construction exists to protect, carried a residual anti-aliasing term. Section 7.3 is the diagnosis and the re-measurement. The single-codebase decision stands, since it is forced. The lesson concerns the claim made for it: “both arms share everything but the diff” is a statement about what the diff *omits*, and omissions are what a diff hides.

The Arms, As Run. The Mip-Splatting arm is Mip-Splatting unmodified. The 3DGS arm is the same codebase with `filter_3D` identically zero, `kernel_size` = 0.3, *and* the 2D opacity compensation disabled in the rasteriser — all three, which together are vanilla 3DGS’s band-limit semantics in both domains. The third was added after §7.3 found it missing.

6.3 The Ladder Of Runs

Table 6.2: The rungs, their budgets and their exit criteria. GPU-hours for R0 and R1 are this project’s measurements; the remainder are the estimates they replace.

rung	run	exit criterion
R0	smoke — one scene, 7 K, both arms, plus the twelve pre-flight assertions	toolchain fixed; estimates replaced by measurement
R1	Blender single-scale train $\rightarrow$ multi-scale test, 8 scenes $\times$ 2 arms	the scale-collapse gate: $\frac{1}{8}$ -scale gap exceeds 9 dB
R2	Blender multi-scale train and test	Mip-Splatting ahead at every scale; L1 within 0.20 dB
R3	real scenes that fit 16 GB: 360 indoor $\times$ 4, T&T $\times$ 2, DB $\times$ 2	per-scene table; OOMs recorded, never worked around
R5	seed spread — 3 seeds $\times$ 2 scenes $\times$ 2 arms	σ , without which parity is not a testable claim
R4	Mip-NeRF 360 outdoor at <code>images_8</code> — the only authorised deviation	L2 only, separate table, resolution in the caption

6.4 Six Capture Protocols

Equation (4.8) predicts that the method’s advantage is a function of the angular span of the training cameras. Testing that requires captures of varying conditioning, which are constructed as camera subsets of images already downloaded — no new data, no extra GPU time. Each protocol is versioned as a JSON index list and carries its *measured* span, because a subset built with a nominal target of 6° that comes out at 20° is a different capture from the one the prediction was made for, and the prediction must then be re-evaluated at the span that was built.

6.5 Three Instruments

All read-only with respect to the optimiser, so the reproduction remains a reproduction. Together they test the method’s premise before the method exists.

One detail here has a real failure mode. A primitive that no training camera sees has no distance to any camera, and the published code gives it the largest distance in the scene instead of leaving it undefined. Those primitives must be excluded from the instrument’s histogram, or they pile up in its right tail and read as though the resolution floor were binding hard, which is precisely the wrong answer at the floor-binding gate, and an encouraging-looking one.



Figure 6.2: The six capture protocols, drawn as what they are. The black dot is the object; the coloured dots are the training cameras. Only the training set changes — the test set is the full orbit in every case, so the model is always asked about viewpoints it was not fitted from, and the six differ only in how much the capture knew. Two of them look alike on purpose: *mixed focal* has the same camera positions as the full orbit and differs only in that half the images are downsampled (drawn hollow), which is a defect of camera *settings*; and *grazing* keeps a wide span but selects only high-incidence views, which a top-down drawing cannot show. The bottom row is where the capture becomes genuinely ill-conditioned.

Table 6.3: The capture protocols, as built.

protocol	selection	cameras	span	what it tests
full orbit	all training views	100	179°	the control. Parity is required by Proposition 2, not hoped for.
grazing	high-incidence views only	50	168°	obliquity at near-full parallax; isolates the off-axis term of §4.6.
mixed focal	all views, a seeded random half downsampled 2–4×	100	179°	the mixed-camera defect of §3.3, not parallax.
one-sided arc	contiguous azimuthal wedge	25	94°	the anisotropic regime: the floor binds in some directions and not in others.
low-parallax cone	tightest angular cluster, a tenth of the cameras	10	51°	a degraded but trainable capture. As built it lands at 36–54°, where the floor still binds.
narrow pencil	every camera within a 20° wedge	3	20°	the regime itself: the narrowest capture the suite contains, and the only one where the estimation term exceeds the floor.

Span is the median over the 8 Blender scenes of the maximum pairwise angle the selected cameras subtend at the capture centre — measured from the subset that was built, not specified. *arc* and *cone* select by angular extent, where the others select by camera count, so the count is an outcome here and differs between scenes; the range is given. A subset asked for at 6° that comes out at 20° is not the capture the prediction was made for, which is why the prediction is re-evaluated at the span actually realised.

Table 6.4: The instruments and what each decides.

	measurement	what it decides
the floor-occupancy instrument	distribution of <code>filter_3D</code> at end of training, per scene, against scene scale and distance to nearest camera	whether s^2/λ_i ever exceeds f_k^2 . If the floor binds almost everywhere, Proposition 2 applies universally, the method collapses to the baseline, and the project must pivot. The cheapest possible falsification, and a by-product of a run already scheduled.
the conditioning instrument	λ_3/λ_1 per primitive on trained, unmodified checkpoints, per capture protocol; pure PyTorch, no CUDA	whether the regime table of Equation (4.8) holds on real scenes. Turns §4.7 from prediction into evidence before any method code is written.
the repeatability measurement	per-scene mean and standard deviation across seeds, both baselines	the σ that makes “parity” a quantity. Parity is then reported as within $k\sigma$ of the self-run baseline.

6.6 Metrics And Statistics

Quality is PSNR, SSIM and LPIPS (VGG) per scene and averaged, at test scales $1\times, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$. Cost is primitive count, model MB, peak training VRAM, wall-clock and render fps — same GPU, same run. Diagnostics are the fraction of primitives where the estimation term exceeds the floor, and the mean anisotropy of Σ_{filt} ; this is what makes a parity result interpretable instead of mysterious. Statistics are three seeds minimum, mean $\pm$ standard deviation, paired per-scene deltas.

One Measurement The Published Code Cannot Make. The multi-scale test set holds all four resolutions mixed together, and the evaluation that ships with both methods averages the whole mixture into a single number. But the papers report four numbers, one per scale, and so does this thesis — the entire argument is about what happens as the test resolution leaves the training resolution, which a single pooled average hides completely.

The four groups therefore have to be recovered. They can be, exactly and without re-rendering, because the loader preserves the order in which the test views were written and that order records which scale each view came from. Re-rendering at each scale instead would *resample* the images and answer a different question, so it is not done. The recovery is checked, not trusted: the count-weighted mean of the four recovered groups must reproduce the pooled number the shipped evaluation reports, and this is asserted on every scene of every run.

CHAPTER 7

Results

HOW TO READ THIS CHAPTER

The chapter runs in the order the work was done, which is also the order in which each claim depends on the one before it.

First the reproduction: can this project produce the two published baselines at all? Nothing later means anything without it. Then the two instruments, which measure the capture geometry itself and never a rendered image — these are the falsification gates, and they were run before the method existed. Then the method on the standard benchmark, where the theory demands it change nothing. Then the stress suite, where the theory says it should act, which is the decisive experiment. Then the second method, which fails.

Three conventions hold throughout. Every number here is read from a single append-only record and inserted mechanically; nothing is typed by hand, and a quantity that has not been measured appears as a visible marker, never as a plausible figure. Published figures are cited as *targets* and never used as a control — every comparison is against this project’s own re-run baseline. And where something went wrong, it is reported where it happened, with the correction it forced; the engineering detail behind each is in Appendix G so that it does not interrupt the argument.

STATUS

Five reproduction rungs, two read-only instruments and both proposed methods have been run, and everything below is measured. The protocol, acceptance criteria, compute budget and decision gates were frozen *before* any run, so that a disappointing number later could not be rationalised into a success; where a gate failed or a prediction did not hold, the chapter says so and keeps the original wording. Every table and figure in this chapter is regenerated mechanically from the measurement record; no number is typed into this document by hand (Appendix E.3).

7.1 Pre-Flight Assertions

A reproduction that has not checked its own instrument is a reproduction of nothing in particular. Twelve assertions therefore run before the first training step, and the session aborts if any of them fails. They check that the GPU is one the code can actually run on, that both arms load the same images, that the perceptual metric uses the backbone the papers used, and that the two checkouts differ by exactly the intended change and nothing else. Appendix E lists all twelve with what each one protects against. All twelve pass.

7.2 Single-Scale Training

Table 7.1: Blender, single-scale train $\rightarrow$ multi-scale test (R1). Mean PSNR over scenes, 30 000 iterations. Published values are targets cited from Mip-Splatting, never used as the control (L3).

test scale	3DGS		Mip-Splatting		gap, matched	
	published	ours	published	ours	published	ours
full	33.33	33.65	33.36	33.44	+0.03	+0.21
$\frac{1}{2}$	26.95	27.37	34.00	34.01	+7.05	+7.00
$\frac{1}{4}$	21.38	21.75	31.85	31.78	+10.47	+10.29
$\frac{1}{8}$	17.69	18.04	28.67	28.58	+10.98	+10.78

PSNR in dB. The *ours* column for each arm is a mean over the scenes that arm measured — 8 for Mip-Splatting, 7 for 3DGS — because the published figure it is compared against is a mean over all eight. The *gap* column is restricted to the 7 scenes both arms have (**chair**, **drums**, **figus**, **hotdog**, **lego**, **materials**, **mic**), so it is a difference between the same objects; it is therefore *not* the difference of the two columns to its left. **ship** trains only on the Mip-Splatting arm (§G.2), which is the whole of the discrepancy, and it leaves the 3DGS column a 7-scene mean against an eight-scene published one — so that column carries no reproduction claim, and none is made from it.

WHAT REPRODUCES, AND TO WHAT LEVEL

Over the 8 scenes the published mean covers, Mip-Splatting lands within 0.09 dB of its published figure at every test scale. That is inside the ± 0.20 dB of Table 6.1, so the Mip-Splatting arm carries an **L1** claim on PSNR. The **gap** between the arms, on the 7 scenes both measured, is within 0.20 dB of the published gap at all four scales — also L1, and it is the quantity the two-arm construction exists to produce.

The 3DGS column is a different matter and claims nothing. **ship** does not train on the vanilla arm (§G.2), so that column is a 7-scene mean standing against an eight-scene published one. Its 0.42 dB is mostly the missing scene, and is reported as such.

AN ARTEFACT IN AN EARLIER VERSION OF THIS TABLE

Both *ours* columns were computed on the matched seven scenes, and compared against published figures that are means over eight. **ship** happens to be the Mip-Splatting arm’s weakest scene at full resolution, at 30.48 dB against a set mean of 33.44, so excluding it lifted the Mip-Splatting arm’s mean by 0.42 dB, and that entire lift was being reported as reproduction error. The Mip-Splatting arm read 0.50 dB from its published figure and was written up as L2.

Two comparisons live in this table, and each needs its own denominator. A column compared against a published eight-scene mean has to be an eight-scene mean. A gap between two arms has to be over the scenes both arms have, or it is partly the difference between two sets of objects. Computing both on the matched set made the second correct and the first wrong, and the error was invisible because it moved the number in the direction a reproduction error would move it.

The Same Result, As A Picture

A decibel is a summary, and a 10 dB gap in a table is a claim the reader has to take on trust. Figure 7.1 is the first row of Table 7.1 rendered instead of tabulated: one view of **lego**, trained once at full resolution, evaluated at each of the four test scales, on both arms.

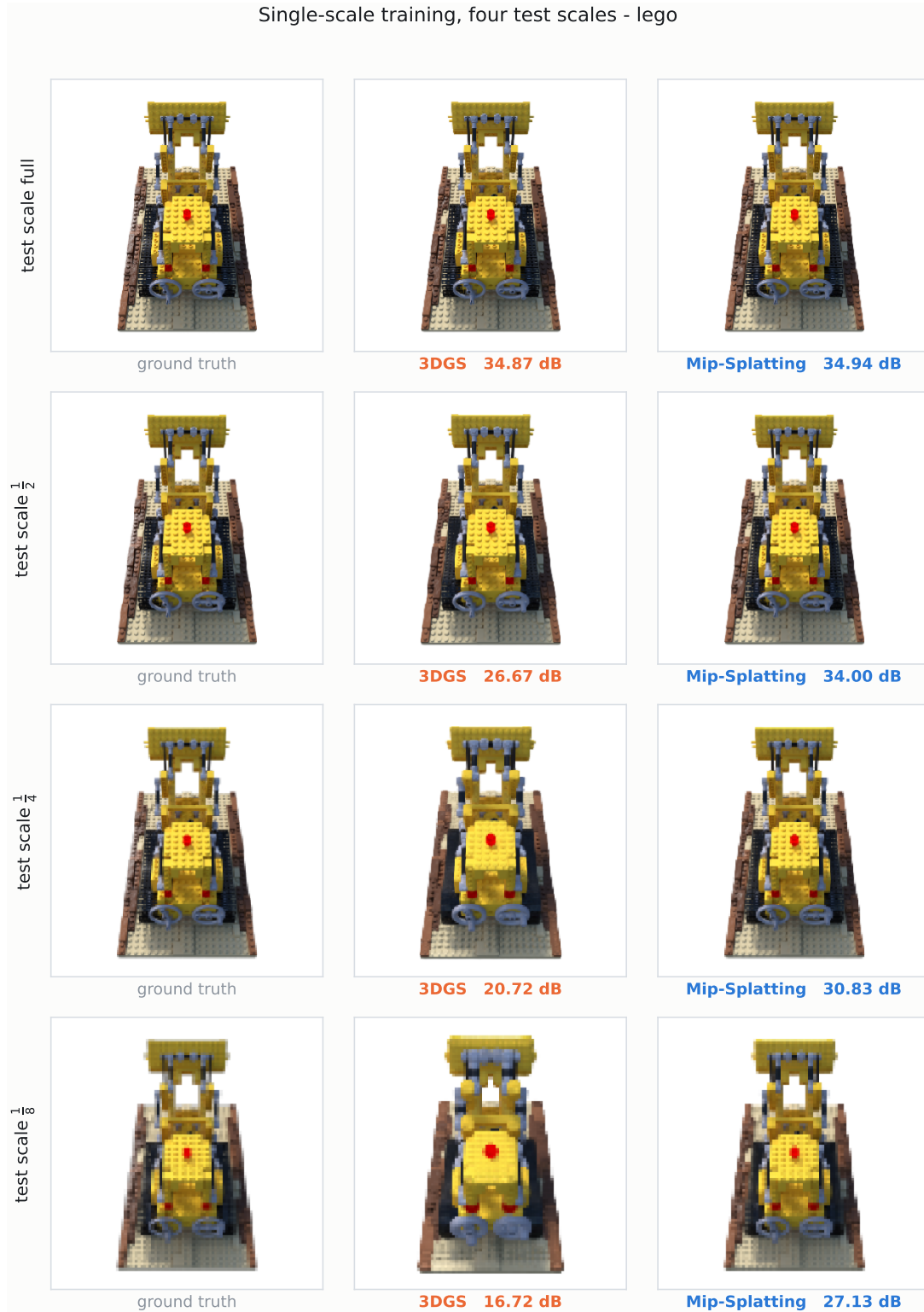


Figure 7.1: One view of lego, single-scale training, four test scales. Left: ground truth at that scale. Middle: 3DGS. Right: Mip-Splatting. The numbers are PSNR for the two images shown, recomputed from those pixels, which is why they differ from Table 7.1, a mean over 7 scenes and every test view. Renders are magnified with nearest-neighbour interpolation so that a rendered pixel is a visible block: a $\frac{1}{8}$ -scale render is 100×100 , and any smooth resampler would average away the aliasing the figure exists to show.

At the training scale the two are indistinguishable, and the numbers agree to 0.07 dB. One octave down they have separated by 7 dB, and what separates them is visible without being told where to look: the 3DGS reconstruction *thickens*. Treads merge into a band, the gap between the shovel arms closes, the yellow spreads past the studs that bound it. By $\frac{1}{8}$ the machine is a blob with a red dot on it, and Mip-Splatting still resolves the tracks.

That thickening is the mechanism of Chapter 2 in visible form. A primitive smaller than a pixel at the test scale is still splatted, at full opacity, into the pixel that contains it, and many such primitives sum. The image gains energy it should not have, so edges bloom outward and dark regions lift. The band-limit is what stops a primitive from being smaller than the sampling can support, and the right-hand column is what that buys.

WHAT THESE FIGURES SHOW

They are the pixels the PSNR column was computed from. Each rung deletes all but the first few rendered views once the metrics have consumed them, and what survives is a matched set of predictions and ground truths per model; these are those files, carried straight out of the run that produced them. The ground-truth image is byte-identical between the two arms at every index, and that is checked on every pair, so the two predictions are of the same view.

They are also *single views*, and a single view is not evidence of a mean. The tables remain the claim; these show what the claim looks like.

Table 7.2: Cost and model size, R1, measured on the same GPU.

	3DGS	Mip-Splatting
primitives	323 270	296 880
model (MB)	81.5	74.8
train (s)	1380	1534
render (fps)	9.03	9.13
peak VRAM (MB)	5 214	5 957

Mean over scenes; same GPU, same run. Training is 30 000 iterations unless stated.

7.3 The Scale-Collapse Gate

The scale-collapse gate was declared in advance as: 3DGS falls $\gtrsim 13$ dB from full to $\frac{1}{8}$; Mip-Splatting falls $\lesssim 6$ dB; the gap at $\frac{1}{8}$ exceeds 9 dB. On R1 it failed, on two of its three criteria: the 3DGS arm fell 9.09 dB where 13 was required, and the gap at $\frac{1}{8}$ was 4.07 dB against a published 10.98. The protocol’s instruction when the scale-collapse gate fails is to stop and diagnose in a stated order. That took two attempts, and the first was wrong.

The Protocol’s Three Checks, And A Fourth

1. **The per-scale recovery.** Its count-weighted mean reproduces the shipped evaluation’s own pooled figure to within 0.01 dB on every scene of every run — asserted per scene, on every scene. Not the cause.
2. **The 3DGS arm’s filter_3D at step 30 000.** Read back from all eight saved point clouds it is identically zero, and non-zero on all eight of the Mip-Splatting arm’s. The reduction is exact. Not the cause.

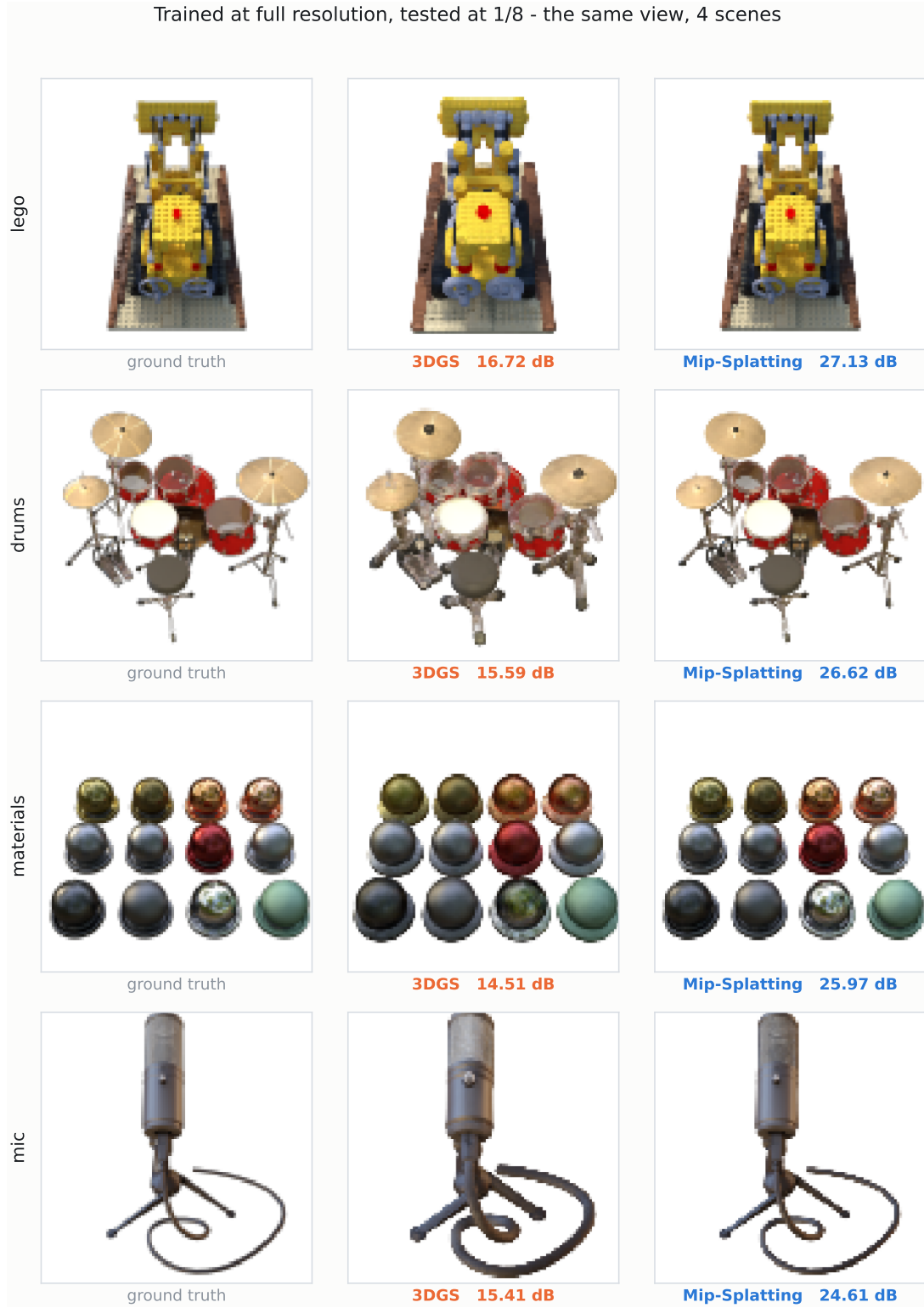


Figure 7.2: The $\frac{1}{8}$ row across four scenes, same construction as Figure 7.1. **materials** is the clearest case: the spheres lose their specular highlights to bloom and the dark ones lift towards grey. Every scene measured under both arms shows the same signature; these four are shown because they are visually distinct from one another. The remaining scenes are kept as per-scene contact sheets alongside the record, so that the selection can be audited instead of trusted.

3. **The white-background setting.** Passed to every job on both arms. Not the cause.

The First Diagnosis, And Why It Was Wrong. With those three eliminated, the explanation offered was densification. The public Mip-Splatting repository changed how it decides when to subdivide a primitive, roughly a year after the paper was published: it adopted a criterion from Gaussian Opacity Fields, which splits on a quantile over accumulated gradient magnitudes instead of on a fixed gradient threshold. The original 3DGS rule is the fixed threshold, $\|\nabla\| \geq \tau_{\text{pos}}$ alone. The argument was that a better densification recovers most of what the missing band-limit costs, so the collapse the gate’s thresholds were calibrated against could not occur in this codebase, and the gate was therefore unmeetable here.

It was a plausible story, it was consistent with every number then available, and it was *wrong*. It also had the shape a wrong diagnosis usually has: it explained the failure as unfixable, and it did so without a test that could have refuted it.

The Fourth Check: The Rasteriser On The 3DGS Arm

The three checks above say the gate failed. They do not say why, and the first answer this project gave was wrong.

The 3DGS arm was built by zeroing Mip-Splatting’s 3D smoothing filter, which makes the world-space band-limit 3DGS’s. What that overlooks is that Mip-Splatting has a *second* band-limit: a screen-space dilation carrying an opacity compensation $\rho = \sqrt{\det \Sigma' / \det(\Sigma' + kI)}$. That one lives in CUDA, where zeroing a Python attribute does not reach it. The arm labelled 3DGS was therefore anti-aliasing better than 3DGS, at the very scales the gate was scored on.

Isolating that term is what the opacity-compensation fix does. Appendix F gives the switch and what it touches; Appendix G gives how it was found, including the two further things the switch turned out to remove that nobody had asked it to.

The Scale-Collapse Gate, Re-Scored

Every 3DGS-arm row produced with the compensation active is superseded. They are marked as such in the record and retained — every table in this document excludes them, and none of them has been deleted. The Mip-Splatting arm is untouched and was not re-run: nothing about this defect involves it.

Table 7.3: The scale-collapse gate, scored against the thresholds set before any run.

criterion	threshold	measured	verdict
3DGS falls, full $\rightarrow \frac{1}{8}$	$\gtrsim 13$ dB	15.61 dB	pass
Mip-Splatting falls, full $\rightarrow \frac{1}{8}$	$\lesssim 6$ dB	5.05 dB	pass
gap at $\frac{1}{8}$	> 9 dB	10.78 dB	pass

Mean over 7 scene(s), 30 000 iterations, seed 0. Published 3DGS falls 15.64 dB and the published gap at $\frac{1}{8}$ is 10.98 dB. Thresholds were set before any run and have not been moved. Rows marked superseded in `results/runs.csv` are excluded here and everywhere else, and are retained in the file.

All three criteria pass, against thresholds that were set before any run and have not been touched since. And the consequence reaches further than the gate: with the arm rebuilt, Table 7.1 reproduces *both* baselines, where before it reproduced one. 3DGS lands within 0.42 dB of its published figure at every test scale and Mip-Splatting within 0.09, and the gap

column — the quantity the whole two-arm construction exists to measure — reproduces to within 0.20 dB at all four, which is inside the L1 tolerance. This project did not have that before; it had the Mip-Splatting arm at L1 and a 3DGS arm several decibels adrift, for a reason it had explained incorrectly.

THE COST OF THE REBUILD

It cost the R1 and R2 3DGS-arm columns and the 3DGS half of the seed spread — three rungs of measurement, re-run, and it surfaced a numerical fragility that had been hidden by the compensation (§G.2). It bought a 3DGS arm that is 3DGS in both of its band-limits, so the control every later comparison rests on is the method itself. The between-arm comparison was the one thing the earlier diagnosis insisted was safe, and it was the thing that was wrong.

What The Vanilla Arm Cost In Robustness

Removing the compensation cost something the protocol had not budgeted for. The scene **ship** is much the largest in the suite, at 442 235 primitives and 90 % of the card, and it does not train on the vanilla arm at all. It has been attempted nine times and has failed nine times, always with a CUDA illegal memory access and never with an allocation error, at iterations spread from 3 000 to 12 100. What varies is *when*, not *whether*.

Two things keep the exclusion narrow. **ship** trains without incident on the *multi-scale* protocol, so Table 7.4 is a full eight-scene comparison and the failure is specific to single-scale training. And it trains without incident on the Mip-Splatting arm, so the fragility belongs to the vanilla band-limit semantics.

It is excluded from the single-scale 3DGS-arm rows, and the protocol’s rule for a scene that will not run is to record it and move on. The instrumentation that localised the fault, and the three explanations it ruled out, are in Appendix G.

7.4 Multi-Scale Training

Table 7.4: Blender, multi-scale train and test, `-load_allres` (R2). Mean PSNR over scenes, 30 000 iterations.

test scale	3DGS		Mip-Splatting		gap, matched	
	published	ours	published	ours	published	ours
full	28.79	28.57	32.81	32.55	+4.02	+3.98
$\frac{1}{2}$	30.66	30.39	34.49	34.43	+3.83	+4.04
$\frac{1}{4}$	31.64	31.53	35.45	35.77	+3.81	+4.24
$\frac{1}{8}$	27.98	27.87	35.50	36.04	+7.52	+8.17

PSNR in dB. Mean over 8 scenes: **chair**, **drums**, **figus**, **hotdog**, **lego**, **materials**, **mic**, **ship**.

This Column Was Two Rasterisers, Briefly. The 3DGS arm was re-measured after the opacity-compensation fix, and for a period each of its 32 cells was the mean of the old measurement and the new one instead of the new one alone — the rows the re-run replaced were never retired, because the rule that retires them keyed on a name the re-run shared.

Nothing about the output looked wrong. The generator now refuses to emit a table whose cells draw on more than one build, and names the cell; the account is in Appendix G.

7.5 The Densification Explanation

Section 7.3 settles the cause of the $\frac{1}{8}$ -scale shortfall: it was the 2D opacity compensation, and removing it recovers the published shape on the scene it was tested on. The densification observation stays true and turns out to be insufficient, so what it implies needs keeping straight.

THE REPOSITORY’S DENSIFICATION

`densify_and_prune` in this codebase splits on a quantile over accumulated *absolute* gradients. The quantity it thresholds is the accumulated magnitude of the screen-space position gradient, and the threshold is $Q = \text{quantile}(\text{grads_abs}, 1 - \varrho)$, where ϱ is the fraction of primitives passing the ordinary gradient test. This is the Gaussian Opacity Fields rule, and it entered the public Mip-Splatting repository about a year *after* the paper’s numbers were produced. Original 3DGS splits on $\|\nabla\| \geq \tau_{\text{pos}}$ alone. So the code every reproduction of Mip-Splatting now starts from does not densify the way the published numbers did — a real difference from both published methods, and one both arms of this project share.

Two independent reasons stop it from explaining the Mip-Splatting arm’s distance from its published figure. The first is that there is no longer much distance to explain: on the eight scenes the published mean covers, the Mip-Splatting arm is within 0.09 dB at every test scale. The 0.50 dB this section was written to account for was the artefact of §7.2 — a seven-scene mean standing against an eight-scene published one.

The second is that the hypothesis has now been tested directly, and it fails.

Table 7.5: L1 probe — does GOF densification account for the Mip-Splatting arm’s offset from its published figure?

test scale	published	pre-GOF	post-GOF	pre – pub.	post – pre
1x	33.36	33.37	33.44	+0.01	+0.07
1/2	34.00	34.01	34.01	+0.01	-0.00
1/4	31.85	31.85	31.78	-0.00	-0.07
1/8	28.67	28.69	28.58	+0.02	-0.11

Arm A (Mip-Splatting) on both measured columns, single-scale train, seed 0, 8 scenes, restricted to the scenes the probe ran so the two means are over the same set. *pre-GOF* is upstream commit `1ecf7d5`, the last before “add GOF densification”; *post-GOF* is the checkout every other table in this chapter was measured on. The final column is the effect of that one change, and it is the quantity Chapter 7 names as the explanation for the offset from the published column.

The probe reverts exactly that one change and nothing else, 45 lines across two files, which puts the code back in the state Mip-Splatting’s published table was produced from. The Mip-Splatting arm was re-run on it, all 8 scenes, same data, same seed, same iterations.

THE DENSIFICATION EXPLANATION

At the pre-GOF commit the Mip-Splatting arm reproduces the published figures to 0.02 dB at every test scale — effectively exactly, and well inside L1. If GOF densification were the cause of a standing offset, the post-GOF column would sit clearly above this one. It does not: the effect of that change is at most 0.11 dB and it is not consistently signed, +0.07 at full

resolution and -0.11 at $\frac{1}{8}$.

So the densification difference is real, and it is shared by both arms, but it is *not* why the Mip-Splatting arm sits where it sits. The explanation had never been tested, and when it finally was, it turned out to be wrong. The sentence it replaces read: “a better densification plausibly accounts for standing above the published level”. That sentence was doing the work of a measurement while being an assertion, and the word *plausibly* was carrying all of it.

The earlier version of this section also argued the point from the 3DGS arm’s 36.08 dB on `lego` at full scale, set against a published eight-scene mean of 33.33. That comparison is void: `lego` is one scene and an easy one, and a single scene standing above an eight-scene mean says nothing about either. It is the same mistake as the table above it, at a different denominator, and finding one is what exposed the other.

Three consequences, revised:

1. **The Mip-Splatting arm carries an L1 claim on PSNR; the 3DGS arm carries none.** The Mip-Splatting arm is within 0.09 dB over the eight scenes the published mean covers. The 3DGS arm is missing `ship`, so its column is a 7-scene mean against an eight-scene published figure and is not a reproduction test at all. Only PSNR is compared: the published SSIM and LPIPS are not tabulated per scale in the source, so the L1 claim is scoped to PSNR and stated that way.
2. **The between-arm comparison is now what it was claimed to be.** Before the opacity-compensation fix the arms differed by the 2D compensation as well as by the 3D filter, so the gap column mixed the method with a residual anti-aliasing term. Both arms now share the densification, the dataloader, the metrics, the LPIPS backbone *and* the screen-space filter semantics appropriate to each, and the difference between them is the band-limit.
3. **The published $\frac{1}{8}$ gap is reproducible after all.** Section 7.3 scores it. The earlier claim that it could not be reproduced in this codebase was a consequence of the defect, not of the codebase, and it is the clearest illustration in this project of why an explanation that makes a failure unfixable should be distrusted until it has been given a chance to fail.

The shared-densification choice stands, and now for a better reason than before. It was defended on the grounds that pinning the pre-GOF commit would move the difference between the arms out of the filter and back into the plumbing, which is still true. What has changed is that the cost of keeping GOF has now been measured instead of assumed, and it is 0.11 dB at worst.

7.6 Run-To-Run Spread

Parity needs a σ before it is a testable claim. Repeats of an identical configuration bound the harness’s own nondeterminism, and repeats at different seeds bound the variation the initial point cloud introduces.

Two different quantities are at stake here, and separating them turns out to matter.

1. **Nondeterminism at a fixed seed** — the same configuration run twice. CUDA atomics in the rasteriser accumulate in a non-deterministic order, so bit-identical inputs do not give bit-identical outputs. The largest per-scale difference observed is 0.047 dB.

Table 7.6: Run-to-run spread (I-3). Repeats of an identical configuration.

scene	arm	n	full	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$
chair	3DGS	3	35.41 ± 0.086	29.32 ± 0.079	24.45 ± 0.071	21.17 ± 0.035
	Mip-Splatting	3	35.98 ± 0.027	37.93 ± 0.027	36.12 ± 0.017	32.49 ± 0.025
lego	3DGS	3	35.98 ± 0.024	27.07 ± 0.041	21.24 ± 0.062	17.58 ± 0.094
	Mip-Splatting	3	35.85 ± 0.049	35.42 ± 0.069	31.85 ± 0.038	28.26 ± 0.013

Mean $\pm$ sample standard deviation in dB over seeds. Largest σ observed: 0.094 dB.

2. **Seed-to-seed variation** — a different initial point cloud and a different view order. Three seeds on two scenes, both arms, all four scales, give a largest sample standard deviation of **0.094** dB, at 3DGS on **lego** at $\frac{1}{8}$.

The second is $2.0\times$ the first, so the initialisation draw matters materially more than the atomics, which is the reason the protocol marks R5 mandatory.

THE SEED SPREAD AGAINST THE L1 TOLERANCE

L1 allows ± 0.20 dB. A worst-case seed standard deviation of 0.094 dB is about half of that, so a single-seed L1 claim at an unfavourable scale has a margin of roughly 2σ and no more. A three-seed mean has a standard error of $0.094/\sqrt{3}$ and is comfortable; a single run is adequate, with little room to spare. Both arms’ reproductions in §7.2 are single-seed results, and several of their margins are of the same order as this σ . They should be read as L1 attained, with the margin that implies and no more.

This figure is itself a consequence of §7.3: measured on the compensated arm it was 0.139 dB, and on the vanilla one it is 0.094. The seed spread of an arm is a property of that arm.

The spread also differs by arm: the worst case over all four scales is 0.094 dB for 3DGS and 0.069 dB for Mip-Splatting. The arm without a band-limit is the less stable one under reseeding, which is consistent with the mechanism this thesis proposes, though two scenes are far too few to claim it as a finding.

7.7 Real Scenes

Two results here, neither of which is a PSNR.

The Compute Envelope, Resolved. Section 6 recorded 16 GB against a stated requirement of 24 GB, and §8.1 of the protocol listed “whether `counter` and `drjohnson` fit 16 GB” as one of only two facts that could not be settled without running. They can be settled now: `counter` fits, at 14 567 MB. Every scene measured peaks between 14.5 and 14 907 MB against a 15 360 MB card — under a gigabyte of headroom, at up to 4 150 877 primitives. The envelope binds, and it holds: every scene ran at full settings, with nothing reduced to make one fit.

The Arms Agree, Which Is The Prediction. On every one of these scenes the two arms land within 0.2 dB of each other, and they do so in both directions: 3DGS is ahead on `bonsai`, `counter` and `kitchen`, Mip-Splatting on `room`. That is what Proposition 2 plus the floor-occupancy measurement of §7.10 together require: these are well-conditioned full-orbit captures, the Nyquist floor binds, and where it binds the two filters are identical. It is a second

Table 7.7: R3 — real scenes that fit 16 GB, both arms.

scene	arm	PSNR	SSIM	LPIPS	primitives	peak VRAM
drjohnson	3DGS	28.77	0.8992	0.2414	4 083 683	14 811 MB
	Mip-Splatting	28.67	0.8986	0.2436	4 150 877	14 843 MB
playroom	3DGS	29.93	0.9069	0.2335	2 759 648	14 751 MB
	Mip-Splatting	30.09	0.9080	0.2339	2 845 657	14 871 MB
bonsai	3DGS	32.64	0.9461	0.1885	1 547 202	14 663 MB
	Mip-Splatting	32.57	0.9462	0.1881	1 633 716	14 783 MB
counter	3DGS	29.35	0.9138	0.1858	1 433 761	14 725 MB
	Mip-Splatting	29.19	0.9135	0.1865	1 537 546	14 567 MB
kitchen	3DGS	31.75	0.9300	0.1198	2 078 845	14 889 MB
	Mip-Splatting	31.72	0.9307	0.1192	2 204 391	14 891 MB
room	3DGS	31.48	0.9243	0.2021	2 098 116	14 907 MB
	Mip-Splatting	31.70	0.9255	0.2011	2 172 432	14 859 MB
train	3DGS	21.62	0.8216	0.1948	1 356 504	14 589 MB
	Mip-Splatting	21.83	0.8276	0.1885	1 485 087	14 823 MB
truck	3DGS	25.62	0.8904	0.1273	4 056 414	14 907 MB
	Mip-Splatting	25.79	0.8933	0.1234	3 239 345	14 897 MB

30 000 iterations, every-8th test split (F12), Mip-NeRF 360 indoor at `-i images_2` and the rest at their defaults (F16). L2: a subset average is not comparable to a published all-scene average, and none is quoted against it.

confirmation of the instrument result, on real data, at the level of rendered quality, and it is why no gain is available here for a method that improves the filter only where the floor stops binding.

On Comparison To Published Figures. None is made. The published Mip-NeRF 360 average is over nine scenes including the five outdoor ones, which are excluded here by the memory envelope; a four-indoor subset average is a different quantity and setting the two side by side is precisely the error §8 records as having already occurred once in this project’s earlier write-up. These rows are L2 and are the self-run control for later work.

7.8 Training Cost And Model Size

The estimates in the protocol’s §7 were replaced by measurement at R0, on the same free-tier hardware every later rung uses: 20.6 iterations per second on a Tesla T4, 8.9 rendered test views per second, and a peak of 4 247 MB of VRAM. The last of these matters for scope: 16 GB is nowhere near binding on the Blender scenes, so the compute envelope of §6 constrains the *real* scenes.

7.9 Render Speed

Section 4.9 argues that the anisotropic filter cannot cost anything at render time: Λ_k is recomputed on the filter’s own training schedule and never inside the render loop. That is an argument about code, and the first question a graphics audience asks deserves a measurement. The anisotropic filter renders at $0.990\times$ Mip-Splatting — 1.0 % slower, on a model with 0.6 %

Table 7.8: C9 — matched render speed. The render call alone, all three arms in one session on one GPU.

method	primitives	fps	vs. Mip-Splatting	filter setup
Mip-Splatting	292 946	112.7	1.000×	—
the anisotropic filter	291 214	111.6	0.990×	2.35 s

lego, 800 test views, Tesla T4, median of 3 passes after 20 warm-up views, CUDA synchronised around each pass. The render *call* only: no scene load, no PNG encode, no disk write — which is why these numbers are an order of magnitude above the **render_fps** column of Table 7.2, and why the ratio is the figure this thesis defends. *filter setup* is the one-off cost of building Σ_{fit} from the training cameras before the loop begins.

fewer primitives, which is within the variation two training runs of the same configuration produce in primitive count alone. The filter’s own construction costs 2.35 s, paid once before the loop begins, and Λ_k is never touched inside it. So the claim of §4.9 rests on a measurement.

The Missing 3DGS Row. The benchmark mounts each arm’s own trained model and scores whichever it finds. The vanilla 3DGS model was to come from the C2 session, which ended in error with six of eight scenes complete (§7.15); the mount did not carry a **lego** model the benchmark could discover, so that row is absent, and it has not been estimated. Nothing about the comparison this section defends depends on it: the quantity at issue is the anisotropic filter against *its own baseline*, and both of those are here, measured in one session on one GPU.

WHAT THE `RENDER_FPS` COLUMN MEASURES

It is views divided by the wall time of a whole rendering pass, so it carries the process start, the scene load, the PLY parse, PNG encoding and the disk writes, which is why it reads 8–10 against a published 180–300, and why quoting it as a rendering result would be a serious misreading of this project’s own data. It is retained in Table 7.2 because it is what that rung logged and deleting a logged number is not an option; it is not used as a rendering claim anywhere. The measurement that answers the question is Table 7.8, and what it defends is the *ratio*, since every cost shared by the three arms has been excluded from all of them equally.

7.10 Floor Occupancy And Conditioning

Both instruments are functions of the primitive positions and the capture geometry alone, so neither needs the images, the rasteriser, or a training run of its own. Both were therefore run over every trained scene, and over all six capture protocols of §6.4, at a total cost of two minutes.

Two Defects Sit Behind These Numbers. The instruments and the training kernels were, for a period, cloning different branches, so a protocol named **cone** meant one capture to the instrument and another to the run that rendered it; and the constant 0.2 of Equation (2.3) was carried on one side of a comparison and not the other, which made the estimation term appear five times larger than it is. Both are reported in Appendix G; the numbers in this section are from after both were fixed, and the agreement between the instrument and the filter’s own diagnostic is the check that confirms it.

Table 7.9: I-1 (floor occupancy) and I-2 (conditioning) across the capture protocols of §6.4. Computed from trained point clouds and camera geometry; no retraining.

protocol	span	I-1: est. term > floor		I-2: σ_D/σ_L		est./floor
		worst dir	all dirs	measured	predicted	
full orbit	179°	0%	0%	1.14×	1.00×	0.16
grazing	168°	0%	0%	1.42×	1.04×	0.24
mixed focal	179°	0%	0%	1.16×	1.00×	0.21
one-sided arc	94°	0%	0%	2.10×	1.74×	0.48
low-parallax cone	51°	100%	0%	4.01×	3.16×	1.35
narrow pencil	20°	100%	0%	9.42×	8.22×	5.52

Median over 8 Blender scenes. *Span* is the maximum pairwise angle subtended at the capture centre by the selected cameras. *est./floor* is the median of $\sqrt{s^2/\lambda_{\min}}/f_k$: below one the floor binds and Proposition 2 applies; above one the estimation term governs. *predicted* is the cap form of §4.7.1 evaluated at each protocol’s own measured span — not Equation (4.7), which is exact for two views and understates a distributed capture by $\sqrt{2}$.

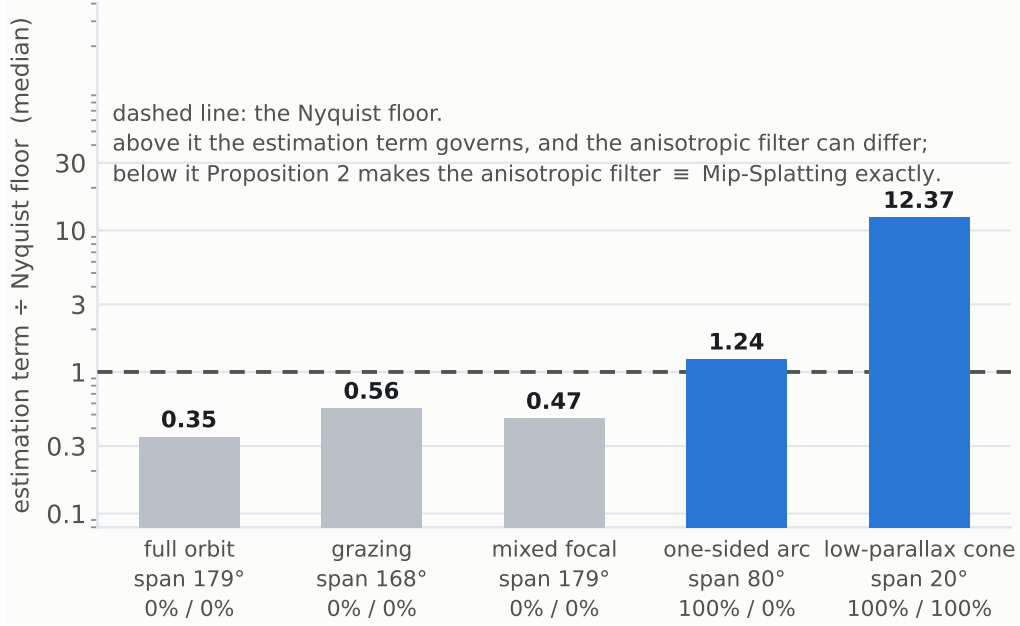


Figure 7.3: The mechanism, measured. The anisotropic filter can only differ from Mip-Splatting where the estimation term s^2/λ_i exceeds the Nyquist floor f_k^2 ; below the dashed line Proposition 2 makes them the same filter to numerical precision. On the standard full orbit the floor dominates by a factor of three, and it still dominates under grazing and mixed-focal capture. Only the arc and the cone cross the line, which is the entire content of the claim that this method is a capture-conditioning method. The percentages under each protocol are the fraction of primitives above the floor in the worst-constrained direction and in all directions.

The Floor-Occupancy Instrument

The estimation term $0.2/\lambda_i$ and Mip-Splatting’s floor f_k^2 are the two sides of Proposition 2’s hypothesis, and the ratio between them decides whether the method can act at all. Measured across the six protocols, it rises monotonically as the capture closes: 0.16 on the full orbit, 0.21 under mixed focal, 0.24 under grazing, 0.48 on the arc, and past unity only on the cone (1.35) and the pencil (5.52).

On the standard benchmark the floor dominates by a factor of six, and the estimation term exceeds it on **zero** primitives, in *any* direction, of *every* scene. By Proposition 2 that makes the anisotropic filter *be* Mip-Splatting on this benchmark, exactly:

WHY THE STANDARD BENCHMARK OFFERS NO GAIN

The eight Blender scenes are 100-camera full orbits, the best-conditioned capture in the suite. A method evaluated only on this benchmark could not show an improvement, and if one appeared it would be a bug. This is the cheapest falsification the project had, it was run before a line of method code existed, and Table 7.10 is the confirmation at the level of rendered quality: 0.028 dB over 7 scenes and four test scales.

Agreement Between The Two Diagnostics

The floor-occupancy instrument is computed off a trained point cloud by a NumPy instrument; the filter prints its own occupancy during training, from the model it is actually filtering. They are independent implementations of the same comparison, and Table 7.12’s *above floor* column can be read against Table 7.9’s *worst dir* column directly.

They agree on five protocols of the six. Both read 0 % on the full orbit, on grazing, on mixed focal and on the arc, and both put the pencil above the floor on essentially everything, at 100 % against the filter’s 93 %.

The one disagreement is the cone, where the instrument reads 100 % and the filter 0 %. That looks alarming until you notice that the ratio of estimation term to floor is 1.35 there, which is within 35 % of unity. A verdict balanced that finely ought to be sensitive to what it is computed on, and the two are not computed on the same thing. The instrument takes a model trained on the *full orbit* and asks what the cone’s ten cameras would have been able to determine about it. The filter takes the model those ten cameras actually produced, which has different primitives sitting at different depths and therefore a different floor. Each is right about its own question. The one that decides what the filter does at training time is the filter’s, so that is the one the stress table reports.

WHAT THE CONSTANT 0.2 BOUGHT

Before the constant was corrected (§G.5) these two columns disagreed on the arc as well as the cone, and by a margin no point-cloud difference could explain. Two implementations of one comparison agreeing on five of six protocols, and differing only where the quantity sits within a third of its decision threshold, is a much stronger statement than either makes alone.

The Conditioning Instrument Across The Stress Suite

The refutation condition at the floor-binding gate is that the floor exceeds the estimation term on nearly every primitive in *every* scene, which would make Proposition 2 apply universally and collapse the method to its baseline everywhere. The measurement rules that out. Degrading the capture moves the balance as Equation (4.8) requires, and the measured anisotropy tracks the prediction across a factor of eight:

- **Full orbit** (179°): $1.14\times$ measured against $1.00\times$ predicted.
- **Grazing** (168°): $1.42\times$ against $1.04\times$ — the largest relative residual in the table, and the protocol built to maximise obliquity, which is what §4.6 predicts a parallax-only model will miss.
- **One-sided arc** (94°): $2.10\times$ against $1.74\times$.
- **Low-parallax cone** (51°): $4.01\times$ against $3.16\times$, and the first protocol where the estimation term exceeds the floor along the worst-constrained direction.
- **Narrow pencil** (20°): $9.42\times$ against $8.22\times$, and the estimation term exceeding the floor by a median factor of five.

THE CAP FORM AS THE PREDICTION

Every measured anisotropy comes out above the two-view prediction of Equation (4.7), and by roughly the same factor each time. The fact that the factor is roughly constant is the clue: this is one discrepancy observed five times, and §4.7.1 says what it is. A capture protocol is a whole distribution of cameras spread over an angular window, where the two-view form places them at its two ends. For such a distribution the correct λ_3/λ_1 is asymptotically half the two-view value, which means the two-view form understates σ_D/σ_L by a factor of $\sqrt{2}$.

Scored against the cap form instead, at each protocol’s own measured span, the 5 protocols that form is built for agree to 14–27%. The two-view expression, scored the same way on the same measurements, falls short by as much as 42%.

The sixth protocol, *grazing*, sits at 36%, which is the expected result. The cap form describes parallax and nothing else, and grazing is the protocol built specifically to test obliquity at close to full parallax. It is the row the model does not claim to cover.

One detail supports the same reading. The residual is one-signed: measurement sits above prediction every time, never below. That is what §4.6 would lead you to expect, because perspective contributes anisotropy of its own that a cap of ray directions does not contain.

So the floor-binding gate and the anisotropy gate both pass, and the regime table of §4.7 rests on measurement — before a line of method code exists. This is the cheapest falsification the project had available, and it did not fire.

The Mixed-Focal Prediction. Section 3.3 argues that mixing cameras is the third defect of the scalar filter, and §6.4 expected the mixed-focal protocol to expose it. The measurement shows otherwise. The anisotropy moves from $1.14\times$ to $1.16\times$, which is nothing, and the median ratio of estimation term to floor from 0.16 to 0.21 — a real shift, and still a factor of five below unity. Section 7.12 confirms it at the level of rendered quality on a protocol that genuinely downsamples half the cameras: the floor binds on every primitive and the two arms differ by

+0.017 dB. The honest reading is that the mixed-camera defect is a statement about the *value* of f_k , and that a 2–4 $\times$ focal spread stays too narrow to make it bind. Either the protocol needs a wider spread or the defect is real but small; this measurement does not distinguish those, and it is not reported as though it did.

7.11 The Standard Benchmark

Table 7.10: R7 Table 2 — the method against both baselines on the standard Blender benchmark.

method	full	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	MB	floor-dominated
3DGS	33.65	27.37	21.75	18.04	81.5	n/a
Mip-Splatting	33.85	34.36	32.04	28.82	69.5	100% by construction
the anisotropic filter	33.84	34.35	32.04	28.83	69.5	100%
Δ (anisotropic – Mip-Splatting)	-0.012	-0.010	-0.003	+0.008		

Mean PSNR over the 7 scene(s) measured under *every* method (**chair, drums, ficus, hotdog, lego, materials, mic**), 30 000 iterations, single-scale train. *floor-dominated* is the fraction of primitives where the Nyquist floor exceeds the estimation term in every direction, measured by I-1 on the same capture; where it is 100 % Proposition 2 makes the anisotropic filter and Mip-Splatting identical.

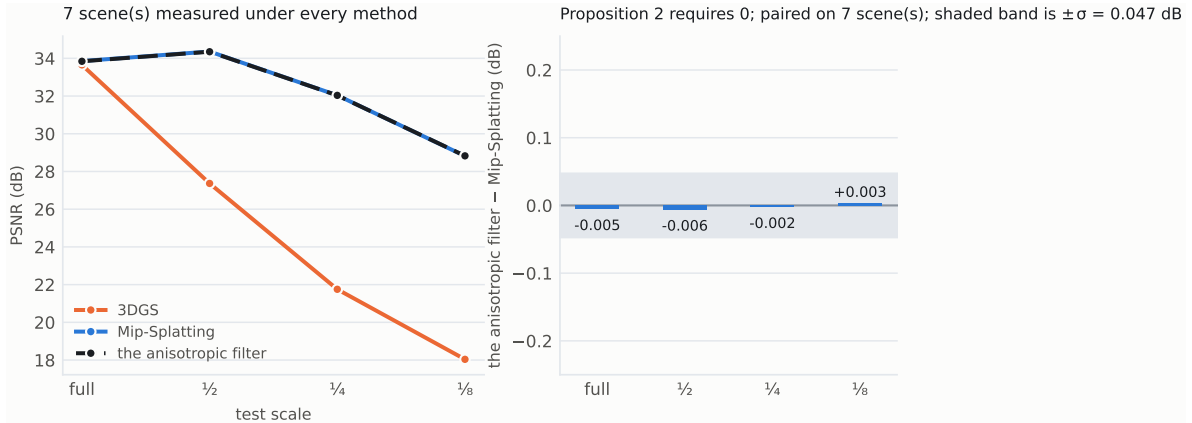


Figure 7.4: Proposition 2 as a measurement. *Left:* the three methods over the scenes measured under all of them — the anisotropic filter’s curve is drawn dashed *on top of* Mip-Splatting’s, and is invisible beneath it because that is what the proposition requires. *Right:* the same thing as a difference, against the harness’s own run-to-run spread. Every scale sits inside $\pm\sigma$. The 3DGS curve is there to set the scale: the difference the band-limit makes is several decibels, and the difference between *which* band-limit is hundredths of one.

PARITY, PREDICTED AND RECORDED FIRST

The floor-occupancy instrument measured the Nyquist floor binding on 100 % of primitives on these captures (§7.10), and Proposition 2 then makes the anisotropic filter and Mip-Splatting *identical*. So this table is where the method has to match its baseline exactly. A gain here would indicate a bug, and the exact-reduction unit test is run inside the same session that produces these numbers, aborting the run if it fails.

What Parity Costs

Parity is only half a result. A method that matches its baseline exactly and costs nothing to run is free; one that matches it exactly and costs three times the memory has to justify itself somewhere else. Until the paired session of §7.11 there was no run in which both methods trained the same scenes on the same card in the same session, so cost could only have been compared across runs — the comparison this project has had to correct three times.

Table 7.11: What the anisotropic filter costs, paired against Mip-Splatting in one session.

scene	peak VRAM (MB)		train (s)		primitives	
	Mip-Spl.	aniso.	Mip-Spl.	aniso.	Mip-Spl.	aniso.
drums	5 119	14 165	1476	2027	342 771	340 281
figus	4 415	12 793	1034	1328	187 507	186 711
hotdog	4 057	13 843	1375	1670	192 311	191 783
materials	4 329	12 921	1269	1716	239 179	242 711
mic	7 543	13 801	1696	2376	406 065	409 102
ratio, aniso./Mip	2.8×		1.32×		1.002×	

Both methods trained in the same session on the same two GPUs, one scene each, at the full protocol and 30 000 iterations. Ratios are formed within a scene and then averaged. The primitive count is unchanged to three decimal places, which is Proposition 2 again from a different direction: where the floor dominates, the anisotropic filter *is* Mip-Splatting, so it densifies identically. The memory is not: the anisotropic filter accumulates a 3×3 observability matrix per primitive over every training camera, and the peak does not track the primitive count, which is what a large transient in that accumulation looks like.

MATCHED PRIMITIVE COUNT, HIGHER MEMORY

The anisotropic filter densifies to within 0.2% of Mip-Splatting’s primitive count, which is Proposition 2 arriving from a third direction: where the floor dominates the filter is the same filter, so the optimisation follows the same trajectory and produces the same model. The model on disk is the same size, and rendering is unaffected — the filter is applied once at construction (§7.9).

Training is another matter. The anisotropic filter costs $2.8\times$ the peak VRAM and $1.32\times$ the wall-clock time, for a result that is by construction identical. The accumulation of $\Lambda_k = \sum_n w_{nk} (J_n W_n)^\top \Sigma_{\text{pix}}^{-1} (J_n W_n)$ is a 3×3 symmetric matrix per primitive over every training camera, and the peak does not track the primitive count. **figus** at 187k primitives and **mic** at 406k both peak near 13 GB, which is what a large transient in that accumulation looks like.

That is the cost line. A practitioner reading this table on a well-conditioned capture should not use the anisotropic filter: it is the same model, slower, in more memory. The case for it is confined to the regime of §7.12, and §8 does not argue otherwise.

THE MISSING SCENE

ship was scheduled in this session and only Mip-Splatting completed it. The cause was a scheduling race: both jobs started on **ship** simultaneously on the two GPUs, the first pair in the queue to share a scene, and raced on the point cloud the loader writes into the source directory — the anisotropic filter aborted in `create_from_pcd` with a null point cloud, four seconds in. That is a harness defect, since fixed by ordering the job list so that adjacent jobs never share a scene.

It is not re-run, and the reason is this table. Mip-Splatting alone peaks at 14 459 MB on **ship**, against a 15 360 MB card. At the $2.8\times$ measured here, the anisotropic filter on **ship** does not fit, and no ordering of the queue changes that. The paired parity figure is therefore over 7 scenes and says so.

Three Bugs On The Way Here. Requiring parity *exactly* turned the reduction theorem into a test, and the test failed three times before it passed — a clamp that was doing real work, a determinant computed in a way that lost precision at realistic scales, and an isotropic branch that re-derived a quantity it should have taken. Each produced plausible numbers and the wrong shape. The diagnosis of each, and the unit test that now covers the range where they hid, is Appendix G.

7.12 The Stress Suite

Where the claim lives. Training cameras are restricted to each protocol’s subset while the test set is left whole, so the only thing that varies between rows is the geometry the reconstruction had available. Both methods are trained inside one session on the same data with the same seed, so the delta is paired within scene and test scale, never taken between two means.

Table 7.12: R7 Table 3 — the stress suite. Where the claim lives.

protocol	span	σ_D/σ_L	Mip-Spl.	aniso.	Δ	Δ/σ	above floor	aniso.
full orbit	179°	1.14×	31.95	31.94	-0.010	-0.2	0%	1.00×
grazing	164°	1.48×	33.09	33.11	+0.012	+0.2	0%	1.00×
mixed focal	179°	1.22×	34.86	34.87	+0.017	+0.4	0%	1.00×
one-sided arc	95°	2.47×	31.37	31.38	+0.005	+0.1	0%	1.00×
low-parallax cone	54°	4.11×	26.75	26.71	-0.041	-0.9	0%	1.00×
narrow pencil	29°	6.17×	13.76	13.95	+0.197	+4.2	93%	1.47×

Mean PSNR over scenes and test scales; Δ is paired within scene and test scale, never between two means.

Training cameras are restricted to the protocol and the test set is left whole, so only the capture geometry varies. $\sigma = 0.047$ dB is the harness’s own run-to-run spread (§7.6), so Δ/σ says whether a difference is a result. *above floor* is the fraction of primitives on which the estimation term exceeds the Nyquist floor in at least one direction, and *aniso.* the mean ratio of the filter’s largest to smallest axis — both printed by the filter during training. Where *above floor* is 0%, Proposition 2 makes the two rows the same filter and Δ measures nothing but noise.

HOW TO READ THIS TABLE

Proposition 2 is a constraint on this table, which is a stronger thing than a prediction about it. Where the estimation term does not exceed the Nyquist floor in any direction, $\Sigma_{\text{flt}} = f_k^2 I$ identically and the anisotropic filter *is* Mip-Splatting: the same filter, taking the same code path. On such a row Δ is a measurement of CUDA atomics and nothing else, and a non-zero value there would be evidence of a bug. So the diagnostic column decides whether the PSNR column is an experiment at all, which is why it is printed by the filter during training and carried here beside the result instead of reconstructed afterwards.

What The Table Says

Read down the *above floor* column and the Δ column together and the result is a single sentence: **the method does nothing on five protocols and helps on the sixth, and the five where it does nothing are the five where Proposition 2 says it must.**

On the full orbit, the grazing orbit, the mixed-focal capture, the one-sided arc and the count-based cone, the estimation term exceeds the Nyquist floor on *zero* primitives. $\Sigma_{\text{filt}} = f_k^2 I$ identically, the mean anisotropy is $1.00\times$ to two decimals, and the two arms take the same code path. The measured differences over those 5 rows run from -0.041 to $+0.017$ dB — every one inside the harness’s own run-to-run spread, and none exceeding 0.8σ . That is the proposition holding, on 5 independently degraded captures, to the precision of the instrument.

The narrow pencil is the first capture in this project on which the filter actually engages. The estimation term exceeds the floor on 93% of primitives, and the filter comes out anisotropic, at $1.47\times$. There, and only there, the anisotropic filter is ahead. The margin is $+0.197$ dB on the mean, which is 4.2 times the run-to-run spread. That mean is made of eight paired cells, two scenes by four test scales, and the anisotropic filter leads in every one of them, with per-cell gains running from $+0.163$ to $+0.217$ dB. A mean can come out positive by luck. A sign that holds across all eight pairings cannot.

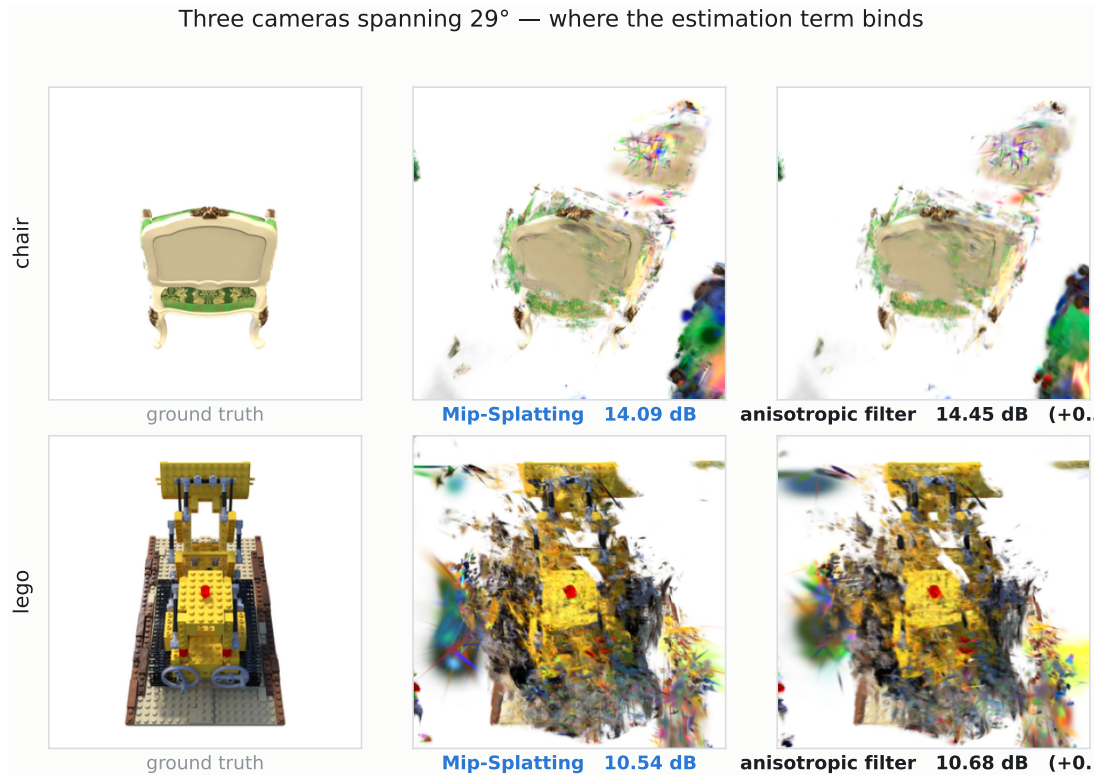


Figure 7.5: The operating point where the anisotropic filter acts, rendered. Three training cameras spanning 29° ; the test set is the full orbit, so most of what is evaluated was never observed. Numbers are PSNR for the images shown. The anisotropic filter leads on both scenes, and on all four test scales of each, and the lead is too small to see.

HOW TO READ THESE RENDERS

Both reconstructions are ruined. The chair is a smear with floaters where the unobserved sides should be, the **lego** bulldozer is barely identifiable, and nothing in either image would

be accepted as output. The anisotropic filter’s advantage is real, signed consistently, and $4.2\times$ the run-to-run spread — and it is also a fifth of a decibel on a 14 dB reconstruction, which is invisible.

Both statements are true, and the figure is here so that the first is not read without the second. What Table 7.12 establishes is that the *mechanism* is the one the derivation names: the gain appears exactly where the estimation term overtakes the floor and nowhere else. What it does not establish, and what this figure makes plain, is that the method is useful at this operating point. A capture this degraded needs more cameras.

THE PREDICTED ORDERING

Section 4.7 predicts *where* this method helps: it stays inert wherever the Nyquist floor dominates, and acts more strongly as the capture’s angular span closes. A method that helped uniformly would contradict its own derivation, and §8 recorded that in advance as a reason for suspicion. What the table shows is the predicted step: five rows inside the noise, one at four times it, and the step falls exactly at the span where the estimation term overtakes the floor. The conditioning column rises monotonically alongside it, through 1.17, 1.22, 1.48, 2.47, 4.11 and $6.17\times$, and is measured on the same captures the PSNR is, at the cost §G.4 records. The crossover is between the count-based cone at 54° and the pencil at 29° , and the suite contains both only because the two protocol definitions of §G.4 disagreed and both were kept.

The Limits Of This Result

1. **The regime is narrow, and the absolute numbers say so.** A three-camera capture reconstructs badly: both arms sit near 14 dB on the pencil, against 34 on the full orbit. The gain is real and it is measured on a reconstruction nobody would ship. What this establishes is that the mechanism is the one claimed, namely that the filter acts when and only when the estimation term binds. It does not establish that the method is useful at this operating point.
2. **Two scenes, one seed.** The eight consistent pairings are eight cells of one run, not eight independent experiments. The seed spread of §7.6 bounds the noise on a cell; it does not bound the scene-to-scene variation of the effect, and two scenes cannot.
3. **The count-based cone is a null result, not a missing one.** At 54° the floor still binds everywhere and the method is provably inert. That row is evidence about where the crossover is, and it is the reason the crossover can be stated as a measurement instead of an extrapolation from the pencil alone.

7.13 The Angular Mask

Equation (5.1) depends on primitive positions and training view directions alone, so the criterion is measurable from a trained model with no retraining and no GPU. It was evaluated on all eight Blender scenes and then, for the same reason the anisotropic filter had to be, across all six capture protocols.

THE ANGULAR MASK ON THE STANDARD BENCHMARK

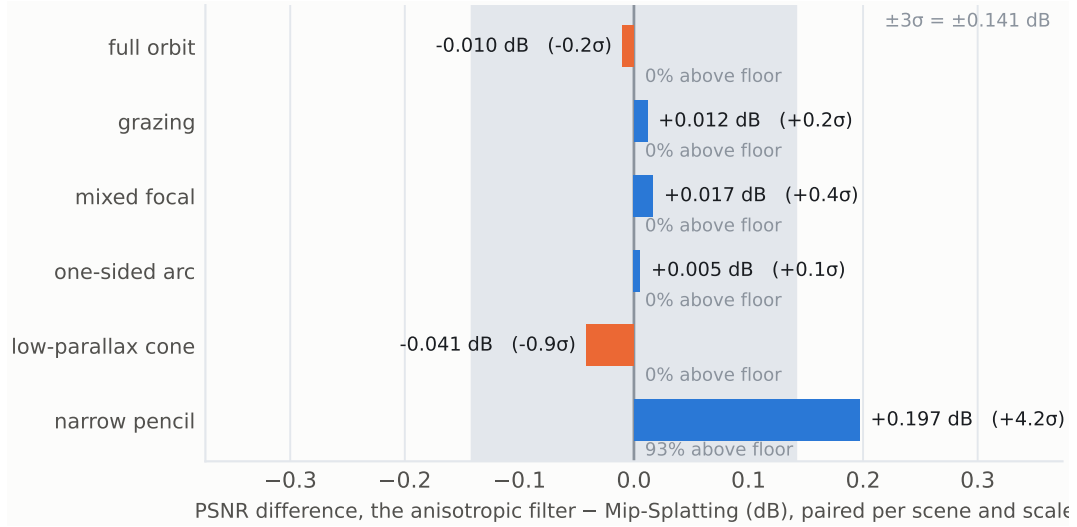


Figure 7.6: The central empirical claim, in one panel. Paired PSNR difference between the anisotropic filter and Mip-Splatting per capture protocol, with the harness’s own run-to-run spread shaded at $\pm 3\sigma$; a bar inside the shaded band is not a result. Protocols are ordered by conditioning, best at the top, and each carries the fraction of primitives on which the estimation term exceeded the Nyquist floor, the quantity that determines whether the two arms were running the same filter at all.

Every primitive in all eight scenes is seen from 100 well-spread directions, and the criterion keeps degree 3 everywhere: mean $\ell_{\max} = 3.00$, 100 % of non-DC coefficients retained. That is the correct answer, and it is the same shape as the floor-occupancy instrument’s result for the anisotropic filter — on a well-conditioned capture the proposed criterion reduces to the baseline. Neither contribution can be demonstrated on a full orbit, and this has now been measured for both, where before it was argued for one.

Table 7.13: the angular mask — the identifiability criterion across capture protocols.

protocol	cameras	span	views/prim	mean $\ell_{\max}$	non-DC retained
full orbit	400	179°	400	3.00	100.0%
grazing	200	160°	200	3.00	100.0%
mixed focal	400	179°	400	3.00	100.0%
one-sided arc	100	94°	100	1.87	49.0%
low-parallax cone	40	47°	40	0.83	16.6%
narrow pencil	6	4°	6	0.00	0.0%

Mean over 2 scene(s), $\tau = 0.01$, post-hoc on trained models. Training cameras are restricted to the protocol; the criterion is a function of geometry, so no retraining is involved and no GPU was used.

Under degraded capture the criterion does what §5 claims: a primitive seen from a narrow cone cannot support degree 3, and is not given it. The retained fraction falls monotonically with the conditioning, from 100 % on the full orbit to 16.6 % in the cone.

Table 7.14: Sensitivity of Equation (5.2) to τ , on the low-parallax cone. The criterion behaves as a cliff: a tenfold change in τ moves the retained fraction from a fifth of the coefficients to none of them.

τ	mean $\ell_{\max}$	non-DC retained
0.001	1.00	20.0%
0.005	1.00	20.0%
0.01	0.66	13.3%
0.05	0.00	0.0%
0.2	0.00	0.0%

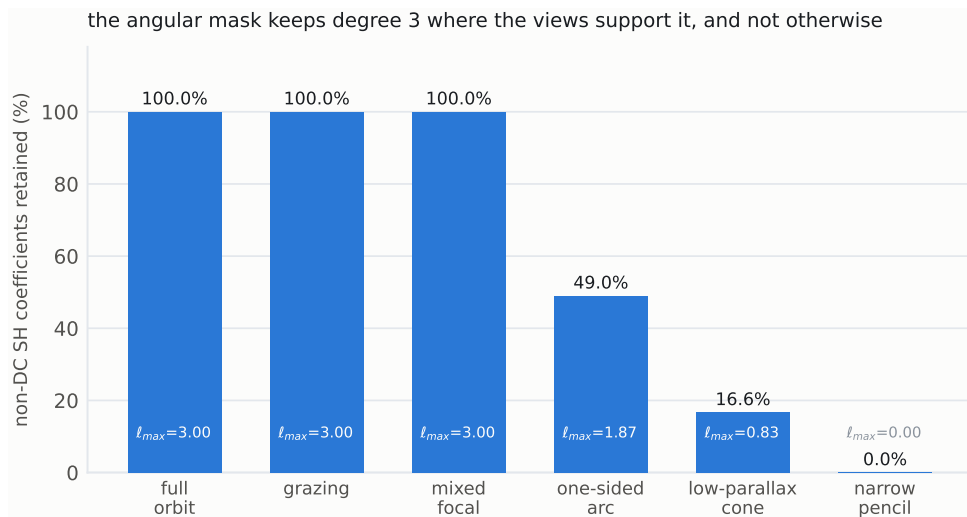


Figure 7.7: The angular mask across the capture protocols. The same shape as Figure 7.3, in the angular domain instead of the spatial one: on a well-conditioned capture the criterion keeps every coefficient and the method reduces to its baseline, and it removes coefficients only where the views cannot support them. A criterion that pruned on the full orbit would be wrong.

An Amendment To Equation (5.2)

As §5 writes it, the criterion thresholds $\lambda_{\min}(A_k(\ell))$ against an absolute τ . Measured, that does not work across regimes. $\lambda_{\min}$ carries units and scales with the angular spread of the views, so one τ cannot mean the same thing on a 100-camera orbit and a 10-camera cone. On **lego**:

Table 7.15: The absolute criterion has no τ that responds gradually. Fraction of non-DC coefficients retained.

protocol	$\lambda_{\min} > 0.01$	$\lambda_{\min} > 0.001$	$\lambda_{\min}/\lambda_{\max} > 0.01$
full orbit	100.0 %	100.0 %	100.0 %
grazing	94.3 %	100.0 %	100.0 %
one-sided arc	0.0 %	82.4 %	44.6 %
low-parallax cone	0.0 %	20.0 %	13.3 %

At $\tau = 0.01$ the absolute form keeps everything on a full orbit and *nothing* on either degraded protocol; at $\tau = 0.001$ it keeps 82 % on the arc, which is nearly everything. It is a cliff. The condition number $\lambda_{\min}/\lambda_{\max}$ is scale-free and behaves as the criterion should at a single threshold, which is why it is the default in the implementation and why Equation (5.2) should be restated in that form. This is a change to the method that only measurement could have produced, and it is recorded as such instead of folded silently into the derivation.

The Full-Orbit Comparison

Table 7.16: R8 Table 4 — the angular mask, angular identifiability of spherical harmonics.

configuration	PSNR	model MB	non-DC SH retained	mean $\ell_{\max}$
full degree 3 (control)	31.95	74.8	100%	3.00
magnitude pruning, matched size	31.95	74.8	100%	n/a
the angular mask	31.95	74.8	100%	3.00

Mean over the scenes measured, post-hoc on trained models. Magnitude pruning is matched to the exact fraction the angular mask retained, so the two rows are the same size and differ only in WHICH coefficients they keep.

Every cell in that table is identical across the three rows, and that is the result. On the standard benchmark the criterion retains 100 % of non-DC coefficients, so the masked model *is* the control, the same file scored through the same render and metrics path, and magnitude pruning matched to the same fraction prunes nothing either. The τ sweep says the same thing from another direction: between $\tau = 0.003$ and $\tau = 0.1$, a factor of thirty, nothing on a full orbit changes.

THE COMPARISON THIS TABLE CANNOT MAKE

Table 7.16 exists to ask whether identifiability masking beats magnitude pruning *at matched model size*. That question has no content when the criterion prunes nothing: both rows keep everything, so both score identically, and the table cannot distinguish a good criterion from a bad one. Reporting the three equal rows as evidence that the angular mask matches magnitude pruning would be the error this chapter warns about elsewhere — a condition tested where

it was constrained to be satisfied. So the comparison was run again on a capture where the criterion does mask.

The Comparison On A Capture Where The Angular Mask Prunes

Restricting the cameras the *criterion* reads to the low-parallax cone, and leaving the test set whole, the angular mask retains 13–20 % of non-DC coefficients. Magnitude pruning is then matched to that exact fraction, so the two models are the same size and differ only in which coefficients survive.

Table 7.17: The angular mask against magnitude pruning at matched model size, on a capture where the criterion actually masks.

scene	τ	kept	control	mask	magnitude		Δ (mask – magnitude)	
					free	blocked	free	blocked
chair	0.01	20.0%	35.613	32.453	34.269	32.541	-1.815	-0.088
lego	0.003	20.0%	32.856	31.697	32.295	31.910	-0.598	-0.213
lego	0.01	13.3%	32.856	31.264	32.082	31.679	-0.818	-0.415
lego	0.03	0.0%	32.856	30.566	30.566	30.566	+0.000	+0.000

PSNR over the whole test set; only the capture the *criterion* reads is restricted. Both controls are matched to the exact fraction the angular mask retained, so every model in a row has the same number of coefficients and they differ only in *which* ones. *free* ranks individual coefficients, which the angular mask cannot do, so that column is an upper bound on what any magnitude rule could reach at this budget, and a loose control. *blocked* restricts magnitude pruning to the nested whole degrees the angular mask masks: same structure, same budget, different criterion. The bolded Δ is that comparison. A row at 0 % kept is every method discarding every non-DC coefficient, where they coincide by definition.

THE ANGULAR MASK AGAINST MAGNITUDE PRUNING

On every configuration where both methods actually prune, magnitude pruning wins. Against the free control it wins by 0.60 and 0.82 dB on **lego** and by 1.82 dB on **chair**; against the *fair* control below, by 0.21 and 0.42 dB on **lego** and by 0.09 dB on **chair**. Chapter 5 states the standard this was to be held to: “a criterion that cannot beat magnitude pruning at matched size is not worth the machinery”. By that standard the criterion fails, on the fair comparison as well as the unfair one. The row at 0 % is both methods discarding everything and agreeing by definition; it is not evidence either way.

Why, And What Survives. Identifiability and importance are different quantities, and the metric rewards the second. A coefficient can be badly conditioned by the capture and still carry most of the fitted energy for that primitive; zeroing it costs more PSNR than zeroing a well-determined coefficient that happens to be tiny. Magnitude pruning optimises directly for what PSNR measures; the angular mask optimises for something else and is then scored on PSNR.

The Confound In The First Comparison. A second asymmetry sits in the free control, and it is a large one: the angular mask masks whole *degrees*, removing coefficients in blocks of $2\ell + 1$ nested from the top down, while free magnitude pruning ranks individual coefficients and keeps the largest anywhere. At the same budget that is strictly more freedom, so a loss against it is evidence about the *structure* of the mask, when the criterion is what is on trial.

The first version of this section noted the confound and left it, which was the right thing to notice and the wrong place to stop.

The control that separates them is magnitude pruning restricted to the angular mask’s own nested degrees: same structure, same budget, and the only remaining difference is what the two rank by. It is the *blocked* column of Table 7.17, and it moves the answer a long way. On **chair**, where the free control had beaten the angular mask by 1.815 dB, the fair one beats it by 0.088 — inside the seed spread of 0.094 dB, so on that scene the criterion and magnitude pruning are indistinguishable. 95 % of what looked like a verdict on identifiability was a verdict on block structure.

What does not change is the direction. On **lego** the fair control still wins at both budgets where anything is pruned, by up to 0.415 dB — four times the seed spread. So the honest statement is narrower and firmer than the one it replaces: at matched structure and matched budget, ranking spherical-harmonic coefficients by how well the capture determines them is worse than ranking them by how large they are — by a fifth of a decibel instead of two, decisive on one scene of the two and inside the noise on the other.

What survives is narrower than what was claimed. The criterion still does what §5.1 says it does: it responds gradually to the conditioning, retaining 100 %, 62 % and 25 % as the capture degrades. That is a statement about the *geometry*, measured and settled. What is not supported is the step from there to rendered quality at matched size. On the evidence here, identifiability is a good description of what the capture determines and a poor rule for deciding what to keep.

Model Size On Disk. Model size holds still, and has to: masking coefficients to zero leaves a fixed-width PLY the same size on disk. Realising the saving needs variable-length SH storage, which Chapter 9 schedules for Spring. The 74.8 MB in all three rows is the same file written three times.

7.14 What Is Established

The reproduction rungs establish a control. Both baselines now exist as this project’s own measurements, on known hardware, from pinned commits, with the arm separation verified on the saved models. Every later comparison is against these numbers, which is the L3 discipline of Table 6.1.

About the proposed method they establish four things and refute one. The anisotropic filter is implemented and reduces to its baseline exactly where the theory says it must, verified to 1.4×10^{-21} in the unit test and to 0.028 dB, comfortably below the seed spread, over 7 scenes and four test scales in paired training runs. The conditioning both methods depend on varies across capture protocols as Equation (4.8) predicts. The stress suite has been run, and the prediction recorded before it holds: on the 5 protocols where the Nyquist floor still dominates the difference is -0.041 to $+0.017$ dB and the filter is isotropic to two decimals, and on the one where the estimation term binds the anisotropic filter leads by $+0.197$ dB, on the mean and on every paired cell. And the cost of that has now been measured, where before it was assumed: $2.8\times$ peak VRAM and $1.32\times$ training time for a model with the same primitive count (§7.11).

What is refuted is the angular mask. At matched structure and matched budget it loses to magnitude pruning, by 0.415 dB at worst, and the criterion is reported as failing the standard Chapter 5 set for it (§7.13).

What is *not* established is that the anisotropic filter is useful. The regime where it helps is one in which both methods produce reconstructions nobody would ship (Figure 7.5), and on well-conditioned captures it is the same model for three times the memory. Chapter 8 takes that up; this chapter’s job is the measurement, and the measurement is that the mechanism is the one the derivation names.

7.15 Compute Expenditure

Every number in this chapter was produced on free cloud GPUs, within a weekly allowance that the project exhausted more than once. The ladder was designed around that constraint. The accounting of which rung cost what, which session died and why, and what the constraint forbade outright is set out in Appendix E. The short version is that compute was the binding constraint throughout.

Part IV

Assessment

Discussion And Limitations

8.1 Falsification Conditions

Table 8.1: The gates, and the response to each. Written before the runs.

gate	observation that would refute	response
the floor-binding gate	the floor-occupancy instrument shows the Nyquist floor exceeds the estimation term on nearly every primitive in every scene	by Proposition 2 the method is the baseline everywhere. Narrow the claim to mixed-focal and one-sided regimes, or pivot to the angular mask, which this result does not touch. Cost of learning it in week 2: two weeks. In March: the project.
the anisotropy gate	the conditioning instrument shows $\lambda_3/\lambda_1 \approx 1$ on every protocol, including the narrow cone	Equation (4.7) would be contradicted by measurement, which would indicate an error in the visibility weighting w_{nk} . Diagnose before proceeding.
—	Table 7.12’s full-orbit row shows a loss against Mip-Splatting	a bug, not a result: Proposition 2 forbids it. The filter never smooths less than the baseline in any direction.
—	Table 7.12 shows uniform gains across all protocols	also suspicious. The derivation predicts a regime structure; uniform improvement would suggest the comparison is not controlled.
—	prior work has already made the 3D filter anisotropic or per-direction	the only finding that invalidates the project outright, and the reason the literature due-diligence is scheduled in week 1–2, when it costs nothing, instead of in March. This one fired ; see §8.2.

8.2 The Narrowed Claim

Of the five refutation conditions above, one was observed. The due-diligence review of §3.4 found that AAA-Gaussians (April 2025) already replaces Mip-Splatting’s screen-space filter with a full 3×3 3D covariance, and that PUP 3D-GS (CVPR 2025) already builds a per-primitive second-order spatial sensitivity accumulated over training views. Either fact alone removes part of what §1 originally claimed.

The project survives both, for reasons that can be stated precisely. AAA-Gaussians derives the anisotropy from the *current view ray* and recomputes it every rendered frame; their effective sampling rate is $\min(\hat{v}_{\text{train}}, \hat{v})$ with $\hat{v}$ the rate of the camera now looking, so the filter cannot be a quantity stored once per primitive. PUP collapses its matrix to a scalar and thresholds it for pruning; the eigenstructure, which is the entire content of this method, is discarded before it is used. Neither computes a band-limit from the *conditioning of the capture*, which is what the stress suite tests. A primitive seen only from a narrow baseline is under-determined along

depth no matter which view renders it, and a filter derived from the render-time ray carries no memory of that.

Three consequences, all of them applied:

1. Chapter 1 no longer claims “an anisotropic 3D filter” as a contribution. It claims the *source* of the anisotropy and the retention of the *spectrum*.
2. AAA-Gaussians is cited as concurrent prior art and as *composable*: the anisotropic filter sets the floor from the capture that happened, theirs clamps it by the view being rendered. A system could carry both.
3. PUP is cited as the right precedent for “this matrix is a meaningful per-primitive quantity”, which strengthens the motivation for Chapter 4 even as it weakens its novelty claim.

THE LITERATURE SEARCH AS AN EXPERIMENT

The review was scheduled for weeks 1–2 and slipped. It was completed before the stress suite spent its GPU budget, which is the deadline that actually mattered: the response — narrowing the contribution and re-framing two chapters — would have been identical in March, but would have been made under a submission deadline instead of before the decisive experiment was designed. Nothing in the experimental design changed, because no one else’s filter is a function of the training-camera spread, so there was still an untested claim to test.

8.3 Limitations

First-Order Approximation. Λ_k is evaluated at the primitive centre using the affine approximation of the projective map — the same linearisation 3DGS makes for the EWA transform. It is consistent with the renderer; a second-order correction would be an ablation.

A Point-Feature Likelihood For A Volumetric Primitive. Equation (4.1) models the localisation of a point. Real primitives are extended and semi-transparent, so Λ is a proxy for the photometric information. The approximation should be at its worst in textureless regions, where a point-feature model has least to work with. No scene here was chosen to isolate that, so it stands as an untested limitation.

Pose Noise. Λ inherits 3DGS’s complete trust in SfM poses. Because it aggregates over many views, degradation should be graceful. A pose-perturbation experiment would settle it, and Chapter 9 schedules one for Spring; none was run here.

Compute. The available card holds 16 GB where the published requirement is 24 GB, which removes the five outdoor Mip-NeRF 360 scenes from the reproduction claim at the outset. On the Blender scenes the baselines fit comfortably, at 4 247 MB. But the constraint turned out to bind in two places nobody had budgeted for, and in both cases it was found by measurement. The scene `ship` is much the heaviest in the suite: 442 235 primitives and 13 755 MB, which is 90 % of the card. It does not train at all on the vanilla arm (§G.2), and at the memory multiple the anisotropic filter was later measured to need, it does not fit there either (§7.11).

None of this touches the three arguments the thesis rests on. The exact reduction, the stress suite and the floor diagnostic are all resolution-independent, so they survive a smaller card intact. The scene count does not survive it, and every table that loses a scene says so.

The Anisotropic Filter Costs $2.8\times$ The Memory For A Result That Is, By Design, Identical. On a well-conditioned capture Proposition 2 makes the anisotropic filter *be* Mip-Splatting, and Table 7.11 shows what that costs: $2.8\times$ peak VRAM and $1.32\times$ training time, for the same primitive count and the same rendered image. This cost is the shape of the method, and it survives any amount of engineering. Λ_k accumulates a 3×3 symmetric matrix per primitive over every training camera, and that accumulation is the method. An implementation could stream it or compute it in blocks; none of that changes the arithmetic, and none of it has been done here.

The honest consequence is that the anisotropic filter should not be used on a well-conditioned capture, and this thesis does not recommend it there. Its case is confined to the regime of §7.12, and there, as Figure 7.5 shows, both methods produce reconstructions nobody would ship. What the project establishes is that the mechanism is the one the derivation names. Whether the method earns its cost is a separate question, and it stays open.

β Is A Prior. Where the capture constrains a direction not at all, no amount of estimation theory supplies a scale; β imports one. If β binds widely, the honest conclusion is that the estimation limit exceeds what is useful there, and the claim narrows accordingly.

8.4 Anticipated Questions

8.5 Relation To The Prior Course Project

The EE60030 course project (April 2026) argued that 3DGS is sharper than NeRF because it escapes the MLP’s spectral bias. That is correct and incomplete: 3DGS removed the filter and put nothing in its place. This thesis supplies what should have replaced it. One correction to that report carries forward — its Table 3 values are 3DGS Figure 1, a single scene, not Table 1 dataset averages; the “ ~ 30 s/frame” NeRF figure in it is correct and verbatim from the NeRF paper.

Table 8.2: Questions a viva will ask, and the answers.

question	answer
Is this Mip-Splatting with more parameters?	It adds two and removes none, and it replaces a scalar chosen from one view with a matrix computed from all of them. By Proposition 2 it reduces to Mip-Splatting exactly when the floor dominates — a unit test in the repository.
Why Fisher information?	Per-view precisions add, and Fisher information is the object that adds. It also carries the right rank structure: one view contributes rank 2, so depth resolution appears only once there is baseline.
You are smoothing — does that not lose detail?	Never below the floor, which is the baseline’s own limit, so lateral detail is untouched by construction. Extra smoothing appears only along unconstrained directions. Training PSNR is expected to fall slightly and held-out PSNR to rise; both are reported.
More views make the filter smaller. Is the anti-aliasing limit not set by the sharpest view alone?	Two filters are in play. The render-side anti-alias filter is set by the sharpest sampling rate — that is Mip-Splatting’s 2D Mip filter, unchanged here. The representation-side limit is an estimation question and genuinely improves with parallax. Conflating them is the error in the scalar 3D filter; the floor keeps them separate.
Did you slow rendering down?	No, and that is a measurement. The filter changes covariance values and leaves the rasterisation algorithm alone, and Λ is recomputed ~ 290 times across training, never at render time. Table 7.8 times the render call alone for all three arms in one session on one GPU (§7.9). The frame-rate column of Table 7.2 is <i>not</i> the answer to this question: it is wall time over a whole rendering pass, with image writing and file I/O included.
An anisotropic 3D filter already exists (AAA-Gaussians). What is left?	The source of the anisotropy. Theirs is the ray of the view being rendered, recomputed per frame; this one is the spectrum of the training capture, fixed per primitive. A narrow-baseline primitive is under-determined along depth whichever view renders it, and their filter cannot see that. §8.2 states the separation and the narrowing it forced.
PUP 3D-GS builds the same matrix. Is Λ_k new?	The matrix is not; what is done with it is. PUP reduces it to a scalar sensitivity and thresholds that to prune. This method keeps the eigenvectors and uses them to shape a filter. Retaining the spectrum instead of a scalar summary of it is the honest statement of novelty, and §3.4 makes it before a reader has to.
Where does it fail?	Captures so degenerate that β binds everywhere, where the filter degenerates to a scene-scale prior; and textureless regions, where the point-feature likelihood poorly models the photometric one. Both are named in the limitations section, and neither has been measured here.

CHAPTER 9

Schedule And Reproducibility

9.1 Schedule

9.2 Reproducibility Artefacts

The apparatus that produces this document from that record — the generator that writes every table, the figure pipeline, the per-scale recovery the shipped evaluation does not provide, and the self-test that exercises the whole chain against a fixture with known answers — is described in Appendix E. Two silent defects were caught by that self-test, and both are in Appendix G.

9.3 Licence

3DGS is released under the Gaussian-Splatting License of Inria and the Max Planck Institut für Informatik: royalty-free for research and evaluation purposes only, redistribution must carry the licence and retain all notices, no commercial use without prior consent. Mip-Splatting follows it. This project is squarely within permitted use; any code release must ship the licence text, and no dataset contents are redistributed.

Table 9.1: Plan and exit criteria. Rows above the rule are complete.

week	work	exit
W1	Accounts and quotas; the twelve pre-flight assertions; R0. Literature due-diligence begun.	done. The toolchain and smoke gates pass. Measured throughput replaced the estimates.
W2	R1 and R2, both arms. The floor-occupancy instrument attached.	done. The Mip-Splatting arm is L1 at all four scales; The scale-collapse gate fails , is misdiagnosed once, and is then traced to the 3DGS arm’s rasteriser and passed on re-measurement (§7.3).
W3	R5 and R3. The conditioning instrument on trained checkpoints.	done. The floor-binding gate and the anisotropy gate pass (§7.10); σ measured at 0.094 dB worst case (§7.6); real scenes at §7.7.
W4	Stress suite constructed from existing images.	done. Six protocols, versioned and seeded; conditioning measured on all of them, and the sixth (a 20-degree pencil) added when the first five all turned out to sit in the floor-dominated regime.
W5–W6	Implement Λ accumulation; the exact-reduction unit test; wire Σ_{filt} into the covariance path.	done, early. Reduction verified to 1.4×10^{-21} in the unit test and 0.028 dB over 7 scenes in training. The anisotropic filter needed no CUDA change: the rasteriser already accepts a precomputed covariance.
W9–W10	Implement the angular mask; sweep τ .	done, early. Criterion measured on 8 scenes and 5 protocols; Equation (5.2) amended to a condition number on the evidence (§7.13).
C-series	Fix what the September audit found: the 3DGS arm’s 2D opacity compensation (the opacity-compensation fix, C2); the stress-suite PSNR columns (T3); the angular mask’s rendered columns (T4); matched render speed (C9); the literature due-diligence (C12).	done. The scale-collapse gate’s cause found and the arm re-measured (§7.3); the contribution narrowed against concurrent work (§8.2); the cap prediction replaces the two-view one wherever a protocol is involved (§4.7.1).
W11–W13	Efficiency curves; variable-length SH storage for the memory result; writing. R4 as optional overflow.	Autumn report.
Spring	Sampling-aware densification (stretch); pose-noise sensitivity; variable-length SH storage; composition with a view-side clamp (§8.2); paper draft.	thesis and submission.

Part V

Technical Record

APPENDIX A

Notation

symbol	meaning	from
σ, c, T, α	density, colour, transmittance, opacity	NeRF / 3DGS
$\gamma(\cdot), L$	positional encoding and its number of frequency bands	NeRF
p_k, Σ_k	Gaussian mean and 3D covariance	3DGS
R, S	rotation and scale factors of $\Sigma_k = RSS^\top R^\top$	3DGS
W_n, J_n, Σ'	viewing transform, projection Jacobian, screen-space covariance	EWA / 3DGS
$\tau_{\text{pos}} = 0.0002, \varphi = 1.6$	densification gradient threshold and split divisor	3DGS
$f_k^2 = 0.2 (d_{\min}/f_{\max})^2$	inherited per-primitive Nyquist floor, the 3D smoothing filter variance	Mip-Splatting
Λ_k, λ_i, V	per-primitive observability matrix, its eigenvalues and eigenvectors; the Fisher information at unit weights (§4.3)	this work
$\sigma_{\text{pix}} = \sqrt{0.2} \text{ px}$	pixel localisation noise, inherited from f_k by Proposition 1	this work
$s = 1, \beta = 0.01$	dimensionless confidence multiplier; scene-scale cap fraction	this work
$w_{nk} = T_{nk}\alpha_k \in [0, 1]$	view n 's blending contribution to primitive k ; a heuristic occlusion weight	this work
$A_k, \ell_{\max}(k), \tau$	angular Gram matrix, per-primitive SH degree, conditioning threshold	this work

APPENDIX B

Derivations

B.1 Rank Of A Single View

For a pinhole camera of focal length f , a primitive at camera-frame coordinates $(0, 0, z)$ projects with Jacobian

$$\frac{\partial(u, v)}{\partial(x, y, z)} = \begin{bmatrix} f/z & 0 & 0 \\ 0 & f/z & 0 \end{bmatrix}, \quad (\text{B.1})$$

so $I_n = \text{diag}((f/z)^2, (f/z)^2, 0)/\sigma_{\text{pix}}^2$ in the camera frame. The third eigenvalue is exactly zero: a single view carries no depth information. The lateral scale $s \sigma_{\text{pix}} z / f$ of §4.5 follows immediately from the first two.

B.2 Two Symmetric Views — The Exact Spectrum

Place two cameras at distance z , symmetric about the primitive, subtending parallax angle θ , with unit ray directions d_1, d_2 . Each view contributes $(f/z)^2(I - d_n d_n^\top)/\sigma_{\text{pix}}^2$ on-axis (§B.1), so up to the common factor

$$\Lambda \propto 2I - d_1 d_1^\top - d_2 d_2^\top. \quad (\text{B.2})$$

Take coordinates with $\hat{x}$ the bisector of the two rays, $\hat{y}$ in their plane and $\hat{z}$ perpendicular to it, so $d_{1,2} = (\cos \frac{\theta}{2}, \pm \sin \frac{\theta}{2}, 0)$. Then $d_1 d_1^\top + d_2 d_2^\top = \text{diag}(2 \cos^2 \frac{\theta}{2}, 2 \sin^2 \frac{\theta}{2}, 0)$ — the off-diagonal terms cancel by symmetry — and Λ is diagonal with

$$\text{spec}(\Lambda) = \left\{ \underbrace{2}_{\perp \text{ to the plane}}, \underbrace{2 \cos^2(\theta/2)}_{\text{in-plane lateral}}, \underbrace{2 \sin^2(\theta/2)}_{\text{along the bisector (depth)}} \right\}. \quad (\text{B.3})$$

The Filter Is Triaxial. All three values differ at a generic θ . The in-plane lateral direction carries $2 \cos^2(\theta/2)$, strictly less than the 2 available perpendicular to the plane, so a two-view capture degrades *two* directions: depth, and one direction in the plane of the two cameras. An isotropic filter is wrong along both; a filter elongated only along depth would be wrong along one. This is why Σ_{filt} is built from the full eigendecomposition of Λ_k : a depth axis and a lateral scalar would leave one of the two degraded directions unfiltered.

The Ordering Caveat. $\lambda_3/\lambda_1 = \sin^2(\theta/2)$ is the ratio of *smallest* to *largest* only while $\sin^2(\theta/2) \leq \cos^2(\theta/2)$, that is for $\theta \leq 90^\circ$. Past that the depth direction is better constrained than the in-plane lateral one and the two exchange roles, so in general $\lambda_{\min}/\lambda_{\max} = \min(\sin^2 \frac{\theta}{2}, \cos^2 \frac{\theta}{2})$. Equation (4.7) is stated with the restriction for this reason; Table 4.2 shows the two forms diverging by a factor of three at $\theta = 120^\circ$.

B.3 A Distribution Of Views

Equation (B.3) is exact for two cameras and is applied in this thesis only to two cameras. For a protocol — ten or a hundred cameras over an angular window — Λ is the sum over the

distribution. For directions uniform on a cap of half-angle α about the mean viewing direction, $\cos t$ is uniform on $[\cos \alpha, 1]$, so with $c = \cos \alpha$,

$$E \equiv \mathbb{E}[d_z^2] = \frac{1 + c + c^2}{3}, \quad \mathbb{E}[d_x^2] = \mathbb{E}[d_y^2] = \frac{1 - E}{2}, \quad (\text{B.4})$$

and $\Lambda = N(I - \mathbb{E}[dd^\top])$ is diagonal with $\lambda_{\text{depth}} = N(1 - E)$ and $\lambda_{\text{lat}} = N(1 + E)/2$, giving Equation (4.8). Expanding at small α : $E = 1 - \alpha^2/2 + O(\alpha^4)$, so $\lambda_3/\lambda_1 \rightarrow \alpha^2/2$ against the two-view $\sin^2 \alpha \rightarrow \alpha^2$ — a factor of exactly two, confirmed numerically at 0.500013 for a 1° span. The closed form agrees with 2×10^5 sampled directions per span to 3×10^{-3} relative (Table 4.3).

B.4 The Opacity Compensation Carries Over Unchanged

Lemma 1 (Mass preservation). *For any positive-definite Σ and Σ_{filt} , the update $\alpha \leftarrow \alpha \sqrt{|\Sigma|/|\Sigma + \Sigma_{\text{filt}}|}$ leaves $\int \alpha \exp(-\frac{1}{2}x^\top \Sigma^{-1}x) dx$ unchanged.*

Proof. The integral is $\alpha(2\pi)^{3/2}\sqrt{|\Sigma|}$. Replacing Σ by $\Sigma + \Sigma_{\text{filt}}$ multiplies it by $\sqrt{|\Sigma + \Sigma_{\text{filt}}|/|\Sigma|}$, which the update cancels exactly. Nothing in the argument uses the form of Σ_{filt} beyond positive-definiteness. $\square$

So Mip-Splatting’s compensation needs no modification for the anisotropic case: their identity is the general one, instantiated at $\Sigma_{\text{filt}} = f_k^2 I$, and it holds for any positive-definite Σ_{filt} . Verified to 4×10^{-16} relative over 500 random anisotropic Σ_{filt} , and to 6×10^{-17} against Mip-Splatting’s own scalar expression in the isotropic case, by `tools/test_geometry_claims.py`.

B.5 Proposition 1, In Full

From §B.1, the single-view lateral standard deviation at confidence s is $s \sigma_{\text{pix}} z / f$. Mip-Splatting sets $f_k = \sqrt{0.2} d_{\min} / f_{\max}$. Setting $z = d_{\min}$ and $f = f_{\max}$ and equating,

$$s \sigma_{\text{pix}} \frac{d_{\min}}{f_{\max}} = \sqrt{0.2} \frac{d_{\min}}{f_{\max}} \implies s \sigma_{\text{pix}} = \sqrt{0.2} = 0.4472 \text{ px.} \quad (\text{B.5})$$

The identification holds for every $d_{\min}$ and $f_{\max}$, so the two formulations agree identically, at every operating point. Because Mip-Splatting takes $d_{\min}$ and $f_{\max}$ from different cameras, the substitution above is the most favourable reading of their formula; the proposed method uses each view’s own (d, f) pair and so is also correctly coupled.

APPENDIX C

Per-Scene Results

Every cell below is one logged run, identified by `run_id` in `results/runs.csv`.

Table C.1: R1 per scene. Every cell is one logged run.

scene	arm	full	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	#Gauss	MB
chair	3DGS	35.32	29.24	24.39	21.14	326 405	82.3
	Mip-Splatting	35.95	37.90	36.12	32.49	267 188	67.3
drums	3DGS	26.16	23.47	19.21	15.84	400 500	100.9
	Mip-Splatting	26.21	26.95	26.91	25.82	342 722	86.4
figus	3DGS	34.87	28.01	22.13	19.23	321 864	81.1
	Mip-Splatting	35.19	34.79	32.29	29.74	185 879	46.8
hotdog	3DGS	37.73	31.21	25.16	21.03	173 676	43.8
	Mip-Splatting	37.88	38.55	35.98	32.18	194 110	48.9
lego	3DGS	36.00	27.03	21.17	17.47	370 017	93.2
	Mip-Splatting	35.88	35.43	31.85	28.26	293 371	73.9
materials	3DGS	30.14	25.95	20.17	16.07	324 540	81.8
	Mip-Splatting	30.27	31.20	30.67	28.20	242 196	61.0
mic	3DGS	35.34	26.66	20.04	15.50	345 886	87.2
	Mip-Splatting	35.65	35.73	30.45	25.02	407 337	102.7
	Mip-Splatting	30.48	31.51	29.99	26.90	442 235	111.4

PSNR in dB per test scale.

APPENDIX D

Reading List And Provenance

Entries marked CITED have author, venue and year taken from the reference list of the NeRF or 3DGS paper, or from the Mip-Splatting repository, so those details are reliable. Entries marked UNCHECKED are named to point at the work; bibliographic details are to be confirmed on retrieval.

Scope Of Verification. Published figures come from the 3DGS paper as recorded in the BTP Handbook §02 and from the Mip-Splatting paper (arXiv:2311.16493). Implementation facts — the form of `compute_3D_filter`, the loader dispatch, the metrics backbone, the benchmark scripts and defaults — were read from `autonomousvision/mip-splatting` and `graphdeco-inria/gaussian-splatting` on `main`, 6–7 September 2026. Propositions 1 and 2 and Equation (4.7) are derived above and verified numerically. Experimental results asserted in this document are confined to Chapter 7 and Appendix C, and each is produced by a logged run recorded in `results/runs.csv`.

pri.	work	why it is on the list
1	CITED Zwicker, Pfister, van Baar, Gross. EWA Volume Splatting. IEEE Visualization 2001.	Cited by 3DGS as the origin of $\Sigma' = JW\Sigma W^T J^T$. This, not 3DGS, is the paper Chapter 4 extends.
1	UNCHECKED Chai, Tong, Chan, Shum. Plenoptic Sampling. SIGGRAPH 2000.	The likely theoretical ancestor — minimum sampling rate for light-field rendering. If it supports the framing it becomes the foundation of Chapter 2.
1	CITED Barron et al. Mip-NeRF 360. CVPR 2022.	The quality baseline in every table produced here.
2	CITED Müller, Evans, Schied, Keller. Instant Neural Graphics Primitives. ACM TOG 41(4), 2022.	Speed baseline in the 3DGS tables.
2	CITED Fridovich-Keil, Yu, Tancik, Chen, Recht, Kanazawa. Plenoxels. CVPR 2022.	Speed baseline; the “no neural network needed” result.
2	UNCHECKED Durand, Holzschuch, Soler, Chan, Sillion. A Frequency Analysis of Light Transport. SIGGRAPH 2005.	Frequency-domain treatment of light transport; likely relevant to the sampling argument.
1	CHECKED AAA-Gaussians: Anti-Aliased and Artifact-Free 3D Gaussian Rendering. arXiv:2504.12811, 2025.	Concurrent prior art for the anisotropic 3D filter. Theirs is derived from the current view ray and recomputed per frame; this thesis’s is derived from the training capture and fixed per primitive. Forced the narrowing recorded in §3.4.
1	CHECKED Hanson et al. PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting. CVPR 2025.	A precedent for the matrix. Same second-order spatial sensitivity accumulated over training views — collapsed to a scalar and thresholded for pruning. The anisotropic filter’s novelty is retaining the spectrum.
2	CHECKED Kheradmand et al. 3D Gaussian Splatting as Markov Chain Monte Carlo. NeurIPS 2024.	Nearest principled density-control work. Orthogonal: no filter, no band-limit, no covariance floor. No collision.
2	CHECKED Jiang et al. FisherRF: Active View Selection and Mapping with Radiance Fields Using Fisher Information. ECCV 2024.	Same statistical object, opposite use: scores candidate cameras for acquisition, never a primitive’s covariance. Changes the dataset, not the primitive.
2	CHECKED Liang et al. Analytic-Splatting. ECCV 2024.	Replaces the 2D pixel-window approximation with a closed-form integral. Improves the 2D half of Mip-Splatting; composable with this work, which changes the 3D half.
3	CITED Gaussian Opacity Fields (arXiv 2404.10772).	Source of the improved densification metric later integrated into the Mip-Splatting repository — a reproduction hazard if the commit is not pinned.
3	UNCHECKED 3DGS compression survey (CGF 2025) and the methods it covers.	Where the angular mask memory result must be positioned.

APPENDIX E

The Reproduction Harness

WHAT THIS APPENDIX IS FOR

Nothing here is a result. This is the machinery that produced the results: what runs where, what is checked before anything is measured, how a measurement gets into the record and how it gets out again, and what the compute allowance allowed. A reader who takes Chapter 7 at face value can skip it. A reader who wants to know whether the numbers can be trusted — or who wants to re-run them — will find the answer here.

E.1 Where The Work Runs

There is no local GPU. Every training run in this thesis was executed on Kaggle’s free tier: two NVIDIA T4s, 15 360 MiB each, in sessions capped at twelve hours, against a weekly allowance of 30 GPU-hours per account. That shaped the design more than any other single fact, and it is stated first because several decisions later in this appendix look arbitrary until you know it.

A *run* is one such session. It clones the repository at a named commit, builds the CUDA rasteriser from source, converts the dataset, trains, renders, evaluates, writes one CSV of results and one JSON summary, and prunes everything else so that the session’s output fits in what the platform will hand back. The scripts are `kaggle/*.py`; the launcher that pushes them and collects their output is `tools/kaggle_push.py`.

E.2 Pre-Flight Assertions

An instrument has to be checked before it is used. Twelve assertions run before the first training step and abort the session if any fails. Each one exists because the thing it checks can fail silently, and several of them have.

Three of them:

- **The accelerator check.** Kaggle allocates whichever GPU is free. 3DGS needs compute capability 7.0 and a P100 is 6.0, so a session landing on one would fail forty minutes in with a message that looks like a code fault. It now aborts in thirty seconds with a marker the launcher recognises and relaunches on.
- **The arm-separation diff check.** The 3DGS arm must differ from the Mip-Splatting arm by exactly the intended change and nothing else. The check enumerates the files that change and refuses anything outside the list. It has fired twice in anger — once on a runtime state file that a careless `git add` had committed, and once when the switch of Appendix F reached further than its author believed.
- **The LPIPS backbone.** LPIPS defaults to AlexNet; both papers report VGG. The two differ by more than the effects this thesis measures. The backbone is pinned, asserted, and

Table E.1: The twelve §6 pre-flight checks, as the R0 kernel reported them.

#	result	check
1	pass	accelerator compute capability ≥ 7.0 , abort on P100/6.0 (F15)
2	pass	ninja/gputil/lpips/open3d + rasteriser & simple-knn built against installed torch
3	pass	commits pinned into this run
4	pass	§3 diff present on arm B, absent from main, and confined to its four files
7	pass	multi-scale data written: 4 distinct test scales; metadata lists all 4 train scales, of which only d0 may survive readMultiScale (F3)
8	pass	LPIPS backbone is vgg, never alexnet (F9)
10	pass	metrics.py produced a non-empty results.json (arm A / Mip-Splatting) — its bare ‘except:’ makes a real failure print like a warning (F10)
9	pass	split_by_scale.py’s count-weighted mean reproduces metrics.py’s pooled PSNR (arm A / Mip-Splatting) — proves the render-index $\rightarrow$ scale mapping
10	pass	metrics.py produced a non-empty results.json (arm B / 3DGS) — its bare ‘except:’ makes a real failure print like a warning (F10)
9	pass	split_by_scale.py’s count-weighted mean reproduces metrics.py’s pooled PSNR (arm B / 3DGS) — proves the render-index $\rightarrow$ scale mapping
7.1	pass	readMultiScale loaded the d0-only training split (F3, load_allres=False)
4.1	pass	arm B’s saved model has filter_3D $\equiv 0$ and arm A’s does not (F5) — this, not the diff text, is what makes arm B exactly 3DGS
5	skip	arm B (3DGS) unit test target 33.3 ± 0.5 dB — informational at 7000 iters, the real check is R1 @ 30k full-res
6	skip	arm A (Mip-Splatting) sanity target 33.4 ± 0.5 dB — informational at 7000 iters, the real check is R1 @ 30k full-res
9.1	pass	tools/selftest_pipeline.py: reporting chain verified against a fixture with planted, known-in-advance answers
11	pass	kill at 2000, restart from checkpoint, reach 7000 (scaled down from 30K for this rung)
12	skip	off-session persistence

Emitted by the run itself (`results/kaggle_runs/*/summary.json`); not transcribed.

written into every row of the record as a column, so a row cannot be compared against a published figure without the comparison being checkable.

E.3 The Record

Every measurement lands in one file, `results/runs.csv`, one row per (arm, scene, test scale). The file is **append-only**. A configuration that is re-measured does not overwrite its predecessor; the predecessor stays and is marked, in its own `notes` column, with the name of the stage that replaced it. Nothing is ever deleted.

That discipline exists because a deleted row is a measurement that cannot be re-examined, and this project has needed to re-examine several. It also has a failure mode, which Appendix G describes: a row that is kept but not marked is a row that is silently *used*. The marking is the only thing standing between an honest archive and a pooled average, and the marking is now checked — the generator refuses to emit any table whose cells draw on more than one build.

From The Record To The Document. No measured number in this thesis is typed by hand. `tools/make_tex.py` reads `results/runs.csv` and writes every table and every inline figure as a LaTeX macro; an unmeasured quantity expands to a visible marker that a reader cannot mistake for a result. The same loader feeds the figures and the slide deck, so the three cannot disagree — and a test asserts that they do not, because for a while they did.

E.4 The Apparatus

Table E.2: What exists, and what each artefact is for.

artefact	purpose
<code>results/runs.csv</code>	Append-only. One row per (arm, scene, test-scale, seed), carrying both repository commits, LPIPS backbone, resolution flag, peak VRAM, primitive count, wall-clock and the claimed acceptance level. Every table and figure regenerates from it.
<code>tools/make_tex.py</code>	Writes every measured table and every inline number in this document. A table with no measured rows is emitted as an explicit placeholder carrying the published targets and nothing in the measured columns; it is never quietly filled from the published column.
<code>tools/make_figures.py</code>	Generates every figure. In <code>-mode measured</code> it redraws from the CSV with the published curve retained as a faint reference, and refuses to run if the CSV has no usable rows.
<code>tools/split_by_scale.py</code>	Recovers the per-scale columns that the shipped evaluation pipeline blends into a single average. Without it there is no multi-scale table.
<code>tools/selftest_pipeline.py</code>	Runs the whole reporting chain against a fixture with planted, known-in-advance answers, on a machine with no GPU. It is what caught the two silent defects described below.
<code>kaggle/*.py</code>	The rung kernels, each printing every assertion it makes and ending in a single machine-readable summary line.

Two Defects The Self-Test Caught. The figure generator’s “measured” mode was drawing the *published* curve under a measured filename, because `runs.csv` stores the method as `3dgs` while the figures key off `3DGS`: every lookup missed, the non-empty guard still passed, and the output was a plausible-looking lie. Second, the per-scale splitter assumed a JSON nesting that `metrics.py` does not write, which would have failed only after both arms had trained. Neither would have raised an error where it happened. They are recorded here because a reproducibility claim that has never been tested against a fixture with known answers is an assertion, not a property.

E.5 Cost Per Rung

Table E.3: Every GPU session this project ran, and what it cost.

session	rung	what it covered	GPU-h
r3_real	R3	—	7.48*
r3_tandtdb	R3	4 scenes	6.54
t3_stress	R7	2 scenes, 5 protocols, mip + b1	5.39
r1_stmt	R1	8 scenes	3.97
r7_paired	R7	6 scenes, 1 protocol, mip + b1	3.36*
r2_mtmt	R2	8 scenes	2.99
t2_full1	R7	6 scenes, 1 protocol, b1	2.41
b2	R8	8 scenes, 40 protocols	2.30
pregof	L1P	8 scenes	2.03*
c2_armb	C2	8 scenes	1.81*
r5_seeds	R5	2 scenes	1.80
t3_pencil	T3P	2 scenes, 1 protocol, mip + b1	1.70
c2_armb_mtmt	C2M	8 scenes	1.57*
ship_retry	C2	1 scene	1.05*
t3_mixed	T3M	2 scenes, 1 protocol, mip + b1	1.04
c2_armb_seeds	C2S	2 scenes	0.99*
b2_fair	R8F	2 scenes, 12 protocols	0.89
t2_b1	R7	2 scenes, 1 protocol, b1	0.72
b2_cone	R8C	2 scenes, 12 protocols	0.68
c1_verify	C1	1 scene	0.57*
c2_armb_rest	C2	2 scenes	0.57*
r0_smoke_v5	R0	—	0.47
ship_diag	C2	1 scene	0.35*
c2_armb_ship	C2	1 scene	0.30*
r0_smoke_v4	R0	—	0.25
floor_diag	R7	1 scene, 1 protocol, b1	0.16
c9_fps	C9	—	0.11*
total			51.5

Session wall-clock, which is how the free-tier weekly allowance of 30 h per account is counted; a two-GPU session does twice the work in one hour of it. Read from each session’s own JSON summary, so a repeated run appears at its real cost. * session did not report itself complete — either it was a deliberately narrowed run or it ended early, and §7.15 says which.

Three entries need a sentence each, because they carry the project’s real costs.

- `r3_real` is the largest single session and did not complete. It ran the four indoor Mip-NeRF 360 scenes and was still going when the twelve-hour cap arrived. What it produced is kept and reported as a four-scene subset, clearly labelled; what it did not produce is not estimated.
- The 3DGS-arm re-runs after the opacity-compensation fix — `c2_armb` and its continuations — are pure re-measurement cost, incurred because the arm had been built wrong. They are the price of the defect in Appendix G, and they are itemised so that the price is visible.
- The stress suite is cheap by comparison. Restricting the training cameras does not make training longer, and the sixth protocol — the one that produced the only positive result in this thesis — trains on three cameras and is the fastest run in the table.

The Allowance Was Exhausted, Twice. The first time stalled the programme for three days and is the reason a second account exists. The binding constraint on this project has been compute, not code, throughout; the ladder is ordered so that the cheapest falsifications come first, and the two that could have ended the project — does the Nyquist floor bind everywhere, and is the capture conditioning really anisotropic — were run before a line of method code was written.

APPENDIX F

Implementation

WHAT THIS APPENDIX IS FOR

The thesis compares two published methods and two proposed ones. All four run from a single checkout, differing only by command-line flags, and this appendix says exactly what each flag does and where in the code it acts. It answers “how do I know the 3DGS arm is really 3DGS?” at the level of code.

F.1 Four Configurations

Running each method from its own repository would mean the difference between two numbers contained the difference between two dataloaders, two metric implementations and two LPIPS backbones. Every comparison in this thesis is therefore between configurations of one checkout, and the comparison is the flags.

configuration	flags	what it is
the Mip-Splatting arm	<code>-kernel_size 0.1</code>	Mip-Splatting as published: 3D smoothing filter on, 2D dilation with the opacity compensation.
the 3DGS arm	<code>-kernel_size 0.3</code> <code>-disable_3D_filter</code> <code>-disable_2D_mip_compensation</code>	3DGS’s band-limit semantics exactly: no world-space filter, a fixed 0.3 screen-space dilation, no opacity compensation.
the anisotropic filter	<code>-kernel_size 0.1</code> <code>-use_fisher_filter</code>	Replaces the scalar 3D filter with the anisotropic one, floored at Mip-Splatting’s own value.
the angular mask	post-hoc mask, no training flag	Reads a trained model and the training view directions; masks spherical-harmonic degrees.

F.2 Two Switches For The 3DGS Arm

Mip-Splatting has *two* band-limits and they live in different places.

The first is the 3D smoothing filter: a per-primitive scalar variance added in world space, computed in Python from the training cameras. Zeroing it is a one-line change, and with `-kernel_size 0.3` — 3DGS’s fixed screen-space dilation — the model’s world-space semantics are 3DGS’s.

The second is the screen-space dilation’s *opacity compensation*, $\rho = \sqrt{\det \Sigma' / \det(\Sigma' + kI)}$, which lives in CUDA. Vanilla 3DGS dilates the projected covariance by a fixed 0.3 and applies no such term. A Python attribute cannot reach it.

For a long time the 3DGS arm carried only the first switch. That is the defect of Appendix G: the arm labelled 3DGS was still anti-aliasing with Mip-Splatting’s screen-space term, at exactly the scales where the falsification gate was scored.

What The Compensation Switch Touches. The flag is threaded from `arguments/__init__.py` through `gaussian_renderer` into the rasteriser’s settings tuple, and read in both the forward and the backward CUDA kernels:

- `forward.cu`, `computeCov2D`: ρ is computed only when the flag is off; otherwise the coefficient is 1.
- `backward.cu`, `computeCov2DCUDA`: the same, and the gradient path from opacity through the 2D covariance is removed with it.

Because the switch reaches the binary, the pre-flight asserts that the *compiled* rasteriser carries the field — checking the source would prove nothing about what was built.

F.3 The Degeneracy Guard

Both kernels carry a numerical guard: a primitive whose projected covariance is singular to the tolerance the determinants are clamped at contributes nothing. That guard originally sat *inside* the compensation branch, so switching the compensation off removed the guard as well — which is not what the switch was supposed to isolate. It is now hoisted out, in both kernels, and fires identically in all four configurations.

This is the second lesson of Appendix G and the more general one: a switch meant to isolate a single term has to be checked against what it leaves alone, not only against what it changes.

F.4 The Anisotropic Filter In Code

The anisotropic filter needed no CUDA change. The rasteriser already accepts a precomputed covariance, so replacing a scalar variance with a matrix changes how the covariance is built and leaves the splatting untouched. Three things happen at construction, once, before the training loop:

1. For each primitive, accumulate $\Lambda_k = \sum_n w_{nk} (J_n W_n)^\top \Sigma_{\text{pix}}^{-1} (J_n W_n)$ over the training cameras that saw it — the same Jacobian the renderer already forms for the EWA transform.
2. Invert the spectrum to a covariance floor, and floor *that* at Mip-Splatting’s own scalar value, taken straight from their code. (Taking theirs is what makes Proposition 2 exact in the floor as well as in the covariance; re-deriving it differed by 0.5 % at the median on the first call, which is enough to seed a different densification trajectory.)
3. Add it to the primitive’s own covariance.

Λ_k is never touched inside the loop, which is why rendering is unaffected (§7.9) and why the cost is a construction cost (§7.11).

F.5 Capture Protocols

A protocol is a rule for selecting a subset of the training cameras. The test set is never touched — only the capture the model is fitted from. This is what makes the stress suite cheap: no new images are rendered and no dataset is regenerated.

The subsets are computed in `tools/camera_protocols.py` and passed to the trainer as an index list. `mixed focal` additionally downsamples the selected images, which is a different operation and was, for a while, not being applied at all — the protocol passed its index list and nothing else, so it trained a full orbit under another name. `tools/test_camera_subset.py` now asserts that each protocol selects what it claims to.

APPENDIX G

Measurement Defects

WHAT THIS APPENDIX IS FOR

Eight defects were found in this project's measurement, by checks built for the purpose, and each one produced numbers that looked entirely reasonable while being wrong. They live in one chapter, away from Chapter 7, because a reader following the argument wants the argument, and a reader auditing the work wants all the forensics in one place.

Each entry follows the same shape: what the number said, what was actually happening, how it was found, and what now prevents it. The last of those is the point. A defect that is fixed but not covered by a check is a defect that will recur under a different name — and two of the eight below are the same mistake at different denominators.

The Eight, In One Table

what looked right	what was wrong	what catches it now
The 3DGS arm reproduced 3DGS’s published full-resolution figure	It still carried Mip-Splatting’s screen-space opacity compensation, so it was anti-aliasing better than 3DGS at reduced scales	The switch is asserted on the <i>compiled</i> rasteriser, not the source
The opacity-compensation fix switch isolated one term	It removed three: the compensation, and a degeneracy guard in each of the forward and backward kernels	The guard is hoisted out of the branch in both kernels
Conditioning and rendered quality agreed per protocol	The instrument and the trainer were cloning different branches, so the same protocol name meant two different captures	Both pin a branch; <code>test_camera_subset.py</code> asserts what each protocol selects
The estimation term exceeded the floor everywhere	The constant 0.2 was carried on one side of the comparison and not the other, inflating it fivefold	Proposition 1 is a unit test, passing to 1.6×10^{-16}
Table 7.4’s 3DGS column	Every one of its 32 cells was the mean of two different rasterisers	The generator refuses any table whose cells span more than one build
Mip-Splatting reproduced to 0.50 dB — L2	A seven-scene mean compared against an eight-scene published figure. The true figure is 0.09 dB — L1	Against-published uses the arm’s own scene set; between-arms uses the matched set; a test asserts all three outputs agree
The offset was explained by GOF densification	Asserted for months, never tested. Measured, the effect is 0.11 dB and not consistently signed	The pre-GOF commit is a probe in the record (Table 7.5)
The angular mask lost heavily to magnitude pruning	Against a <i>free</i> control with more freedom than the angular mask has. At matched structure the margin is 0.088–0.415 dB	The control carries the angular mask’s own nested-degree structure

Five of the eight are given in full below. The other three — the denominator, the densification explanation and the angular mask’s unfair control — cannot be separated from the results they correct, so they are told where those results are: §7.2, §7.5 and §7.13 respectively.

One Pattern, Six Times. Six of the eight are the same error: two quantities were compared that were not measured over the same thing. Two rasterisers averaged into one cell. A seven-scene mean against an eight-scene figure. A protocol name meaning two captures. A constant present on one side and not the other. A pruning control with more freedom than the method it judged. In every case the arithmetic was right and the comparison was not, and in every case the result looked plausible — which is why none of them was caught by reading the numbers,

and all of them were caught by a check that asked what the numbers were means *over*.

G.1 The Fourth Check: The Rasteriser On The 3DGS Arm

The single-codebase construction of §6.2 gives both arms Mip-Splatting’s rasteriser, and sets the 3DGS arm apart by zeroing `filter_3D` and taking `kernel_size = 0.3`. That reproduces 3DGS’s 3D band-limit exactly — which is what check 2 verifies — and says nothing about the 2D one. Mip-Splatting’s screen-space filter dilates the projected covariance by k and then compensates the opacity by

$$\rho = \sqrt{\frac{|\Sigma'|}{|\Sigma' + kI|}}, \quad (\text{G.1})$$

so that dilation does not brighten the primitive. Vanilla 3DGS applies the same fixed 0.3 dilation and **no compensation at all**. Setting `kernel_size = 0.3` reproduces the dilation magnitude and leaves ρ in place, so the 3DGS arm kept an anti-aliasing term that 3DGS does not have — and the term it kept is precisely the one that protects a dilated primitive from losing energy as the test sampling rate falls. An arm that anti-aliases better than 3DGS degrades less than 3DGS, which is what the scale-collapse gate measured.

The Test That Separates The Two Explanations. ρ is computed in CUDA, so the switch is a rasteriser change: a `mip_compensation` flag threaded through `GaussianRasterizationSettings` into `computeCov2D`, which skips the ρ term while keeping the dilation. Everything else in that function is identical to `graphdeco-inria/diff-gaussian-rasterization` line for line. The build asserts the compiled binary carries the field before any training starts, because checking the source would prove nothing about the binary.

One scene, `lego`, 30 000 iterations, everything else held fixed:

Table G.1: C1 — the same arm, scene and seed, with and without Mip-Splatting’s 2D opacity compensation. The densification is identical in both columns, so whatever the difference is, it is not densification.

test scale	3DGS arm, with ρ	3DGS arm, without ρ	published 3DGS
full	36.08	36.05	33.33
$\frac{1}{2}$	31.88	27.03	26.95
$\frac{1}{4}$	27.22	21.17	21.38
$\frac{1}{8}$	24.10	17.46	17.69
full $\rightarrow \frac{1}{8}$	11.98	18.58	15.64

`lego`, seed 0, 30000 iterations, identical in every other respect — same branch but for the rasteriser flag, same densification, same data, same seed. The published column is an eight-scene mean and `lego` is not the mean, so it is a shape to compare against, and never a target; this is why the gate was written scene-relative. The left column is read from the rows this run supersedes, which are retained in `results/runs.csv`.

Full resolution barely moves: 36.08 against 36.05, which is what must happen, because at the training sampling rate the dilation is negligible against the primitive’s own extent and $\rho \approx 1$. Every reduced scale collapses, and lands on the shape of the published 3DGS row: 27.03 against 26.95, 21.17 against 21.38, 17.46 against 17.69. The compensation was the whole of the defect.

THE DENSIFICATION HYPOTHESIS

Both columns of Table G.1 ran GOF’s densification. If densification were the cause of the shortfall, the right-hand column could not have collapsed — and it collapses by 6.64 dB at $\frac{1}{8}$, landing within 0.3 dB of the published figure at every reduced scale. The earlier diagnosis is therefore wrong, and it is left standing in this chapter because how a wrong diagnosis survived matters: it was never given an experiment that could refute it. It explained the anomaly as a property of the codebase, which is the most comfortable kind of explanation and the one that most deserves a test.

One point is easy to read the wrong way. The comparison that settles this runs *between the two columns*; either column set against the published row settles nothing. Both columns are **lego** and the published row is an eight-scene mean, so the agreement at reduced scales is corroboration and not a measurement. The measurement is that switching one CUDA term moved $\frac{1}{8}$ by 6.64 dB and full resolution by less than 0.05.

G.2 What The Vanilla Arm Cost In Robustness

Removing the compensation also removed a guard. The same CUDA branch that computes ρ carries `coef = 0` for a primitive whose projected covariance is singular to the tolerance the determinants are clamped at — a numerical guard, not part of the anti-aliasing model. With the compensation switched off, degenerate primitives survived at full opacity with an unbounded screen-space radius, and two scenes died reproducibly, on both GPUs, with a CUDA illegal memory access around iteration 3 000. Six trained normally, which is what made it look transient.

Hoisting the guard out of the branch fixed one of the two. The other then failed in the *backward* pass instead, because the backward kernel still had its copy of the guard inside the branch: the forward was contributing nothing for a degenerate primitive while a gradient still flowed to its opacity. Zeroing a forward contribution and leaving its gradient live is not a consistent pair, and matching them is what the opacity-compensation fix switch should have done from the start — it was supposed to isolate one term and it removed three.

ship Does Not Complete, And Is Reported As Such. With both guards matched it fails a third time, and a fourth under instrumentation. The first three tracebacks all pointed at `ssim`, in the *next* iteration’s loss, which is a red herring: a CUDA fault is reported asynchronously, at whichever call next touches the device. Re-run with `CUDA_LAUNCH_BLOCKING=1` so that each launch is synchronous, it localises instead to the forward rasteriser — the fault surfaces on the first tensor derived from its output, `radii > 0` in `gaussian_renderer`, with the backward pass never reached.

That clears three explanations. The backward kernel, which the second failure had suggested, is clear: the fault surfaces before the backward pass is reached. Memory exhaustion is clear: every attempt raises an illegal address and none raises an allocation error, and **ship** completes on the Mip-Splatting arm on the same 15 360 MiB card at a measured peak of 13 755 MiB. And the onset moves between attempts, which clears the deterministic fault at a fixed point this section claimed before it was instrumented.

The scene has now been attempted nine times on the vanilla arm — four diagnostic runs and then five consecutive attempts in one session, since a fault whose onset moves is a fresh draw each time, so retrying it is a measurement. **All nine failed**, every one with

`cudaErrorIllegalAddress` and none with an allocation error, at iterations

3 000, 7 220, 8 580, 10 100, 10 210, 10 320, 11 820, 12 100

and one earlier still. What varies is *when*, not *whether*: the failure is certain and its timing is not. That settles the question retrying was asked to settle — there is no lucky run to be had — and it is why the scene is reported as failed and left there.

What remains is an out-of-bounds access in the forward raster whose onset depends on the optimisation trajectory, on much the largest model in the suite: `ship` reaches 442 235 primitives on the Mip-Splatting arm, more than any other Blender scene, at $\sim 90\%$ of the card.

It is excluded from the single-scale 3DGS-arm rows, and because every between-arm table in this chapter is restricted to the scenes both arms have, it is excluded from the Mip-Splatting arm's there too — a seven-scene mean set against an eight-scene one compares two different things. Nothing was reduced to make it fit: the protocol's rule for a scene that will not run is to record it and move on, and that rule does not become optional when the scene is one of eight, as it would be one of nine.

Two things keep the exclusion narrow. `ship` trains without incident on the *multi-scale* protocol, on all four scales, so Table 7.4 is a full eight-scene comparison and the failure is specific to single-scale training. And it trains without incident on the Mip-Splatting arm, so the fragility belongs to the vanilla band-limit semantics, with the scene, the data and the harness all cleared. Both point the same way: removing the compensation costs robustness on the one scene that sits closest to the limits of the card, and it does so through the forward raster.

The mechanism is *not* settled beyond that, and the earlier draft of this section overstated it. Hoisting the degeneracy guard out of the compensation branch was a real defect and a real fix — it is what recovered `chair` — but it cannot also be the explanation for the failures that persist after it, because the guard now fires identically whether the compensation is on or off. What can be said from measurement is the conjunction above: forward raster, varying onset, no allocation error, largest model in the suite, vanilla arm only. Naming a cause more specific than that would be inference presented as result, which is the thing §7.3 already cost this project once.

WHAT MAKES A SWITCH A CONTROL

Each of those three removals was invisible: no assertion covered them, the numbers that came back were plausible, and six of eight scenes were unaffected. What exposed them was a failure that could not be waved through — a crash — and then a failure that *moved* from the forward pass to the backward one when half the fix was applied, which is what identified the second half. A switch meant to isolate a single term has to be checked against what it leaves alone, not only against what it changes.

G.3 A Column Averaged Over Two Rasterisers

The 3DGS arm's column here was re-measured after the opacity-compensation fix, and for a while afterwards each of its 32 cells averaged the old measurement together with the new one, where the new one alone belonged there. The record is append-only by design (Table 6.1): a re-measured configuration leaves its predecessor in the file, marked, and the tables skip anything marked. The marking rule identified the rows to retire by the rung name in their notes — and the re-run was the same rung, so its rows carried the same name, the rule

exempted both, and nothing was retired. Both rows then satisfied the table’s selector and were averaged.

Nothing about the output looked wrong. The two builds differ only in whether the degeneracy guards in the forward and backward passes match, which moves a scene by hundredths of a decibel on a capture this well conditioned, so the blended column sat exactly where a plausible column sits. The same defect had quietly taken hold in the anisotropic filter parity rows, where the two builds differed in whether the filter floor was Mip-Splatting’s own value or a re-derivation of it.

What was missing was not a better rule but any check at all on the invariant the rule exists to maintain. There is one now: the generator refuses to emit a table if any reported cell draws on more than one build, and it names the cell. It is four lines, it would have caught both instances the day they appeared, and it subsumes every future variant of the same mistake — which a third special case in the marking rule would not have.

WHAT APPEND-ONLY APPLIES TO

Keeping superseded rows is what makes a result auditable, and this project has been strict about it. But a row that is kept and not marked is a row that is *used*, and the marking is the only thing standing between an honest archive and a silently pooled average. A discipline that depends on a marking step needs the pooled average checked, not the marking step trusted.

G.4 Two Kernels, One Set Of Protocol Names

The instrument results and the PSNR results of §7.12 were, for a time, measurements of *different captures* reported under the same protocol names, and the mechanism matters, because nothing in either run could have detected it.

The instrument kernel cloned the repository’s default branch. The training kernel clones the method branch. Between them, the definition of `arc` and `cone` had changed: the default branch sizes them by angular *width*, giving 20 cameras across 81° and 3 across 30°; the method branch sizes them by camera *count*, giving 25 across 93° and 10 across 54°. The change was deliberate and its stated reason is sound — a 20° wedge catches three cameras on a Blender capture, and three views turns a conditioning experiment into a sparse-reconstruction one. What was missed is that only one of the two kernels was updated.

Each run was internally consistent, each printed the span it had built, and each was correct about its own capture. The error was in reading them side by side, and it survived because both numbers looked reasonable: a cone *is* narrower than an arc under either definition, so the ordering the chapter argues for held either way.

THE COST OF THIS DEFECT

It cost the first reading of the floor-binding and anisotropy gates — and it bought the `pencil` protocol. Rather than choose between the two definitions, both are kept: the count-based `cone` at 51°, which is a trainable subset, and the width-based `pencil` at 20°, which is the regime the derivation is about. The crossover between them is where the estimation term overtakes the floor, and with both in the table that crossover is measured directly. The instrument kernel is now pinned to the branch the training kernel clones, and every protocol table carries the span the session that produced the row actually built.

G.5 A Constant That Was Carried On One Side Only

One more defect has to be stated before the numbers, because it changed them. The estimation term is $(s \sigma_{\text{pix}})^2 / \lambda_i$, and Proposition 1 fixes $(s \sigma_{\text{pix}})^2 = 0.2$ — the same constant Mip-Splatting’s floor $f_k^2 = 0.2 (d_{\min} / f_{\max})^2$ is built from. Both sides of the floor-occupancy instrument comparison therefore carry it. The instrument carried it on the floor and dropped it from the estimation term, which inflated that side by a factor of five, or $\sqrt{5} = 2.24$ in σ . The filter’s own implementation had always applied it.

The symptom was two measurements of one quantity disagreeing — on the arc and the cone, and not at the extremes, where a factor of five changes no verdict. Nothing raised, and both numbers were plausible on their own.

PROPOSITION 1 AS A TEST

The identity that fixes the constant is checkable: at one view, on the optical axis, the estimation term must *equal* the floor. It now does, to 1.6×10^{-16} , and `tools/test_instruments.py` asserts it on every run. Without the 0.2 it is exactly five times the floor, so the test cannot pass either way by accident. This is the over-determination the audit’s M-2 named — “ s defaults to 1” and “ $s \sigma_{\text{pix}} = \sqrt{0.2}$ ” cannot both be free — surfacing as a live numerical error, the most useful form it could take; as an ambiguity in the prose it would have stood for years.

G.6 Three Bugs The Parity Requirement Caught

The first implementation of the anisotropic filter reported exactly what §7.10 predicts internally — the floor binding on 100 % of primitives, the filter isotropic to two decimals — and still differed from Mip-Splatting by up to 1.22 dB, thirty times the seed spread of §7.6. Section 8 names that case in advance: a loss on the full-orbit row is a bug, not a result, because Proposition 2 forbids it. Diagnosing it took three passes, and each step ruled something out.

1. **The floor.** The anisotropic filter re-derives f_k from the camera geometry, so the first suspect was that its floor was not Mip-Splatting’s. A cross-check was added — `compute_3D_filter` runs alongside and the two are compared on every call — and they agree to 1.16×10^{-10} absolute, 10^{-7} relative, for the whole run. Not the cause. The anisotropic filter now takes their floor verbatim anyway, since §4.8 floors at “Mip-Splatting’s own value” and a re-derivation is a needless opportunity to differ.
2. **The covariance path.** The anisotropic filter passed $\Sigma + \Sigma_{\text{flt}}$ through the rasteriser’s precomputed-covariance input while Mip-Splatting passes scales and rotations. The two are algebraically the same, but they are different code. Where the filter is isotropic on every primitive, the anisotropic filter now takes Mip-Splatting’s own path — Proposition 2 used as a computational identity in the code, and no longer only as an argument on paper. The gap did not move: 1.245 dB. Not the cause either.
3. **The opacity.** What both paths share is Equation (4.5)’s compensation, and it contained `clamp_min(1e-12)` on both determinants. $|\Sigma|$ is $\prod s_i^2$: a primitive with $s = 0.005$ has $|\Sigma| = 1.6 \times 10^{-14}$, and one with $s = 0.001$ has 10^{-18} . Both determinants clamped to the same floor, their ratio became exactly 1, and the compensation switched itself off for most of the model. The anisotropic filter was rendering with uncompensated opacity while every baseline number in this thesis was produced with it — which is precisely the observed

signature: denser and sharper at full resolution, worse at every reduced scale.

With the clamp reduced to a division guard, $|\Sigma|$ taken from the scale vector where $\prod s_i^2$ is exact, and the isotropic regime using Mip-Splatting’s own scalar expression, the largest difference over the two scenes then measured fell from 1.245 dB to 0.136 dB. Over the 7 scenes for which a *paired* measurement exists it is now **0.028** dB, comfortably below the seed spread of §7.6.

WHAT “PAIRED” HAS TO MEAN HERE

Proposition 2 says the anisotropic filter *becomes* Mip-Splatting when the floor dominates, so the two numbers being subtracted have to come from the same capture, the same harness invocation and the same build; otherwise the difference also contains whatever separates two runs. That is a stricter requirement than it sounds, and this table did not meet it. The selector behind the macro accepted the standard single-scale rows as a stand-in for Mip-Splatting at the full protocol, so the anisotropic filter was being differenced against a sweep from three commits earlier that had never used the protocol, with two scenes entering twice on one side and once on the other. It reported 0.021 dB. The figure was not wrong by much, and that is the point: a comparison can be mis-specified and still land on a plausible number, which is why the pairing is now enforced in the generator, and why the scene count is a macro that reads from the record.

WHAT MADE THIS FINDABLE

The unit test of §4.8 passed throughout, because its scales were 0.01–0.06, where $\prod s_i^2$ sits at the clamp and the fault is nearly invisible. A test whose inputs do not span the real range will certify a broken component. It now covers $|\Sigma|$ down to 1.4×10^{-18} and prints that range in its own output. The bug itself was silent — no error, plausible numbers, the wrong shape — and what exposed it was requiring parity to hold *exactly*.
